

Operations Project Authorization Form

Approved on June 25, 2026 EPAC

[Link to Meeting Minutes](#)

[Non-redacted detailed PAF link](#)

Project Title: Hyde Park Battery Energy Storage System – Rev 3	Project Number: 21334 (DSS) & 24211 (D-Line)
Date Prepared: 3/25/2024 (Rev 0), 01/12/2024 (Rev 1), 01/27/2026 (Rev 2), 6/12/2026 (Rev 3)	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: D SS D Line
Project Initiator: Amsa Mangga	Project Category: Stations - Batteries
Managing Organization: Project Mgt and Construction	Schedule ID#: 80089926
Project Manager: Adam Kelley	Project Type: Specific
Project Sponsor: Juan Martinez	Capital Investment part of original Op Plan: Yes
Estimated in service date: December 2028	Emergency Related Request? No
Eng./Constr. Resources Budgeted? Yes	Facility Type (check all that apply): ☐ PTF ☐ non-PTF ☐ local PTF ☒ Distribution ☐ N/A (General Plant)
EJ Community: Yes	Authorization Type: Partial Funding
Total Capital Request: D SS (21334) - \$8,914K D Line (24211) - \$2,085K	

[1] Executive Summary

Rev 3 - This revised partial funding Project Authorization Form (PAF) is a request for of \$8,914K for 21334 for continued effort towards completion of preliminary and detailed engineering, Project Management, scheduling, Siting and permitting, Environmental and project services support, preparation of a Full Funding (FF) PAF package, external Engineering costs for Engineer of Choice (EOC) Vendor support up to Issued for Construction (IFC), Internal Engineering costs up to IFC, Strategic Sourcing Partner (SSP) vendor preconstruction support, Siting Legal Support, and procurement of the remaining long lead materials which include the Freestanding Breakers and the Generator Step-Up (GSU) Transformers. The free standing breakers have replaced the metal clad switchgear as part of the recent Value Engineering Effort. The project team will return to SDC before seeking final funding. The Full Funding PAF is expected to be presented to Eversource Project Approval Committee (EPAC) in November 2026. No additional funding is being requested for 24211, which currently has partial funding of \$2,085K approved in PowerPlan on October 1, 2024.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

Partial funding of \$5,767K (\$5,767K for 21334) was approved in PowerPlan on February 18, 2026 (Reference 9). To date, \$3,828K (\$3,828K for 21334; \$0 for 24211) has been spent primarily on internal engineering, project support activities (Siting/Environmental/Outreach), external project management support and related overheads and Allowance for Funds Used During Construction (AFUDC). Conceptual level engineering has been completed; preliminary engineering is expected to be completed in September 2026.

Rev 2:

This revised partial funding PAF is a request for an additional \$1,758K and a new total request of \$5,767K for completion of preliminary engineering, a Limited Notice to Proceed (LNTD) and 10% down for procurement of engineering drawings for the Battery System, project management, siting, environmental and project services support and preparation of a full funding PAF package. Once the items included in this PAF are completed, the next partial funding phase will commence to cover the procurement of the remaining long lead materials which include the switchgear and the Generator Step-Up (GSU) Transformer. The next PAF request would be similar to this request which will incorporate a 10% LNTD for the procurement of engineering drawings of the switchgear and the GSU Transformer, project management, siting, environmental, and project services support.

The requirements for the Article 80 Siting filing **have been better defined with Subject Matter Expert (SME) input on deliverables and durations**. Additional partial funding will be requested in June 2026 for the remaining long lead materials. The full funding PAF is anticipated to be presented in January 2028 and the conceptual level estimate for the full funding request is approximately \$43,926K.

Partial funding of \$4,009K was approved in PowerPlan on July 10, 2024 (Reference 9). To date, \$2,579K has been spent primarily on internal completion of Conceptual Engineering, project support activities, external project management support and related overheads and AFUDC. Engineering has been involved in the initial and rebid of the BESS System Procurement process. Procurement for a Battery System Vendor required a re-bid process to secure a Technically Qualified bidder. This was a four (4)-month delay to the schedule, pushing the in service date (ISD) from August 2028 to December 2028.

Schedule development with SMEs found that the approved shop drawings for several long lead materials are required for both Civil and Substation Electrical to complete Preliminary engineering. **Delays to Civil & Substation Electrical Preliminary Engineering also impact P&C Engineering, Distribution Preliminary Engineering and Siting; the breakers and GSU specs have not been finalized, and the drawings will be available following Vendor onboarding.**

Current long lead material lead times have increased since the last funding request. These lead times for Manufacturer shop drawing needs, fabrication lead time, and the logic changes based on SME needs for Engineering deliverables are still under review by the project team to maintain the targeted December 2028 ISD.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

This project is to construct a 10MW/20MWh Battery Energy Storage System (BESS) at 401-433 Cummins Highway (future Hyde Park BESS Station #359) in the Hyde Park section of the City of Boston to help relieve the heavy loading condition of Hyde Park Substation #496 in Hyde Park, MA.

The key drivers for this revised funding request are as follows.

- Procurement Package #1 (Battery System)
Includes Batteries, Racks, Containers, Disconnects, Inverters, All container auxiliaries, Ground Banks, Fire Protection, Local Controller 1-4, UPS, HVAC, BMS 1-4, PCS 1-4, Gas and Fire Safety Panel.
 - Procurement Package estimated total - \$12,106,850.00 (This partial funding request includes 10% payment (\$1,211K).

The Hyde Park Substation #496 peak load forecast for year 2025 exceeded the station firm capacity and keeps increasing incrementally. A mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project will be in service. Partial Funding (Rev 2) was approved at the Eversource Project Approval Committee (EPAC) meeting on February 27, 2025, for preliminary engineering (Reference 1). The expected in service date for the long-term project is March 2033. This proposed BESS project is the mid-term solution.

The project 22345 - 23375 - Hyde Park-Dorchester Area Plan Land Acquisition (Reference 2) was approved by EPAC on August 3, 2023, for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston has been purchased to locate the transmission substation in the project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 – Hyde Park-Dorchester Area Supply Initiatives project. A separate project 25172 - Hyde Park - Dorchester Area Plan: 200 New England Ave Acquisition (Reference 3) was approved at EPAC on April 11, 2025, for the purchase of an adjacent property to locate the distribution substation in project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project.

This Hyde Park BESS project had initial funding of \$200K (Reference 4) approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR (Reference 5) for \$300K approved in PowerPlan on May 25, 2023. The initial funding was used to complete conceptual level engineering and estimates and development of a partial funding PAF for \$4,009K and Solution Selection Form (SSF) (Reference 6). The partial funding PAF and SSF were presented to and conditionally approved at the Solution Design Committee (SDC) on February 15, 2024. The partial funding was to fund the preliminary engineering, siting, and permitting for the BESS.

In the meantime, another Partial Funding PAF (Rev 0) for \$500K was approved in PowerPlan on April 15, 2024 (Reference 7) to continue engineering and siting until approval of the partial funding PAF presented at SDC in February 2024. The SSF and partial funding PAF presented to the SDC on February 15, 2024, received final approval in May and July 2024, respectively.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Partial Funding of \$2,085K for the D Line portion of the project under project number 24211 was approved by the Eversource Massachusetts Project Approval Committee (MPAC) on August 5, 2024 (Reference 8) for completion of D Line preliminary engineering and permitting. This project was approved by SDC on February 15, 2024.

This Rev 2 revised partial funding of \$5,767K will be used for the following:

- 10% LNTP for the Battery System (Procurement Package #1) – \$1,211K
- Preliminary Engineering and Design of Procurement Package #1 – \$500K
- Continuation of Environmental, Siting and Outreach evaluations – \$200K
- Indirects and AFUDC – \$994K
- Project Management Support – \$283K
- Actuals to Date – \$2,579K

Rev 1: This is a partial funding request of \$4.009M to fund the preliminary engineering, siting, and permitting for the 10 MW/20 MWh utility-scale battery storage system (BESS) at 427 Cummins Highway near Hyde Park Substation #496 in Hyde Park, MA.

The primary objective of the proposed project is to help relieve the heavy loading condition of Hyde Park Substation #496.

Hyde Park Substation #496 is heavily loaded above 90% of the substation load carrying capability (LCC) and will continue to increase each year. A mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192 - Hyde Park-Dorchester Area Supply Long-Term Plan, Partial Funding PAF was approved on August 14, 2023 but will be resubmitted to EPAC to accommodate for a 15-year station plan versus a 10-year one) will be in service. The expected in service date for the long-term project is December 2031. The proposed BESS is the proposed mid-term solution.

The project 22345 – Hyde Park-Dorchester Area Plan: Land Acquisition was approved by EPAC on September 11th, 2023, for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston has been purchased to support project 21192 – Hyde Park-Dorchester Area Supply Long Term Plan.

This project had initial funding of \$200K approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR for \$300K approved in PowerPlan on May 25, 2023. This Partial Funding PAF is being submitted concurrently with the project 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Solution Selection Form).

The amount of \$4.009M will support the project up to the completion of preliminary engineering, at which point full funding will be requested.

Rev 0: The Hyde Park Substation #496 peak load forecast for 2025 exceeded the station firm capacity and keeps increasing incrementally. A mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192-23386-23387-23388-25155-25156-25157-25159-25160-

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

25161 - Hyde Park-Dorchester Area Supply Initiatives project will be in service. Partial Funding (Rev 2) was approved at the Eversource Project Approval Committee (EPAC) meeting on February 27, 2025, for preliminary engineering (Reference 1). The expected in service date for the long-term project is March 2033. This proposed BESS project is the mid-term solution.

The project 22345 - 23375 - Hyde Park-Dorchester Area Plan Land Acquisition (Reference 2) was approved by EPAC on August 14, 2023, for the amount of \$6.275M and the property was purchased in October 2023. The property at 161-191 New England Avenue, Boston has been purchased to locate the transmission substation in the project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 – Hyde Park-Dorchester Area Supply Initiatives project. A separate project 25172 - Hyde Park - Dorchester Area Plan: 200 New England Ave Acquisition (Reference 3) was approved at EPAC on April 11, 2025, for the purchase of an adjacent property to locate the distribution substation in project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project.

This Hyde Park BESS project had initial funding of \$200K (Reference 4) approved in PowerPlan on June 16, 2021, and a revised 21334 - Hyde Park BESS – Battery Energy Storage System Solution, Rev 1 IFR (Reference 5) for \$300K approved in PowerPlan on May 25, 2023. The initial funding was used to complete conceptual level engineering and estimates and development of a partial funding PAF for \$4,009K and Solution Selection Form (SSF) (Reference 6). The partial funding PAF and SSF were presented to and conditionally approved at the Solution Design Committee (SDC) on February 15, 2024. The partial funding was to fund the preliminary engineering, siting, and permitting for the BESS.

In the meantime, another Partial Funding PAF (Rev 0) for \$500K was approved in PowerPlan on April 15, 2024 (Reference 7) to continue engineering and siting until approval of the partial funding PAF presented at SDC in February 2024. The SSF and partial funding PAF presented to the SDC on February 15, 2024, received final approval in May and July 2024, respectively.

Partial Funding of \$2,085K for the D Line portion of the project under project number 24211 was approved by the Eversource Massachusetts Project Approval Committee (MPAC) on August 5, 2024 (Reference 8) for completion of D Line preliminary engineering and permitting. This project was approved by SDC on February 15, 2024.

This Rev. 3 revised partial funding of \$8,914K (\$8,914K for 21334 and \$0K for 24211) will be used for the following

Key drivers:

- **LNTF Payment for Freestanding Breakers (\$100K)**
- **LNTF Payment for GSU Battery Transformers (\$400K)**
- **Engineering External Support vendors for Prelim for P&C & S/S Civil, Fire protection and Stormwater analysis in 2026 (Vendor 1 \$1,500K) & Sound studies, Photo Simulations and Landscaping design in 2026 (Vendor 2 \$59K)**

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

- **SSP Vendor Support for early contractor engagement in support of the development of Engineering & Siting Deliverables, Detailed Project Estimates, Constructability Reviews, Design reviews, Preliminary Schedule, along with the execution of both standard and BESS specific system components to meet the December 2028 in-service date. (\$607K)**
- **Siting External Support Vendor for Legal Services in support of establishing an approach for new Siting reform, assistance with Siting Strategy Meetings for on-going Site Layout, Site Suitability Analysis, permitting and Application Matrix (\$100K)**
- **Engineering Internal Support for Engineering Disciplines (Station Design, Communications Engineering, P&C Engineering, Substation Technical Engineering and Capital Projects) through IFC completion (\$402K)**
- **Project Support by Project Management and Scheduling Team to support project through Full Funding approval (\$248K)**
- **Internal Siting and Permitting Support through Full Funding approval (\$25K)**

Costs not shown above are: Indirects/overhead (\$1,278K), other costs including taxes & CWIP (\$72K), Allowance for Funds Used During Construction (AFUDC) (\$159K), escalation costs factored into estimate (\$135K) and Actuals to date (\$3,828K).

The estimated Full Funding request will be approximately \$43,926K.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

[2] Project Costs Summary

21334 – DSS

Note: Dollar values are in thousands and rounded to nearest thousand

Line item Category	Prior Authorized	Actuals to Date	2026 to Go	Total Request
1. ROW / Easements / Land Acquisition	\$16	\$3	-	\$3
2. Environmental Approvals / Permits	\$65	\$18	-	\$18
3. Outreach	\$110	\$64	-	\$64
4. Siting Approvals / Permits	\$135	\$70	\$129	\$199
5. Engineering / Design	\$843	\$292	\$2,646	\$2,938
6. Materials (Eversource purchased)	\$1,213	\$608	\$546	\$1,154
7. Construction (incl materials by contractors)	-	\$14	-	\$14
8. Testing / Commissioning	\$25	-	-	-
9. Project Mgmt. Team	\$860	\$880	\$255	\$1,135
10. Removals	-	\$14	-	\$14
11. Other	\$24	\$38	\$72	\$110
12. Risks	-	-	-	-
PROJECT DIRECTS TOTAL W/ RISKS	\$3,290	\$2,001	\$3,648	\$5,649
13. Indirects/Overhead	\$2,099	\$1,557	\$1,278	\$2,835
14. AFUDC	\$378	\$271	\$159	\$430
PROJECT TOTAL – BASELINE BUDGET	\$5,767	\$3,828	\$5,085	\$8,914
15. Contingency		-	-	-
PROJECT TOTAL – APPROVAL LEVEL	\$5,767	\$3,828	\$5,085	\$8,914
16. Reimbursables/Customer Contribution		-	-	-
PROJECT TOTAL (LESS REIMBURSABLES)	\$5,767	\$3,828	\$5,085	\$8,914
O&M		-	-	-
TOTAL REQUEST	\$5,767	\$3,828	\$5,085	\$8,914

24211 – D line: no additional funding is being requested for this project number at this time. Current approved funding is \$2,085K.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Detail other costs, project risk items and their allocations:
Other (\$110K): Taxes & Construction Work in Progress (CWIP)

Risk	Allocation
1. Risks will be accounted for in Full Funding Request	\$
2.	\$
Total	\$0

Explain unique payment provisions, if applicable:

Rev 3: The following is a list for the Hyde Park BESS System payment comment for pre/post full funding.

Payment Milestones PRE/POST Full Funding

Payment Milestones <u>PRE/POST AFTER</u> Full Funding						
Equipment	Milestone Payment after Full Funding / Description	PF Rev 2 Payment	PF Rev 3 Payment	Milestone Payment Due (After Full Funding)	Expected Payment Date	Vendor Balance due (After Full Funding)
Batteries (Procurement Package #1)	Submittal of Preliminary Project Schedule (LNTP). – 5% Payment	<u>\$605,342.50 (LNTP)</u>	<u>N/A</u>	<u>N/A</u>	5/15/26	<u>\$11,501,507.50</u>
Batteries (Procurement Package #1)	Design Drawings reviewed and finalized, and Shop Drawings Approved (LNTP). – 5% Payment	<u>\$605,342.50 (LNTP)</u>	<u>N/A</u>	<u>N/A</u>	8/15/26	<u>\$10,896,165</u>
Batteries (Procurement Package #1)	Issuance of Final Notice to Proceed (FNTP). – 10% Payment	<u>N/A</u>	<u>N/A</u>	<u>\$1,210,685 (FNTP)</u>	12/15/26	<u>\$9,685,480</u>
Batteries (Procurement Package #1)	Equipment Factory Acceptance Testing (FAT) (at the last piece of equipment). (FNTP). – 30% Payment	<u>N/A</u>	<u>N/A</u>	<u>\$3,632,055 (FNTP)</u>	11/26/27	<u>\$6,053,425</u>
Batteries (Procurement Package #1)	Equipment Delivery to Site (at the last piece of equipment). (FNTP). – 40% Payment	<u>N/A</u>	<u>N/A</u>	<u>\$4,842,740 (FNTP)</u>	1/28/28	<u>\$1,210,685</u>
Batteries (Procurement Package #1)	Final Acceptance (FNTP) – 10% Payment	<u>N/A</u>	<u>N/A</u>	<u>\$1,210,685 (FNTP)</u>	<u>8/14/28</u>	<u>\$0</u>

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Freestanding Breakers (Procurement Package #3)	Fabrication (FNTF) – 75% Payment	<u>N/A</u>	<u>\$100,000 (25% LNTF)</u>	<u>\$300,000 (75% FNTF)</u>	3/9/27	<u>\$300,000</u>
GSU Battery Transformers (Procurement Package #6)	Fabrication (FNTF) – 80% Payment	<u>N/A</u>	<u>\$400,000 (20% LNTF)</u>	<u>\$1,600,000 (80% FNTF)</u>	4/9/27	<u>\$1,600,000</u>

- Battery Energy Storage System Procurement Package includes:
 - Freestanding Breakers (Siemens SDV7)
 - GSU Battery Transformers

Rev 2:

The following material will need to be ordered using the Material Order Exception Request (MOER) (Reference 10) process prior to full funding to meet the current in-service date due to material lead times, which are 15 months after receipt of order, and drawing lead times, which are 12-16 weeks after receipt of order. Drawings are required to complete preliminary design deliverables for EPAC.

- Battery Energy Storage System
 - DC Power Blocks and Tier 1 controls with required HVAC and Fire protection

The following is a list for the Hyde Park BESS System payment comment for pre and post full funding.

- Milestone Payment Commitments Prior to Full funding: Battery Energy Storage System
 - 10% Payment (\$1,210,685) to support drawing development and reserve manufacturing slot.

Identify and detail any unique O&M impacts resulting from this project:

The Project will incorporate new equipment not normally maintained by Eversource, including battery modules, inverters, and other energy storage system specific equipment. Contractual language and requirements have been placed in the Battery Energy Storage System Vendor Request for Proposal to require training of Eversource personnel, material warranty and spares, and O&M manuals for all non-standard equipment.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

[3] Technical Justification

Project Need, Objective and Justification

Project Need

This project is triggered because of the increasing load growth at the Hyde Park #496 Substation as a result of step loads and electrification. Hyde Park Substation #496 2021 peak load was 157 MVA, which exceeded the 2021 Load Forecast by 7 MVA; it is 108% or 11 MVA above Substation firm capacity of 146 MVA. The substation firm capacity is the remaining capacity with the loss of a Hyde Park Substation 115/13.8kV 75/100/140 MVA transformer. In 2021, 7.7 MVA was transferred permanently off of Hyde Park Substation after the Substation peaked for that year. The 2022 post load transfer peak load was still 152 MVA, 104% of the Substation firm capacity of 146 MVA. About an additional 5 MVA of load was transferred in 2022 after the Substation peaked for that year. About 2.5 MVA was transferred from Hyde Park Substation #496 to Colburn St Substation #350 in 2024. The substation was loaded to about 97% of its firm capacity in 2024. Even with these load transfers, the Hyde Park Substation #496 peak load for year 2025 exceeded the station firm capacity and will keep increasing incrementally.

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving Substation. Building a new Substation to address this thermal constraint is expected to take up to March 2033 (at least) and is a long-term solution to address the capacity constraint in the area. Therefore, a near-term mitigation plan is urgently needed to relieve this thermal Substation constraint until the long-term plan is in service.

Further transfers can only be done temporarily to relieve the Hyde Park Substation #496 load if the station is loaded above its firm capacity under loss of one (1) of the station 115/13.8kV transformers. As noted above, these transfers are not favorable.

Project Objective

The objective of the project is to alleviate Hyde Park Substation #496 heavily loaded condition, to prevent the Substation from being overloaded before the long-term solution is in place. The long-term solution is the new Hyde Park Substation #454 (New England Ave Substation).

Four (4) 2.5MW / 5 MWh battery units will be installed at 401-433 Cummins Highway, each connected to a Hyde Park 13.8kV distribution feeder. The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the Substation peak. The battery can operate through multi-day peak load events.

Hyde Park substation load forecast continues to gradually increase beyond its firm capacity. The forecast will continue to be monitored to evaluate whether the capacity required would need to be delivered by other options until the BESS is in service. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours typically from 6 PM to 8 PM. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Project Scope

The Hyde Park BESS project consists of a 10MW/20MWh energy storage system. The Project will be located at 401-433 Cummins Highway in Boston on Eversource-owned property. The Hyde Park BESS will be operated in parallel with Eversource's (the Owner's) electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system(s), pad mounted transformers, and Freestanding Breakers. The BESS will be located and interconnected to Eversource's existing 13.8kV distribution infrastructure adjacent to the BESS facility. Constructability reviews were performed on 7/20/2023 and 10/31/2023 as part of the partial funding Rev 1 PAF (Attachment C).

This Rev. 3 revised partial funding will be used for the following Key drivers:

- **LNTF Payment for Freestanding Breakers (\$100K LNTF payment is 25% of the total cost)**
- **LNTF Payment for GSU Battery Transformers (\$400K LNTF payment is 20% of the total cost)**
- **Engineering External Support vendors for Prelim for P&C & S/S Civil, Fire protection and Stormwater analysis in 2026 & Sound studies, Photo Simulations and Landscaping design in 2026**
- **SSP Vendor Support for early contractor engagement in support of the development of Engineering & Siting Deliverables, Detailed Project Estimates, Constructability Reviews, Design reviews, Preliminary Schedule, along with the execution of both standard and BESS specific system components to meet the December 2028 in-service date.**
- **Siting External Support Vendor for Legal Services in support of establishing an approach for new Siting reform, assistance with Siting Strategy Meetings for on-going Site Layout, Site Suitability Analysis, permitting and Application Matrix**
- **Engineering Internal Support for Engineering Disciplines (Station Design, Communications Engineering, P&C Engineering, Substation Technical Engineering and Capital Projects) through IFC completion**
- **Project Support by Project Management and Scheduling Team to support project through Full Funding approval**
- **Internal Siting and Permitting Support through Full Funding approval**

Scope of Work

- **D Line (Street) work:** The BESS facility is proposed to interconnect to the following 13.8kV circuits supplied from Hyde Park Substation #496:
 - 441-1401H
 - 362-1207H
 - 362-1209H
 - 496-H7
- **BESS Preliminary Scope of Distribution Line Work as referenced in Attachment D include:**
 - **Eversource Property: Manhole Installation:** three (3) concrete manholes with inside dimension of 6'x10'x8'
 - **Eversource Property: Conduit Installation (1,270')** 6-5" PVC type EB encased in concrete from MH12362 to BESS via three (3) new manholes

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

- Street work: Primary Underground Cable Removal (3,646') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (ten (10) sections)
- Street work: Primary Underground Cable Installation (9,940') of 3-700 FS / 3-1000 FS 15kV primary cable (29 sections)
- Street work: Interconnection to Circuit 496-H7
 - Install 1,270 feet of 3-700 FS (four (4) sections) from MH12362 to new BESS and make joints.
- Street work: Interconnection to Circuit 441-1401H
 - Cut and remove 186 feet of 800/3 cable from MH15729 to MH12365 (one (1) section).
 - Install new cable from MH15729 to MH12362 with 1,400 feet of 3-700 FS (four (4) sections) and make joints.
 - Install 1,270 feet of 3-700 FS (four (4) sections) from MH12362 to new BESS and make joints.
 - Note: one (1) single 3-way joint in MH15729 and three (3) single multiple joints in MH12365.
- Street work: Interconnection to Circuit 362-1207H
 - Cut and remove cable from MH15729 to MH12361 (five (5) sections).
 - Install 1,810 feet of 3-500 FS (five (5) sections) from MH15729 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (four (4) sections) from MH12362 to new BESS and make joints.
- Street work: Interconnection to Circuit 362-1209H
 - Cut and remove cable from MH12365 to MH12361 (four (4) sections).
 - Install 1,650 feet of 3-700 FS (four (4) sections) from MH12365 to MH12361 and make joints.
 - Install 1,270 feet of 3-700 FS (four (4) sections) from MH12362 to new BESS and make joints.
 - Note: one (1) single 3-way joint in MH12361 and three (3) single multiple joints in MH12362.

Rev 2: The partial funding (Rev 2) being requested will be used for material only for 10% down on the Battery System procurement package, as well as for the following:

Distribution System Planning

- Coordinate with the various stakeholders to carry the project up until full funding.
- Conduct Power Systems Computer Aided Design (PSCAD) dynamic study using the specific battery inverter that will be procured for the project.

Transmission System Planning

- Perform transmission system impact study and obtain approvals of Proposed Plan Applications (PPA) from ISO-NE.
 - Develop study scope and report for the study
 - Completed
 - Conduct steady state, stability, short circuit and EMT analysis for the System Impact Study.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

- Completed
- Conduct project inverter model validation in PSCAD.
- Have weekly technical discussion with ISO-NE on the study.

Capital Project Engineering

- Coordinate the overall engineering effort for the external and internal disciplines below.
 - Battery Energy Storage System Vendor
 - Substation Engineering
 - Protection & Control Engineering
 - Telecom Engineering
 - Distribution Engineering
- Finalize technical specifications for the Battery Energy Storage System Vendor Request for Proposal. Engineering has decided to bring on an engineering vendor to support this tech spec development.
 - Completed
- Procurement of Battery Energy Storage System vendor to develop conceptual and preliminary packages required for the installation and interconnection of the battery system as outlined above.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Project Management

- Overall project planning and scheduling
- Project reporting, document control and management, and financial projections
- Battery Energy Storage System Vendor procurement coordination

Siting

- Coordinate and lead siting effort which includes pre-filing meetings with the City of Boston for the Article 80 Small Project siting path. Coordination with Eversource legal and outside legal counsel to assist with the submittal for Small Project Review, along with assistance throughout the duration of the permitting process and securing approval at the Boston Zoning Board of Appeals.

Project Engagement

- Continue the community outreach and engagement strategy required for the project through the Project Engagement and Community Relations teams. This outreach strategy has been reviewed and communicated with the City of Boston and Boston Planning and Development Authority and has been aligned and coordinated with the outreach strategy for project 21192 – Hyde Park - Dorchester Area Supply Initiatives.
 - Project Collateral, i.e., posters and printing services – for community outreach, mailings and public meetings, door to door outreach
 - Cost of tabling events – can vary from free to \$500
 - Translation services – Translations into target languages of community for all collateral, live translations, up to five (5) languages.
 - Branded collateral – totes, coloring books, etc.
 - Food and refreshments for community events

Equity and Environmental Justice (EJ)

- Identify and engage EJ communities, populations, Community Based Organizations (CBOs), influencers, advocates, etc. impacted by or who are relevant for outreach regarding this Project. Identify languages and procure services for language translation/interpretation in EJ stakeholder communities to enhance strategic communication with multilingual customers. Identify and communicate with EJ communities and stakeholders in a broad geographic area to understand historical engagement, determine potential Project, Program, or policy impact, and assist in resolution efforts to open doors, build trust and enable fair outcomes.

Environmental

- Internal project support from the Environmental Affairs Department for Licensing & Permitting and Remediation groups to carry the project up until full funding.
- External project support by the environmental licensing & permitting vendor for:
 - Development of a Soil and Groundwater Management Plan for implementation during project construction that dictates regulatory requirements for construction phase handling, management, and disposal of contaminated soils and groundwater regulated under the Massachusetts Contingency Plan.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

- Development/acquisition of necessary environmental-related permits from Massachusetts Department of Environmental Protection (MassDEP) Solid Waste Bureau for any geotechnical investigations proposed on the capped landfill upon which the BESS facility is being proposed to be built.
- Coordination with MassDEP on any pre-application meetings associated with disturbance/alteration of the capped landfill upon which the BESS facility is being proposed to be built.
- Development/acquisition of necessary environmental-related permits from applicable environmental regulatory agencies for:
 - SW36 Landfill Post-Closure Use Permit from MassDEP Bureau of Solid Waste
 - National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) authorization and Stormwater Pollution Prevention Plan (SWPPP) from U.S. Environmental Protection Agency (USEPA)
 - NPDES Dewatering and Remediation General Permit (DRGP) authorization and Best Management Practices Plan (BMPP) from USEPA (if necessary)

Substation Equipment (Hyde Park BESS Station #359)

Device Additions <i>include manufacturer</i>	Device Position / Nomenclature	Device Removals <i>include manufacturer</i>	Voltage	Quantity
Battery System Nomad Energy	Voyager Eagle LFP	N/A	1000V DC	12
AC Station Service Transformer TBD	TBD	N/A	13.8kV/480V	2
Freestanding Breakers Siemens	SDV7	N/A	15kV Class	4
GSU Transformers TBD	TBD	N/A	480V/13.8kV	4
Metering Package	TBD	N/A	13.8kV	4

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Background

The 13.8kV Hyde Park Substation is forecasted to have a steady increase in electrification load. The system needs for the northern area can be classified into several broad categories:

Capacity Deficiency: Under an N-1 contingency of a Hyde Park 115/13.8kV transformer, the remaining Hyde Park Substation #496 transformer is overloaded. Load is typically transferred during a Substation transformer contingency via distribution circuit ties utilizing automatic field switches. However, with this load transfer, the Hyde Park Substation #496 load is still forecasted to contribute to overloads under an N-1 contingency. 2025 summer peak load was over the Station Firm capacity and future years forecast show an increase in the load. Hyde Park Substation #496 2021 peak load exceeded the Substation firm capacity and almost exceeded the Substation Load Carrying Capability (LCC). The firm capacity is the capacity based on the loss of a Hyde Park Substation #496 transformer, and the LCC is the firm capacity in addition to the amount of load that can be transferred to nearby Substations utilizing distribution circuit ties. In 2021 and 2022, load was transferred permanently off of Hyde Park Substation #496 after the Substation peaked in those years. However, even with these load transfers, the Hyde Park Substation #496 is forecasted to exceed the firm capacity of the Substation and reach 95% of the Hyde Park Substation #496 LCC by 2024. An additional load transfer permanently off of Hyde Park Substation #496 is expected in 2026. Circuit 496-H7 runs 80-90% of its summer normal ampacity and circuit 362-1209H runs 60-90% of its summer normal ampacity in the summer months. Circuit 496-H7 supplies Milton and Dorchester while circuit 362-1209 supplies Milton.

By July of 2027, the capacity to be delivered by project 21334 will need to be delivered by other means. Those options may include deployment of spot generation, voltage reduction, and load shedding during peak load hours during the summer timeframe typically from 18:00 PM to 20:00 PM. Based on the 2026 forecast, one (1) generator of one (1) MW capacity would be needed in 2027 and two (2) generators of one (1) MW capacity each would be needed in 2028, and three (3) generators of one (1) MW capacity each would be needed in 2029 until the BESS is installed. Six (6) MW of generation would be needed in the worst-case scenario in 2030 where the BESS does not get installed until the station is at 100% of LCC. Additional incremental generation, increasing to 11 MW, would be required through 2033. Spot generation would be required to supply load during peak load hours and can remain on standby for the remainder of the time. System Planning will monitor the annual forecast that may shift the plan laid out above.

Transmission Reliability: Various N-1 transmission contingencies may result in loss of the 115kV line supplying the Hyde Park 115/13.8kV transformer including: faults on the 115kV Baker St - Hyde Park lines; or 115kV bus faults at Hyde Park or Baker St; or 115kV breaker failures at Hyde Park or Baker St. These would cause the loss of a Hyde Park 115/13.8kV transformer, which would leave Hyde Park Substation #496 supplied by only one (1) remaining transformer. This results in capacity issues explained in the section above.

Distribution Reliability: Circuits 496-H7 and 362-1209H run up to 90% of their summer normal ampacity. This thermal constraint will be explained further in sections below.

Asset Condition: Both the 110A and 110B transformers at Hyde Park Substation #496 are over 50 years old (installed 1973) and are already in a five (5) year long-term plan for replacement.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Project Background

The Substations surrounding Hyde Park Substation #496 are also heavily loaded including Dewar Street and Andrew Square Substations. Dewar St Substation was forecasted to be above 100% of the firm capacity from 2026 onwards and Andrew Square Substation is forecasted to be above 100% of firm capacity by 2031. Additional capacity is needed in the area to resolve the Hyde Park Substation #496 overload and nearby Substations' heavily loaded conditions. A new Substation is considered as the long-term solution to address the area capacity issues. The project 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project, which addresses the ten (10)-year long-term solution had a partial funding PAF approved; the PAF was then revised and approved to include the 15-year plan.

Below is the Hyde Park ten (10)-year load forecast and percentage of Firm Capacity.

Table 1. Hyde Park Load Forecast

	Station		2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
496	Hyde Park	Forecast Load %age of Firm	157 105%	158 105%	159 106%	161 107%	163 109%	165 110%	167 111%	168 112%	169 113%	172 115%

Load transfers are not a feasible option because they reduce the existing circuit transfer capability, increase the existing distribution lines' exposure, and reduce the spare capacity at the receiving Substation. Building a new Substation to address this thermal constraint is expected to take up to March 2033 (at least) and is a long-term solution to address the capacity constraint in the area. Therefore, a near-term mitigation plan is urgently needed to relieve this thermal Substation constraint until the long-term plan is in service.

Further transfers can only be done temporarily to relieve the Hyde Park Substation #496 load if the station is loaded above its firm capacity under loss of one (1) of the station's 115/15kV transformers. As noted above, these transfers are not favorable.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

[4] Alternatives Considered

Alternative 1 (Preferred Solution): 10MW/20MWh Battery Energy Storage System: Conceptual Level Estimate \$43,926K Preferred

The proposed BESS system is expected to perform peak load shave functionality to relieve the Hyde Park Substation #496 heavy loading condition. The solution was evaluated based on the Dynamic & Steady State Analysis conducted (see Attachment F).

See Figure 1 for the proposed BESS interconnection one line diagram.

Advantages:

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

- BESS will help relieve heavily loaded Hyde Park Substation.
- Construction is confined to a single site where the batteries can each interconnect to four (4) different 13.8kV Hyde Park distribution feeders and provide substation bus diversity.
- Can be completed in a minimum of three (3) years.
- This option has the least overall construction impact on the communities.
- Shorter timeline for Outreach as compared to other alternatives since discussions with the City of Boston is already underway. As part of the overall 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives project and the BESS, outreach and community engagement have already begun with active engagement and participation from the City of Boston and Hyde Park neighborhoods. There has been early involvement with the city for this alternative and the site selection, which helps to smooth the path for siting the BESS with city support and reduces opposition.
- After the five (5)-year window between the BESS installation and the long-term project (new substation installation), the BESS will be redeployed to Dedham #20 Substation.

Disadvantages:

- This alternative will require a City of Boston Article 80E Small Project Review and a conditional use permit from the Boston Zoning Board of Appeals. The estimated timeframe for approval is 12 months.
- If the Article 80E and Zoning Relief process is not successful, a zoning exemption would be required from the Energy Facilities Siting Board (EFSB) which would add significant time to the project.
- Local and neighborhood opposition and concerns have been expected for the BESS development. Project Services and Community Relations have been engaging with the City on this issue and is performing targeted community outreach for this project and the Hyde Park solutions as a whole.

Alternative 2: Addition of Two (2) Distribution Feeders: Not Preferred

Bring additional capacity to Hyde Park area by constructing two 13.8kV distribution lines from Needham Substation. See Attachment E.

Advantages:

- This alternative would relieve the heavily loaded Hyde Park Substation.
- A filing with the EFSB or Department of Public Utilities (DPU) is not anticipated since this is a distribution related solution.

Disadvantages:

- Construction will be crossing multiple towns, busy highways and bridges, having an adverse impact on the community.

This alternative will either take the same amount of time as Alternative 1 or more, due to the extensive in-street work required for this Alternative. Project Services and Community Relations would have to undertake substantial community outreach and dialogue with City officials, which has not commenced. Expected

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

timeline for outreach is approximately 18 – 24 months and continue ongoing thereafter. Extensive permitting would also be needed in order to cross multiple towns.

- Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected.
- Routing options would need to be discussed with cities and other stakeholders.
- Rights would need to be obtained from the towns/cities to install the conduits. Grant of location would be needed for Needham, Dedham and Boston.
- Permits would be needed to cross Route 128 and the bridges, and other entities such as Massachusetts Water Resources Authority (MWRA) or Department of Conservation and Resources (DCR); this can take up to a year.
- The following caveats pose as disadvantages as well and can affect the cost estimates for this alternative:
 - South St in Needham has been recently paved, so it is uncertain whether curb to curb paving would be required. Costs to mill and pave this section are included in the estimate.
 - Two (2) overhead lines will be installed to cross Route 128. This may result in increased labor costs and also decreases reliability of these circuits.
 - Conduit/pipes need to be installed over a bridge crossing at the Dedham/Needham line.
- Expect significant mitigation requests for roadwork restoration, such as curb-to-curb/gutter-to-gutter paving, the Americans with Disabilities Act (ADA) ramp restoration, new sidewalks, and/or traffic calming measures like speed humps, and other community benefits, which could include monetary payments for trees or other types of added green space.
- Depending on the extent of work required at the Needham and Hyde Park substations (not yet known), zoning relief may be required from the Town of Needham and the City of Boston.
- Based on the current forecast, this alternative would not remain as useful after the five (5)-year window between the construction of this alternative and the long-term project; new 115kV/13.8kV substation construction.

Alternative Summary

Although the cost for the Alternative 2 is lower than the preferred alternative, it is critical to note the hurdles and timing delay that would be faced if this alternative were to be implemented as explained above. Opposition in Needham is expected since a project was just completed through Needham. If residents object or create organized opposition, significant delays to approvals are expected. There are also potential hurdles involved in gaining rights and permits for alternative 2. Alternative 2 may take significantly longer than the BESS to implement as explained above. Outreach for the BESS Alternative 1 is already ongoing which helps to smooth the path for siting the BESS with City support and reduces opposition whilst there has been no outreach for Alternative 2.

Two (2) of the feeders that the BESS would connect to would also benefit from the BESS, Circuit 496-H7 supplied from Bus 1 historically runs 80-95% of its summer normal ampacity in the summer. The circuit 362-1209 supplied from Bus 4 historically runs 60-90% of its summer normal in the summer. The BESS would help to reduce the loading on these circuits.

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

The BESS can be redeployed to other stations where it can be useful after the five (5)-year window between the BESS installation and the long-term project construction. However, based on the current forecast, Alternative 2 would not be as useful after the five (5)-year window when the long-term project or new 115/13.8kV substation is in service to relieve load from Hyde Park Substation #496 and surrounding substations.

There was a River Street BESS solution considered but it was rejected as it did not include as much flexibility in connecting to different distribution circuits and would entail further System Planning studies and construction work, which would delay the project significantly. Such a delay would defeat the purpose of having a short-term solution to the capacity issue at this substation. Both alternatives would involve supplying a minority Environmental Justice zone.

Non-Wires Solution Alternatives

A pre-screening assessment was performed based on the scope identified. Projects that answer 'No' to one or more item of the questionnaire do not need to have a non-wires alternative (NWA) solution screening performed. As shown in Table 2 below, all responses are 'Yes' and an NWA screening has been performed.

NWA Pre-Screening Criteria	Response
Preferred Solution Capital Cost > \$3M	Yes
Capacity Need > 3+ Years	Yes
Main Project Driver is Capacity	Yes

Table 2. Non-Wires Alternative pre-screening assessment

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

[5] Project Schedule

Milestone	Scheduled Completion Date
IFR Approval	June 2021 May 2023 (Rev 1)
SDC Approval (for applicable projects)	February 2024
Conversion of Tx development costs, 183 to 107 (applicable to projects that require SSF approval and/or ISO preferred alternative selection)	N/A
EPAC Partial Funding Approval (for applicable projects)	April 2024 (Rec 0) July 2024 (Rev 1) February 2026 (Rev 2) June 2026 (Rev 3)
Preliminary Design Completion (or latest pre-full funding design phase)	September 2026
EPAC Full Funding Approval	November 2026
Siting Approval (for applicable projects)	February 2028
Start of Construction	February 2028
Project In-Service Date (ISD)	December 2028
Project Need Date (for applicable projects)	Present
ISO Related Milestones: (for applicable projects)	
• ISO Preferred Alternative Selected	N/A
• PAC Presentation	N/A
• TCA Submittal	N/A

See Attachment B for the P6 schedule.

[6] Regulatory Approvals

Regulatory Approvals will be included in the Full Funding PAF.

- The installation of the BESS is expected to require Article 80 Small Project Review from the Boston Planning Department and a Conditional Use Permit from the Boston Zoning Board of Appeal. Siting, with the assistance of outside legal counsel, would file for and obtain the required zoning relief.
- If the City is unsupportive of providing zoning relief, then Siting would file for a zoning exemption request to the EFSB.
- The expected timeframe for a zoning exemption would be about 18 months to 24 months from date of filing to a decision.
- Necessary environmental permit approvals include:
 - o Environmental Protection Agency (EPA) National Pollution Discharge Elimination System (NPDES) Construction General Permit (CGP) and Stormwater Pollution Prevention Plan (SWPPP)
 - ☐ Requires development/engineering of structural stormwater management system for facility and development of Operations and Maintenance (O&M) Plan for long term operation of stormwater management system, including semi-annual inspection, maintenance, and record keeping requirements
 - ☐ Requires regular construction phase erosion and sediment control inspections and reporting
 - o Massachusetts Department of Environmental Protection (MassDEP) Landfill Post-Closure Use Permit (PCUP)
 - ☐ Requires both construction phase and long-term operation inspection, monitoring, and reporting requirements
 - ☐ May require installation of sophisticated gas monitoring system inside facility building to send real-time alert to Operations group for any buildup of combustible and/or toxic gases within enclosed spaces
 - o Soil and Groundwater Management Plan
 - ☐ Dictates regulatory requirements for construction phase handling, management, and disposal of contaminated soils and groundwater regulated under Massachusetts Contingency Plan
 - o Spill Prevention, Control, and Countermeasure Plan (SPCC) and Tier II Reporting to EPA, MassDEP, Massachusetts Emergency Management Agency (MEMA), Local Fire Dept / Emergency Response Agency
 - ☐ Required for any new oil filled equipment with capacity >1,320 gal.
 - ☐ Required for onsite storage/use of large quantities of hazardous materials (batteries)

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

EJ Community Considerations:

EJ Mapping

Identify EJ Communities and populations relevant to Project/Program, through creation of an EJ Map. EJ Map will identify, by Census Block:

- Neighborhoods with English isolation
- Low-income areas
- Areas with concentrations of racial/ethnic minorities
- Areas with vulnerable populations (including those with high existing environmental burdens)
- Areas with combinations of any of the above
- Distressed municipalities and any sub-areas with concentrations of EJ communities
- Areas containing sensitive infrastructure or critical facilities (e.g., public housing, nursing/elder care homes or facilities, parks, hospitals, schools)
- Areas containing high residency of low to moderate income renters

Stakeholders Mapping & Engagement

Identify and communicate with communities and stakeholders in a broad geographic area to determine historical engagement and determine potential Project, Program, or policy impact, including local community-based organizations and leaders, state-wide Non-Governmental Organizations (NGOs) and indigenous tribes.

Language Translation Services

Identify languages spoken in stakeholder communities to enhance strategic communication with multilingual customers. Language identification data shall be used to inform and secure interpreters, generate appropriate translations of materials, conversations, dialogue, and in-community events.

Public Engagement

Host public engagement processes, including through non-traditional and community-based communication channels, to solicit public input. This may include, as applicable:

- Determining the level of community interest
- Conducting analysis of stakeholders who will be immediately impacted by the delivery or implementation of the program/project/services
- Creating and implementing engagement & communication strategies
- Hosting in person and or virtual public feedback sessions to clearly define program/project/services scopes of and timelines
- Identify additional stakeholders to attend future public engagement activities and follow-up on feedback received

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Feedback

Establish a database to track all outreach events, interactions, materials, communications received from communities, individuals, and organizations; record all responses, all changes/modifications to projects or programs, and all remediation, mitigation, and community benefit arrangements made to or with respect to relevant EJ communities.

[7] Risk and Risk Mitigation Plans

Contingency:

Contingency will be included in the full funding PAF.

Risk Mitigation:

Risk	Mitigation
1. Risks and associated mitigation plans will be included in the full funding PAF.	
2.	
3.	
4.	
5.	

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

[8] References

References
1. 21192-23386-23387-23388-25155-25156-25157-25159-25160-25161 - Hyde Park-Dorchester Area Supply Initiatives Partial Funding - PAF Rev 2, approved 2/27/2025
2. 22345 - 23375 - Hyde Park-Dorchester Area Plan Land Acquisition Full Funding – PAF Rev 2, approved 8/14/2025
3. 25172 - Hyde Park - Dorchester Area Plan: 200 New England Ave Acquisition Full Funding – PAF, approved 4/11/2025
4. 21334 – Hyde Park BESS – Battery Energy Storage System Solution - IFR, approved 6/11/2021
5. 21334 - Hyde Park BESS – Battery Energy Storage System Solution IFR Rev 1, approved 5/9/2023
6. 21334 – Hyde Park BESS – Battery Energy Storage System Solution, Solution Selection Form, approved 2/15/2024
7. 21334 - Hyde Park Battery Eng Storage Sys - Partial Funding – PAF, approved 3/28/2024
8. 21334 - 24211 - Hyde Park Batter Energy Storage System Partial Funding – PAF, MPAC approved 08/05/2024
9. 21334, 24211 - Hyde Park BESS - Battery Eng Stgr Sys - Partial Funding – PAF (Rev 2), approved 2/5/2026
10. 21334-Hyde Park BESS Material Order Exception Request (MOER) - Package #1 only_Final_pjm (002) _tpr

[9] Attachments (One-Line Diagrams, Images, etc.)

Attachment	Included in Submittal?
A1. 21334 E-24-149 Hyde Park D Station Overhead_AFUDC Calculation REV 25.xlsm R2 (Rev 0)	YES
A2. 21334 E-24-197 Hyde Park BESS (D Station) EST Rev EXT (Rev 1)	YES
A3. ESF - Hyde Park BESS (PF) r0-2 (Rev 2)	YES
A4. ESF – Hyde Park BESS – PF 3 Rev01	Yes
B1. Hyde Park Battery Storage Project - Full Remaining Activities Schedule 03-06-24 (Rev 0)	YES
B2. Hyde Park Battery Storage Project - All Activities Schedule 06-05-24 (Rev 1)	YES
B3. 21334 - Hyde Park - All Remaining Activities - 6-08-26 Rev 3	YES
C. 21334 Constructability Review Form (Rev 1)	YES
D1. 21334 Hyde Park BESS Scope of Work Dist (002) (Rev 1)	YES
D2. Hyde Park BESS Scope of work (Rev 2)	YES
E. New Needham 13.8kV Feeder One Lines (Rev 1)	YES
F. Hyde Park BESS Dynamic & Steady State Study Report (Rev 2) - CEII	YES

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

EVERSOURCE

Project Authorization Form

INSTRUCTIONS:

It is the responsibility of the initiator to contact the subject matter experts in the listed area disciplines to determine if the project considerations contained in this list are applicable to their project. They should fill out the checklist and determine a transition plan for the purpose of project execution. See the PAF Guide for additional instructions.

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park Battery Energy Storage System – Rev 3	Project Number: 21334 (D SS), 24211 (D Line)
Facility Type: ☐ BPS ☐ BES ☐ PTF ☐ local PTF ☐ non-PTF ☐ CIP ☒ Distribution ☐ General Plant ☐ CONVEX ☐ SCADA	
PLANNING	
Is a NX-9 required?	No
Is an ISO-NE PAC presentation required?	No
ISO-NE Presentation Date (if completed):	N/A
Cost in ISO-NE Presentation (\$M) (if completed):	N/A
Is a PPA required?	Yes
ISO-NE Approval Date (if completed):	10/23/2025
Is a TCA Application Required?	No
TCA Submittal Date (if submitted):	N/A
Cost in TCA Submittal (\$M) (if submitted):	N/A
Enter ISO-NE RSP / Asset Condition ID, if assigned:	ES-23-LSP-032
PLANNING/PROTECTION & CONTROLS	
Are RAS/SPS/UVLs affected?	No
OPERATIONS	
Outage Required?	☐ Primary Equipment (Power Transfer) ☒ Secondary Equipment (P&C only) ☐ Outage Not Required
Do SCLL Conditions Exist?	No
Has an outage schedule been approved?	No
Are Operations & Maintenance procedures/training required?	Yes
STANDARDS	
Does the project include standard equipment and designs?	Yes
SUBSTATION ENGINEERING	
Does this impact Revenue Metering	No
Is preliminary short circuit/ breaker duty analysis required?	No
Are there any changes to the baseline audible noise?	Yes
Is there an impact to the existing ground grid?	No
Is a Transient Over Voltage (TOV) analysis required?	Yes

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

EVERSOURCE

Project Authorization Form

Project Checklist - Transmission & Substation Capital Project	
Project Name: Hyde Park Battery Energy Storage System – Rev 3	Project Number: 21334 (D SS), 24211 (D Line)
P&C ENGINEERING	
OP-22 - Are PMUs and DDR required?	No
If BPS, is an NPCC Directory #4 presentation required?	No
Are ISO-NE OP-24 Appendix B updates necessary?	No
TRANSMISSION LINE ENGINEERING	
Are there any changes that affect the baseline EMF?	No
Are there any changes that affect the baseline EMI?	No
SITING	
Has Siting reviewed this project?	Yes
Name of Siting contact:	Eamon McGilligan
Is a Siting filing required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PERMITTING	
Is there any permitting required? <i>(If yes, list in regulatory approvals section)</i>	Yes
PROJECT SERVICES (OUTREACH) and ROW WORK COORDINATION	
Has Project Services reviewed this project?	Yes
Name of Project Services contact:	Gaby Quintero
What is the level of outreach expected?	High
Has ROW Coordination reviewed this project?	Yes
Is this project coordinating with other existing scheduled work?	No
ENVIRONMENTAL JUSTICE COMMUNITIES	
Has the Equity and Environmental Justice team reviewed this project?	Yes
Name of Equity and Environmental Justice contact:	Alicine Francois
Is any part of this project in an Environmental Justice Community?	Yes
Was the Equity Framework Checklist completed?	No
Has enhanced outreach required per State Regulatory requirements been completed?	No
INITIATOR	
Has a field constructability review been completed?	Yes

Project Checklist - Transmission & Substation Capital Project
--

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

EVERSOURCE

Project Authorization Form

Project Name: Hyde Park Battery Energy Storage System – Rev 3	Project Number: 21334 (D SS), 24211 (D Line)
INVESTMENT RECOVERY	
Does the project require development of an Investment Recovery plan?	No _____
COST ESTIMATING	
What team/firm prepared the cost estimate?	Estimate was prepared by the Eversource Estimating Team _____
Name of the person who prepared the estimate:	Brian Roche _____
Was the estimate reviewed by Eversource Estimating?	Yes - Full Review _____

This document contains Critical Energy Infrastructure Information (CEII) which has been redacted.

Form Template Revised 03/13/2024

APS 1 – Project Authorization Policy Issued 12/31/21 Rev. 6

Policy Sponsor: EVP and CFO

JA-AM-2001-A Rev. 7

ATTACHMENT A1 - A4

Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

And on years e version; rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$3						\$8
3. Outreach		\$12	\$3						\$15
4. Siting Approvals / Permits		\$12	\$9						\$21
5. Engineering / Design		\$17	\$7						\$24
6. Materials (Eversource purchased)		\$2							\$2
7. Construction (incl matl's by contractors)		\$11							\$11
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$88	\$3						\$91
10. Removals									
11. Other		-\$1							-\$1
12. Risks									
Project Directs Total w/Risks		\$148	\$23						\$170
13. Indirects/Overhead		\$222	\$47						\$269
14. AFUDC		\$30	\$30						\$60
Project Total - Baseline Budget		\$400	\$100						\$500
15. Contingency									
#N/A									
Project Total - Funding Approval Level		\$400	\$100						\$500
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)		\$400	\$100						\$500
Total O&M									
TOTAL REQUEST	\$300	\$400	\$100						\$500

Breakout costs incl above

ST Labor		\$108	\$17						\$125
OT Labor									
Outside services - Labor		\$38	\$6						\$44
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		-\$1							-\$1
Indirects		\$222	\$47						\$269
AFUDC		\$30	\$30						\$60
TOTAL CAPITAL REQUEST - with out Reimbursables	\$300	\$400	\$100						\$500

- | Worksheet: WBS-076 | | Worksheet: WBS-076 | | | | | | | | | |
|---|-----------------------------------|--------------------|----------------|------|---------|------|--------|------|------------------|----|---------|
| Direct Costs | | | | | | | | | D - Station Work | | |
| WBS Code | WBS Description | Prior Authorized | Actual to Date | 2024 | 2025 | 2026 | 2027 | 2028 | Total | | |
| Costs | | | | | | | | | | | |
| Eversource Internal | | | | | | | | | | | |
| 1 | RDM / Eversource / Land purchase | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 2 | Environmental Approvals / Permits | \$ | 5,532 | \$ | 2,000 | | | | | | |
| 3 | Outreach | \$ | - | \$ | 8,400 | \$ | 2,000 | \$ | 12,932 | | |
| 4 | Siting Approvals / Permits | \$ | 12,649 | \$ | 6,000 | | | | | | |
| 5 | Engineering / Design | \$ | 5,509 | \$ | 2,000 | | | | | | |
| 6 | Materials | \$ | 2,248 | | | | | | | | |
| 7 | Construction | \$ | 10,963 | | | | | | | | |
| 8 | Traffic / Contaminating | \$ | 43,831 | \$ | 1,000 | | | | | | |
| 9 | FMT Project Mgmt Team | | | | | | | | | | |
| 10 | Removals | | | | | | | | | | |
| 11 | Other | | (1,037) | | | | | | | | |
| 12 | Risk | | | | | | | | | | |
| 13 | Contingency | | | | | | | | | | |
| 15 | Contingency | | | | | | | | | | |
| | | \$ | - | \$ | 147,344 | \$ | 22,550 | \$ | - | \$ | 176,844 |
| Outside vendor or contractor | | | | | | | | | | | |
| 1 | RDM / Eversource / Land purchase | | | | | | | | | | |
| 2 | Environmental Approvals / Permits | | | | | | | | | | |
| 3 | Outreach | | | | | | | | | | |
| 4 | Siting Approvals / Permits | | 2,423 | | - | | | | | | |
| 5 | Engineering / Design | | 11,155 | | 4,500 | | | | | | |
| 6 | Materials | | | | | | | | | | |
| 7 | Construction | | | | | | | | | | |
| 8 | Traffic / Contaminating | | | | | | | | | | |
| 9 | FMT Project Mgmt Team | | 24,633 | | 1,500 | | | | | | |
| 10 | Removals | | | | | | | | | | |
| 11 | Other | | | | | | | | | | |
| 12 | Risk | | | | | | | | | | |
| 13 | Contingency | | | | | | | | | | |
| | | \$ | - | \$ | 147,344 | \$ | 22,550 | \$ | - | \$ | 176,844 |
| Escalation (if not included above) | | | | | | | | | | | |
| Escalation rates - Eversource Labor | | | | | | 3.0% | 5.1% | 5.2% | 12.0% | | |
| Escalation rates - subcontract labor | | | | | | 3.0% | 5.1% | 5.2% | 12.0% | | |
| Escalation rates - Materials | | | | | | 1.0% | 4.0% | 4.1% | 6.1% | | |
| Eversource Internal | | | | | | | | | | | |
| 1 | RDM / Eversource / Land purchase | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 2 | Environmental Approvals / Permits | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 3 | Outreach | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 4 | Siting Approvals / Permits | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 5 | Engineering / Design | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 6 | Materials | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 7 | Construction | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 8 | Traffic / Contaminating | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 9 | FMT Project Mgmt Team | | | | | | | | | | |
| 10 | Removals | | | | | | | | | | |
| 11 | Other | | | | | | | | | | |
| 12 | Risk | | | | | | | | | | |
| 13 | Contingency | | | | | | | | | | |
| | | \$ | - | \$ | - | \$ | - | \$ | - | \$ | - |
| Outside vendor or contractor | | | | | | | | | | | |
| 1 | RDM / Eversource / Land purchase | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 2 | Environmental Approvals / Permits | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 3 | Outreach | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 4 | Siting Approvals / Permits | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 5 | Engineering / Design | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 6 | Materials | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 7 | Construction | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 8 | Traffic / Contaminating | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 9 | FMT Project Mgmt Team | | | | | | | | | | |
| 10 | Removals | | | | | | | | | | |
| 11 | Other | | | | | | | | | | |
| 12 | Risk | | | | | | | | | | |
| 13 | Contingency | | | | | | | | | | |
| | | \$ | - | \$ | - | \$ | - | \$ | - | \$ | - |
| Escalated costs | | | | | | | | | | | |
| Eversource Internal | | | | | | | | | | | |
| 1 | RDM / Eversource / Land purchase | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 2 | Environmental Approvals / Permits | \$ | - | \$ | - | \$ | - | \$ | - | | |
| 3 | Outreach | \$ | - | \$ | - | \$ | - | | | | |

Project Number:

Project Title:

D- Station

PAF (dollars in thousands)

And on years e version; rev25

Line item Category	Prior Authorized	Actuals to Date	2024 to Go	2025	2026	2027	2028	2029	Total
1. ROW / Easements / Land purchases									
2. Environmental Approvals / Permits		\$6	\$156	\$161					\$323
3. Outreach		\$14	\$13	\$13					\$39
4. Siting Approvals / Permits		\$13	\$114	\$117					\$244
5. Engineering / Design		\$30	\$169	\$174					\$374
6. Materials (Eversource purchased)									
7. Construction (incl matl's by contractors)		\$14	\$11	\$11					\$36
8. Testing / Commissioning									
9. PMT Project Mgmt Team		\$70	\$154	\$158					\$382
10. Removals									
11. Other		\$4							\$4
12. Risks									
Project Directs Total w/Risks		\$151	\$616	\$635					\$1,401
13. Indirects/Overhead		\$235	\$1,008	\$1,038					\$2,282
14. AFUDC		\$33	\$86	\$208					\$326
Project Total - Baseline Budget		\$418	\$1,710	\$1,880					\$4,009
15. Contingency									
#N/A									
Project Total - Funding Approval Level	\$500	\$418	\$1,710	\$1,880					\$4,009
16. Reimbursables/Customer Contribution									
#N/A									
Project Total (less reimbursables)	\$500	\$418	\$1,710	\$1,880					\$4,009
Total O&M									
TOTAL REQUEST	\$500	\$418	\$1,710	\$1,880					\$4,009

Breakout costs incl above

ST Labor		\$114	\$301	\$310					\$726
OT Labor									
Outside services - Labor		\$30	\$315	\$324					\$670
Materials (All materials incl. outside service purchased which is shown in constr. above)		\$2							\$2
Removals									
Other		\$4							\$4
Indirects		\$235	\$1,008	\$1,038					\$2,282
AFUDC		\$33	\$86	\$208					\$326
TOTAL CAPITAL REQUEST - with out Reimbursables	\$500	\$418	\$1,710	\$1,880					\$4,009

\$ 4,000,144

Project name **Hyde Park BESS (PF)**

Project no **21334**

Project manager **ADAM KELLEY**

Project estimator **Muthana Al-Quraishi**

Business unit **Distribution**

Ext accuracy

			Hyde Park BESS (PF)
Description	% Directs	% Total	REV01
ROW / Easements / Land purchases	0.5%	0.3%	16,100
Environmental Approvals / Permits	2.0%	1.1%	65,000
Project Services	3.3%	1.9%	109,500
Siting Approvals / Permits	4.1%	2.3%	134,500
Engineering / Design	25.6%	14.6%	843,000
Materials (Eversource purchased)	36.9%	21.0%	1,213,000
Construction (inc contractor sourced material)	0.0%	0.0%	-
Testing / Commissioning	0.8%	0.4%	24,900
PMT Project Mgmt Team	26.1%	14.9%	859,500
Removals	0.0%	0.0%	-
Other (including CWIP & Taxes)	0.7%	0.4%	24,200
Risks	0.0%	0.0%	-
Project Directs (including risks)	100.0%	57.0%	3,289,700
Indirects/Overhead		36.4%	2,098,900
AFUDC		6.6%	377,800
Project Total - Baseline Budget		100.0%	5,766,400
Contingency		0.0%	-
Project Total - Approval Level		100.0%	5,766,400
Reimbursables/Customer Contribution		0.0%	-
Project Total (less reimbursables)			5,766,400
Metrics			
Quantity			1.00 EA
Cost/Quantity			5,766,400 \$/EA
Escalation (included above)			-
Miscellaneous			
Risks & Contingency / Project Total - Approval level (less Risks & Contingency)			0.00%
Estimate Range			
Low			4,324,800
Expected			5,766,400
High			7,208,000

CBS1 > 2 > 3 > 4	Description	Total Amount	Actual Amount	Labor Hours	Labor Amount	Material Amount	Subcontract Amount	Equipment Amount	Other Amount
	Hyde Park BESS (PF)	5,766,328	2,578,757	-	850,000	1,210,685	-	-	1,126,886
	Hyde Park BESS (PF) - REV01	5,766,328	2,578,757	-	850,000	1,210,685	-	-	1,126,886
01	ROW / Easements / Land purchases	16,066	16,066	-	-	-	-	-	-
02	Environmental Approvals / Permits	64,968	14,968	-	50,000	-	-	-	-
03	SCS Outreach	109,503	59,503	-	50,000	-	-	-	-
04	Siting Approvals / Permits	134,484	34,484	-	100,000	-	-	-	-
05	Engineering / Design	843,016	343,016	-	500,000	-	-	-	-
06	Equipment /Materials (Eversource purchased)	1,213,043	2,358	-	-	1,210,685	-	-	-
08	Testing / Commissioning	24,915	24,915	-	-	-	-	-	-
09	PMT Project Mgmt Team	859,455	576,658	-	150,000	-	-	-	132,797
11	Other	24,196	24,196	-	-	-	-	-	-
13	Indirects/Overhead	2,098,928	1,280,192	-	-	-	-	-	818,736
14	AFUDC	377,754	202,401	-	-	-	-	-	175,353
Grand Total		5,766,328	2,578,757	-	850,000	1,210,685	-	-	1,126,886

Project name **Hyde Park BESS - PF 3**Project no **21334**Project manager **ADAM KELLEY**Business unit **Transmission**Project estimator **Brian Roche**

Ext accuracy

Description	% Directs	% Total	Hyde Park PF 3
ROW / Easements / Land purchases	0.1%	0.0%	3,100
Environmental Approvals / Permits	0.3%	0.2%	18,000
Project Services	1.1%	0.7%	64,000
Siting Approvals / Permits	3.5%	2.2%	198,500
Engineering / Design	52.0%	33.0%	2,937,800
Materials (Eversource purchased)	20.4%	12.9%	1,154,100
Construction (inc contractor sourced material)	0.2%	0.2%	14,000
Testing / Commissioning	0.0%	0.0%	-
PMT Project Mgmt Team	20.1%	12.7%	1,135,100
Removals	0.2%	0.2%	14,100
Other (including CWIP & Taxes)	1.9%	1.2%	109,900
Risks	0.0%	0.0%	-
Project Directs (including risks)	100.0%	63.4%	5,648,600
Indirects/Overhead		31.8%	2,834,800
AFUDC		4.8%	430,200
Project Total - Baseline Budget		100.0%	8,913,600
Contingency		0.0%	-
Project Total - Approval Level		100.0%	8,913,600
Reimbursables/Customer Contribution		0.0%	-
Project Total (less reimbursables)			8,913,600
Metrics			
Quantity			1.00 LS
Cost/Quantity			8,913,600 \$/LS
Escalation (included above)			180,600
Miscellaneous			
Risks & Contingency / Project Total - Approval level (less Risks & Contingency)			0.00%
Estimate Range			NA
Low			#N/A
Expected			8,913,600
High			#N/A

CBS1 > 2 > 3 > 4		Total Amount	Actual Amount	Labor Hours	Labor Amount	Material Amount	Subcontract Amount	Equipment Amount	Other Amount
Description									
Hyde Park BESS - PF 3		8,913,710	3,828,338	-	3,030,203	546,350	-	-	1,508,818
Hyde Park PF 3		8,913,710	3,828,338	-	3,030,203	546,350	-	-	1,508,818
01	ROW / Easements / Land purchases	3,072	3,072	-	-	-	-	-	-
02	Environmental Approvals / Permits	18,049	18,049	-	-	-	-	-	-
03	SCS Outreach	63,973	63,973	-	-	-	-	-	-
04	Siting Approvals / Permits	198,532	69,782	-	128,750	-	-	-	-
05	Engineering / Design	2,937,833	291,820	-	2,646,013	-	-	-	-
06	Equipment /Materials (Eversource purchased)	1,154,050	607,700	-	-	546,350	-	-	-
07	Construction (incl mat'l's by contractors)	14,044	14,044	-	-	-	-	-	-
09	PMT Project Mgmt Team	1,135,145	879,705	-	255,440	-	-	-	-
10	Removals	14,138	14,138	-	-	-	-	-	-
11	Other	109,879	38,348	-	-	-	-	-	71,531
13	Indirects/Overhead	2,834,786	1,556,507	-	-	-	-	-	1,278,279
14	AFUDC	430,208	271,200	-	-	-	-	-	159,008
Grand Total		8,913,710	3,828,338	-	3,030,203	546,350	-	-	1,508,818

ATTACHMENT A

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24

BESS - Remaining TJP

06-Mar-24 15:55

Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028
Project Lead: Adam Kelley Adam S. Kelley		1037d	15-Jun-21 A	09-Mar-28											
80089926 Hyde Park BESS - Battery Energy Storage System Solution 03-06-24		1037d	15-Jun-21 A	09-Mar-28											
80089926.01 PROJECT PLANNING & FUNDING		686d	15-Jun-21 A	05-Nov-26											
PPF1000	Project Planning & Funding	686d	15-Jun-21 A	05-Nov-26	In Progress										
80089926.01.12 T-Minus Milestones		661d	29-Mar-24	05-Nov-26											
AT2605	Preliminary Design / Engineering Completed	0d		04-Jun-25	Not Started	ENG-PR	ENG.ST.SS		DE2120, DE2100, DE2140, DE2060, DE2040, DE2160						
AT1811	Detail Design/Engineering Completed	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS		COM1000, PC1110, UG1280, OH1430, EE1410, EE1420						
AT0805	Final Engineering Issued for Construction W/BOM	0d		09-Jan-26	Not Started	ENG-IFC	ENG.ST.SS		COM1030, PC1330, UG1370, OH1690, EE1410, EE1420	ISD					
80089926.01.12.01 Civil T-Minus Milestones		930d	29-Mar-24	15-Oct-26											
AT3001	Scope Document Approved	0d		29-Mar-24	Not Started	ENG-SC	ENG.ST.SS		PE1720, ATCGE1010	AT00					
BT2615	Civil Preliminary Design Complete	0d		08-Apr-25	Not Started	ENG-PR	ENG.ST.SS		DE2160	BD1010, PP1000, PHZ1010					
AT1030	ISO I.3.9 Approval	0d		30-May-25	Not Started		ENG.SYS.S		ISO1190	AT00					
BT2625	Civil Detail Design Complete	0d		18-Dec-25	Not Started	ENG-DT	ENG.ST.SS		EE1330	AT00					
AT0805C	IFC - Civil (BG)	0d		09-Jan-26	Not Started	AG-IFC	ENG.ST.SS		EE1350	AT00					
AT0401	Civil Long Lead Items Delivered	0d		19-Feb-26	Not Started		PURC		CMAT1010	AT00					
AT0814	Civil Construction Contracts Awarded - Civil BG	0d		17-Mar-26	Not Started		PURC		BCN2150	AT00					
AT0203	Civil Constr Auth Notice (CAN) completed [go/ no go]	0d		25-Jun-26	Not Started		PM			AT00, BCN2090					
AT00	T-00 Construction Start Civil	0d	26-Jun-26*		Not Started		PM		AT0401, AT0203, AT0805C, BT2625, AT0814, AT0801, AT0203	BCN2090					
AT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM		PP1000, BD1000, BD1010	AT00					
80089926.01.12.02 Electrical T-Minus Milestones		533d	20-May-25	05-Nov-26											
BT2605	Electrical Preliminary Design Complete	0d		20-May-25	Not Started	ENG-PR	ENG.ST.SS		DE2040	BT00, BD1010, PP1000					
BT2404	Assign Sta Tech (Commissioning) Engineer	0d		20-May-25	Not Started		ENG.ST.SL		DE2040	CT00					
BT1811	Electrical Detail Design Complete	0d		17-Oct-25	Not Started	ENG-DT	ENG.ST.SS		EE1210	BT00					
BT0805E	IFC - Electrical (AG)	0d		07-Nov-25	Not Started	AG-IFC	ENG.ST.SS		EE1310	BT00					
BT0814	Electrical Construction Contracts Awarded - Electrical AG	0d		26-Dec-25	Not Started		PURC		BCN2050	BT00					
BT0801	Environmental & Local Permits Received	0d		15-Oct-26	Not Started		PM		PRE1010, PP1000, BD1000, BD1010	BT00					
BT0203	Electrical Constr Auth Notice (CAN) completed [go/ no go]	0d		04-Nov-26	Not Started		PM			BT00					
BT0401	Electrical Long Lead Items Delivered	0d		04-Nov-26	Not Started		PURC		EMAT1230	BT00					
BT00	T-00 Construction Start Electrical	0d	05-Nov-26		Not Started		PM		BT0401, BT0203, BT0805E, BT1811, BT171000, BT2605	BCN1050, BCN1030, ECN1040					
80089926.01.12.06 P&C T-Minus Milestones		390d	22-Apr-25	29-Oct-26											

Actual Work

Remaining Work

Critical Remaining Work

Milestone

Summary

Page 1 of 16

TASK filter: EMA REPORT - Remaining TJP.

© Oracle Corporation

[illegible]

[illegible]

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24					BESS - Remaining TJP						06-Mar-24 15:55				
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028
	80089926.04.03.02 Utility Commission	200d	01-Sep-23 A	09-Dec-24											
	UC1150	3d	01-Sep-23 A	28-Feb-24	In Progress		PM	Adam S. Kelley	UC1070	UC1080					
	UC1080	47d	29-Feb-24	03-May-24	Not Started		PM	Adam S. Kelley	UC1150	UC1120					
	UC1120	150d	06-May-24	09-Dec-24	Not Started		PM	Adam S. Kelley	UC1080						
	80089926.04.3 Local Approvals And Permits	400d	26-Mar-25	15-Oct-26											
	80089926.04.3.01 Building Department	125d	26-Mar-25	19-Sep-25											
	BD1000	125d	26-Mar-25	19-Sep-25	Not Started		PM	Adam S. Kelley	PP1000, DE2150	PHZ1000, BT0801, FINCO1000, AT0801					
	80089926.04.3.09 Special Use Article 80	320d	18-Jul-25	15-Oct-26											
	BD1010	320d	18-Jul-25*	15-Oct-26	Not Started		PM	Adam S. Kelley	BT2605, BT2615, DT2605, ET2605	BT0801, FINCO1000, AT0801, BCN2090					
	80089926.05 ENGINEERING DETAILED / IFC DESIGN	151d	31-Jul-22 A	09-Jan-26											
	ENG1000	151d	31-Jul-22 A	09-Jan-26	In Progress										
	80089926.05.01 Civil Engineering	92d	28-Aug-25	09-Jan-26											
	80089926.05.01.01 Civil Engineering Detailed Design	35d	29-Oct-25	18-Dec-25											
	EE1320	25d	29-Oct-25	04-Dec-25	Not Started		ENG.ST.SS	Shodhan Patel	DE2160, CV1030, CV1040, PF1010, FF1000	EE1330					
	EE1330	10d	05-Dec-25	18-Dec-25	Not Started	ES-DTL	ENG.ST.SS	Shodhan Patel	EE1320	EE1340, BT2625, AT1811					
	80089926.05.01.1 Civil Engineering Pre-IFC Design	42d	28-Aug-25	28-Oct-25											
	CV1010	10d	28-Aug-25	11-Sep-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	PI1050, PF1010, PF1010, FF1000	CV1020					
	CV1020	10d	12-Sep-25	25-Sep-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1010	CV1030, CV1040					
	CV1030	22d	26-Sep-25	28-Oct-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1020	EE1320					
	CV1040	22d	26-Sep-25	28-Oct-25	Not Started	V-DTL	ENG.ST.SS	Shodhan Patel	CV1020	EE1320					
	80089926.05.01.02 Civil Engineering IFC Design	15d	19-Dec-25	09-Jan-26											
	EE1340	10d	19-Dec-25	02-Jan-26	Not Started		ENG.ST.SS	Shodhan Patel	EE1330	EE1350					
	EE1350	5d	05-Jan-26	09-Jan-26	Not Started	ES-IFC	ENG.ST.SS	Shodhan Patel	EE1340	AT0805C, BCN2100, AT0805					
	80089926.05.02 Electrical Engineering	50d	28-Aug-25	07-Nov-25											
	CSD1000	5d	03-Nov-25	07-Nov-25	Not Started	CASCADE	ENG.ST.SS	Shodhan Patel	EE1210, EE1310	CSD1050					
	80089926.05.02.01 Electrical Engineering Detailed Design	35d	28-Aug-25	17-Oct-25											
	EE1200	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.SS	Shodhan Patel	DE1900, DE2040, FF1000	EE1210, OPERM1000					
	EE1210	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.SS	Shodhan Patel	EE1200	A3360, AT1811, BCN2000, BT1811, CSD1000, EE1200					
	80089926.05.02.02 Electrical Engineering Pre-IFC Design	15d	20-Oct-25	07-Nov-25											
	EE1300	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.SS	Shodhan Patel	EE1210	EE1310					
	EE1310	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.SS	Shodhan Patel	EE1300	AT0805, BT0805E, CSD1000, APP1000					
	80089926.05.06 P&C Engineering	70d	28-Aug-25	09-Dec-25											
	OP1010	2d	28-Aug-25	29-Aug-25	Not Started		ENG.ST.P&	P&C Engineer	OP241000, PF1010, FF1000	OP1020					
					Page 6 of 16						TASK filter: EMA REPORT - Remaining TJP.				
											© Oracle Corporation				

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56																				
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028										
			OP1020	P&C Updates OP-24 Appendix B	4d	02-Sep-25	05-Sep-25	Not Started		ENG.ST.P8 P&C Engineer	OP1010	OP1030										P&C Updates OP-24 Appendix B									
			OP1030	P&C Submits OP-24 Appendix B to ISO-NE	1d	08-Sep-25	08-Sep-25	Not Started	OP-24	ENG.ST.P8 P&C Engineer	OP1020	ET0901											P&C Submits OP-24 Appendix B to ISO-NE								
			CSD1030	PE Initiate CASCADE Forms (Appendix A) for Construction Package	5d	03-Nov-25	07-Nov-25	Not Started	CASCADE	ENG.ST.P8 P&C Engineer	PC1330, PC1110	CSD1050												PE Initiate CASCADE Forms (Appendix A) for Construction Package							
			80089926.05.06.01 P&C Engineering Detailed Design				35d	28-Aug-25	17-Oct-25															17-Oct-25, 80089926.05.06.01 P&C Engineering Detailed Design							
			PC1100	Engineer Develop P&C Detail Design	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.P8 P&C Engineer	DE2100, FF1000, FF1000, PF1010	PC1110, OPERM1000												Engineer Develop P&C Detail Design							
			PC1110	ES Engineering Review P&C Detail Design	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.P8 P&C Engineer	PC1100	AT1811, ECN2000, CSD1030, PC1320, ET1044												ES Engineering Review P&C Detail Design							
			80089926.05.06.02 P&C Engineering Pre-IFC Design				35d	20-Oct-25	09-Dec-25															09-Dec-25, 80089926.05.06.02 P&C Engineering Pre-IFC Design							
			80089926.05.06.02.1 P&C Relay Settings Development				20d	10-Nov-25	09-Dec-25															09-Dec-25, 80089926.05.06.02.1 P&C Relay Settings Development							
			PC1400	Complete P&C Relay Settings	20d	10-Nov-25	09-Dec-25	Not Started	ES-RSE	ENG.ST.P8 P&C Engineer	PC1330	ET0404												Complete P&C Relay Settings							
			80089926.05.06.02.03 P&C Engineering IFC Design				15d	20-Oct-25	07-Nov-25															07-Nov-25, 80089926.05.06.02.03 P&C Engineering IFC Design							
			PC1320	Engineer Develop P&C IFC	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.P8 P&C Engineer	PC1110	PC1330												Engineer Develop P&C IFC							
			PC1330	ES Engineering Review & Issue P&C IFC	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.P8 P&C Engineer	PC1320	AT0805, A-DFR1010, A-DFR1060, ET0805, PC1400, CSD1030												ES Engineering Review & Issue P&C IFC							
			80089926.05.07 Telecom Engineering				114d	05-Jun-25	17-Nov-25																17-Nov-25, 80089926.05.07 Telecom Engineering						
			CSD1040	PE Initiate CASCADE Forms (Appendix A) for Construction Package	5d	10-Nov-25	17-Nov-25	Not Started	CASCADE	ENG.ST.SL Bechir El Khoury Hanna	COM1000, COM1030	CSD1050													PE Initiate CASCADE Forms (Appendix A) for Construction Package						
			80089926.05.07.01 Telecomm Engineering Detailed Design				35d	28-Aug-25	17-Oct-25																17-Oct-25, 80089926.05.07.01 Telecomm Engineering Detailed Design						
			COM1040	Engineer Develop Communications Detail Design	25d	28-Aug-25	02-Oct-25	Not Started		ENG.ST.SL Demetrios A.Sakellaris	DE2120, FF1000, PF1010	COM1000													Engineer Develop Communications Detail Design						
			COM1000	ES Engineering Review Communications Detail Design	10d	03-Oct-25	17-Oct-25	Not Started	ES-DTL	ENG.ST.SL Bechir El Khoury Hanna	COM1040	COM1050, AT1811, FCN2000, CSD1040, COMMAT1000												ES Engineering Review Communications Detail Design							
			80089926.05.07.02 Telecomm Engineering Pre-IFC Design				109d	05-Jun-25	07-Nov-25																07-Nov-25, 80089926.05.07.02 Telecomm Engineering Pre-IFC Design						
			COM1010	Confirm Duct Bank/ Poles/ Inner Duct for Fiber is Complete	21d	05-Jun-25	03-Jul-25	Not Started		ENG.ST.SL Bechir El Khoury Hanna	DE2120	COM1020													Confirm Duct Bank/ Poles/ Inner Duct for Fiber is Complete						
			COM1020	Comm Constructability Review	21d	07-Jul-25	04-Aug-25	Not Started		ENG.ENGF Field Communication North	COM1010	FT00, FT0805M, FT0203, FCN1010, FCN1020												Comm Constructability Review							
			COM1050	Engineer Develop Communications IFC	10d	20-Oct-25	31-Oct-25	Not Started		ENG.ST.SL Bechir El Khoury Hanna	COM1000	COM1030													Engineer Develop Communications IFC						
			COM1030	ES Engineering Review & Issue Communications IFC	5d	03-Nov-25	07-Nov-25	Not Started	ES-IFC	ENG.ST.SL Bechir El Khoury Hanna	COM1050	AT0805, CSD1040													ES Engineering Review & Issue Communications IFC						
80089926.06 PROCUREMENT LONG LEAD MATERIAL				305d	28-Aug-25	04-Nov-26																04-Nov-26, 80089926.06 PROCUREMENT LONG LEAD MATERIAL									
PLLM1000	Procurement Long Lead Material	305d	28-Aug-25	04-Nov-26	Not Started																	Procurement Long Lead Material									
80089926.06.01 Civil Long Lead Material				121d	28-Aug-25	19-Feb-26																19-Feb-26, 80089926.06.01 Civil Long Lead Material									
CMAT1020	Civil (BG) Material DR Complete	0d		28-Aug-25	Not Started		ENG.ST.SS		CMAT1000, PF1010, EOC1040	CMAT1010												Civil (BG) Material DR Complete									
CMAT1000	Prepare Civil Material DR	1d	28-Aug-25	28-Aug-25	Not Started	DR-M	ENG.ST.SS Shodhan Patel	FF1000	CMAT1020, CMAT1010													Prepare Civil Material DR									
CMAT1010	Civil Material Fab & Delivery	120d	29-Aug-25	19-Feb-26	Not Started	M-DELIV	PURC Purchasing Department	CMAT1000, CMAT1020, PF1010	AT0401, BCN2090													Civil Material Fab & Delivery									
80089926.06.02 Material Procurement - Batteries & Inverters				305d	28-Aug-25	04-Nov-26																04-Nov-26, 80089926.06.02 Material Procurement - Batteries & Inverters									
EMAT1100	Issue Spec. - Material Procurement - Batteries & Inverters	15d	28-Aug-25	18-Sep-25	Not Started		ENG.ST.SS Shodhan Patel	DE2040, FF1000, EOC1040	EMAT1110, EMAT1210, EMAT1180													Issue Spec. - Material Procurement - Batteries & Inverters									
EMAT1180	Material Vendor Submits Shop Drawings - Material Procurement - Batteries & Inverters	10d	19-Sep-25	02-Oct-25	Not Started		PURC Purchasing Department	EMAT1100	EMAT1190													Material Vendor Submits Shop Drawings - Material Procurement - Batteries & Inverters									
EMAT1190	ES Reviews Shop Drawings - Material Procurement - Batteries & Inverters	10d	03-Oct-25	17-Oct-25	Not Started		ENG.ST.SS Shodhan Patel	EMAT1180	EMAT1200													ES Reviews Shop Drawings - Material Procurement - Batteries & Inverters									
Actual Work		Critical Remaining Work		Summary																											
Remaining Work		Milestone																													
Page 7 of 16												TASK filter: EMA REPORT - Remaining TJP.										© Oracle Corporation									

[illegible]

[illegible]

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP										06-Mar-24 15:56												
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	20242025202620272028												
80089926	FCN1070	Constr Compl	0d		28-Sep-27	Not Started		ENG.ENG			FCN1060	COMM1000, COMPL1000	◆ Constr												
	80089926.07.07.02 Telecom Wiring / Terminating		30d	18-Aug-27	28-Sep-27									▼ 28-Sep											
	BCN2190	Install Communication Wiring and Terminate	30d	18-Aug-27	28-Sep-27	Not Started		CONTR	O&M Southwest		BCN1040, BCN1030, BCN2080, ECN2080, FCN1000	COMPL1020, COMPL1000	■ Install C												
	80089926.08.01 ISD		0d	28-Sep-27	28-Sep-27									▼ 28-Sep											
	COMPL1000	In Service Date	0d		28-Sep-27	Not Started	ISD	PM			ECN1030, BCN1020, DCN1020, FCN1070, BCN2100, FCN1010	COMPL1010	◆ In Serv												
	ISD	ISD - In Service Date	0d	28-Sep-27	28-Sep-27	Not Started					AT0805			ISD --Ir											
	80089926.09 TEST & COMMISSIONING		504d	20-Oct-25	28-Sep-27									▼ 28-Sep											
	TC1000	Test & Commissioning	504d	20-Oct-25	28-Sep-27	Not Started								■ Test &											
	80089926.09.1 Isolation Plan		293d	20-Oct-25	07-Dec-26									▼ 07-Dec-26, 800899											
	A3360	PM to Request Lead Commissioning Engineer GO/NO-GO {Preliminary or Detail design is often the predecessor}	2d	20-Oct-25	21-Oct-25	Not Started	PM	PM	Adam S. Kelley		EE1210	A2930		PM to Request Lead Commissioning											
	A2930	Assign Lead Commissioning Engineer	2d	22-Oct-25	23-Oct-25	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		A3360	A3300, A2950, A3350, A1730, A2750, A3260		Assign Lead Commissioning Engine											
	A3350	Evaluate Drawings and Plan for Construction Outages	20d	24-Oct-25	21-Nov-25	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		A2930	A1730		■ Evaluate Drawings and Plan for Co											
	A2950	Develop Test and Commissioning Plan	20d	05-Nov-26	02-Dec-26	Not Started	V-COMP	TEST	LEAD COMMISSIONII ENGINEER		A3500, A3540, A3260, A2930	A3300		■ Develop Test and C											
	A3300	LCE Review and Approve Vendor's Test and Commissioning Plan	3d	03-Dec-26	07-Dec-26	Not Started	ES-COM	ENG.ST.S	LEAD COMMISSIONII ENGINEER		A2930, A2950	A3270		LCE Review and A											
	80089926.09.04 Testing Activities		236d	03-Nov-26	28-Sep-27									▼ 28-Sep											
	A3260	Mobilize Commissioning Engineer and Testing Vendor Prior to Construction	2d	03-Nov-26	05-Nov-26	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		CT0401, A2930, BCN1050	A2950, A3270, A3280		Mobilize Commission											
	A3280	Support Project/Testing & Execute test plan	234d	05-Nov-26	28-Sep-27	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		A3260	LCE1000, EV1000		■ Support											
	BCN2200	System Commissioning	81d	08-Jun-27*	28-Sep-27	Not Started		CONTR	O&M Southwest		BCN1040, BCN1030, BCN2080, ECN2080, FCN1000	COMPL1020, COMPL1000		■ System											
	80089926.09.03 LCE AUTHORIZATION (Remove)		21d	24-Oct-25	24-Nov-25									▼ 24-Nov-25, 80089926.09.03 LCE											
	A2750	Review and approve Outage Sequence Document (Sign-off)	1d	24-Oct-25	24-Oct-25	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		A2930	A3270		Review and approve Outage Seque											
	A1730	LCE Review TOA / Applications (Pred=TOA#, Dwgs and Const Plan, has T18.01 successor	1d	24-Nov-25	24-Nov-25	Not Started		ENG.ST.S	LEAD COMMISSIONII ENGINEER		A2930, A3350	BT00		LCE Review TOA / Applications (Pr											
	80089926.10 PROJECT CLOSE OUT		365d	16-Oct-26	09-Mar-28									▼ 09-Mar-28											
	PCO1000	Project Close Out	365d	16-Oct-26	09-Mar-28	Not Started								■ Project Close Out											
	80089926.10.02 Electrical Close Out		117d	29-Sep-27	09-Mar-28									▼ 09-Mar-28											
	EV1000	Prepare Electrical Field Mark Ups	21d	06-Oct-27	03-Nov-27	Not Started	V-FLDAS	ENG.ENG	LEAD COMMISSIONII ENGINEER		LCE1000, A3280	EV1100		■ Prepa											
	EV1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Package to Vendor	5d	04-Nov-27	10-Nov-27	Not Started	ES-ASB	ENG.ST.S	Shodhan Patel		EV1000	EV1200		Evers											
	EV1200	Vendor Engineering Verifies Electrical Mark Ups and Updates As-Built Drawing Package. Submits Final Package to ES	30d	11-Nov-27	22-Dec-27	Not Started	V-ASB	ENG.ST.S	Shodhan Patel		EV1100	EV1300		■ Ver											
	EV1300	Eversource Engineering Verifies Quality of Submitted Electrical Drawings	20d	23-Dec-27	19-Jan-28	Not Started	ES-ASB	ENG.ST.S	Shodhan Patel		EV1200	EV1400		■ EV											
EV1400	Electrical Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5d	20-Jan-28	26-Jan-28	Not Started	ES-ASB	ENG.ST.S	Shodhan Patel		COMM1300, EV1300	PM1100		El												
PM1100	PM to Document Electrical As-Built Drawings Complete (Electrical, P&C, Communications)	1d	27-Jan-28	27-Jan-28	Not Started	PM	PM	Adam S. Kelley		COMM1400, PC1310, EV1400	PC1500, COMM1500		PM												
COMPL1010	Final Inspection and Punchlist Work	90d	29-Sep-27	01-Feb-28	Not Started		PM	Adam S. Kelley		COMPL1000	COMPL1020		■ Fi												
Actual Work Critical Remaining Work Summary			Page 12 of 16								TASK filter: EMA REPORT - Remaining TJP.														
Remaining Work Milestone			© Oracle Corporation																						

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP							06-Mar-24 15:56									
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028					
	COMPL1020	Project Complete	0d		09-Mar-28	Not Started		PM		BCN1020, PC1500, COMM1500, CSD1060, COMPL1010										
	80089926.10.03 P&C Close Out		112d	06-Oct-27	09-Mar-28															
	PC1000	Prepare P&C Field Mark Ups	21d	06-Oct-27	03-Nov-27	Not Started	V-FLDAS	ENG.ENGF	LEAD COMMISSIONING ENGINEER	LCE1000	PC1510									
	PC1510	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5d	04-Nov-27	10-Nov-27	Not Started	ES-ASB	ENG.ST.P8	P&C Engineer	PC1000	PC1200									
	PC1200	Vendor Engineering Verifies P&C Mark Ups and Updates As-Built Drawing Package. Submits Final Package to ES	30d	11-Nov-27	22-Dec-27	Not Started	V-ASB	ENG.ST.P8	P&C Engineer	PC1510	PC1300									
	PC1300	Eversource Engineering Verifies Quality of Submitted P&C Drawings	20d	23-Dec-27	19-Jan-28	Not Started	ES-ASB	ENG.ST.P8	P&C Engineer	PC1200	PC1310									
	PC1310	P&C Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5d	20-Jan-28	26-Jan-28	Not Started	ES-ASB	ENG.ST.P8	P&C Engineer	PC1300	PM1100									
	PC1500	FE Files P&C Design Drawings at Station	30d	28-Jan-28	09-Mar-28	Not Started	ES-FLSA	ENG.ST.P8	P&C Engineer	PM1100	COMPL1020									
	80089926.10.04 Telecomm Close Out		117d	29-Sep-27	09-Mar-28															
	COMM1000	Prepare Communication Field Mark Ups	21d	29-Sep-27	27-Oct-27	Not Started	V-FLDAS	ENG.ENGF	Field Communication North	FCN1070	COMM1100									
	COMM1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor	5d	28-Oct-27	03-Nov-27	Not Started	ES-ASB	ENG.ST.SL	Bechir El Khoury Hanna	COMM1000	COMM1200									
	COMM1200	Vendor Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Package. Submits Final Package to ES	30d	04-Nov-27	15-Dec-27	Not Started	V-ASB	ENG.ST.SL	Bechir El Khoury Hanna	COMM1100	COMM1300									
	COMM1300	Eversource Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Package	20d	16-Dec-27	12-Jan-28	Not Started	ES-ASB	ENG.ST.SL	Bechir El Khoury Hanna	COMM1200	COMM1400, EV1400									
	COMM1400	Communications Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field	5d	13-Jan-28	19-Jan-28	Not Started	ES-ASB	ENG.ST.SL	Bechir El Khoury Hanna	COMM1300	PM1100									
	COMM1500	Operations Files Communication Design Drawings at Station	30d	28-Jan-28	09-Mar-28	Not Started	ES-FLSA	ENG.ENGF	Field Communication North	PM1100	COMPL1020									
	80089926.10.11 LCE Closeout		5d	29-Sep-27	05-Oct-27															
	LCE1000	LCE Review Test Data & Issue Commissioning Report (Preds=LCE Oversight of Test Contrctor / Support Project/Testing Plan)	5d	29-Sep-27	05-Oct-27	Not Started		ENG.ST.SL	LEAD COMMISSIONING ENGINEER	A3280, A3270	PC1000, EV1000									
	80089926.10.09 Work Order / Financial Close Out		10d	16-Oct-26	29-Oct-26															
	FINCO1000	PM/ Engr/ Plant Acct Develop Prelim UOP & UOR List(s)	5d	16-Oct-26	22-Oct-26	Not Started	PM	PM	Adam S. Kelley	ISO1190, PP1000, BD1000, PRE1010, PP1000, PP1010	FINCO1010									
	FINCO1010	Review UOP, compl forms, fwd to Plant Acctng, enter in wrkbk	5d	23-Oct-26	29-Oct-26	Not Started		PM	Michael P. Rosenberg, Adam S. Kelley	FINCO1000	DT0602, ET0602, ECN1010, ECN1020, CT0602									
	80089926.10.10 Project Management Close Out		40d	17-Aug-27	11-Oct-27															
	CSD1050	Construction Rep Finalize CASCADE Forms (within 14 days following ISD)	10d	17-Aug-27	30-Aug-27	Not Started		PM	Adam S. Kelley	ECN1030, CSD1030, CSD1000, CSD1040, CSD1010, CSD1020	CSD1060									
	CSD1060	Asset Management Complete Asset Update(s) in CASCADE	30d	31-Aug-27	11-Oct-27	Not Started		PM	Adam S. Kelley	CSD1050	COMPL1020									
	80089926.13 NOT REQUIRED ACTIVITIES		479d	03-Jan-24 A	20-May-26															
	TMP1150	N/R 12-7-23 MR Engineering of Choice (EOC) - Vendor Responds with Proposal (OE/CE)	0d	03-Jan-24 A	24-Jan-24 A	Comple		PM	Adam S. Kelley	TMP1140	TMP1200									
	TMP1200	N/R 12-7-23 MR Engineering of Choice (EOC) - Proposal Evaluations (OE/CE)	0d	25-Jan-24 A	31-Jan-24 A	Comple		ENG.EPC.I	Tania Y. White	TMP1150	TMP1180									
	A3520	N/R 12-7-23 MR TEST SERVICES - BID - Contradors Evaluate RFP and Provide Proposal	0d	19-Jan-24 A	08-Feb-24 A	Comple		PURC	Purchasing Department	A3510	A3530									
	TMP1180	N/R 12-7-23 MR Engineering of Choice (EOC) - EOC Committee Reviews at Monthly Meeting & Approves (OE/CE)	0d	01-Feb-24 A	22-Feb-24 A	Comple		ENG.EPC.I	Tania Y. White	TMP1200	TMP1220									
	A3530	N/R 12-7-23 MR TEST SERVICES - BID - Eversource reviews contractor proposals	0d	09-Feb-24 A	01-Mar-24 A	Comple		PURC	Purchasing Department	A3520	A3540									
	A-DFR-CN1020	Program DAU - DFR at S/S N/R	10d	26-Feb-24*	08-Mar-24	Not Started		ENG.ST.P8	P&C Engineer	A-DFR-CN1010										
Actual Work Remaining Work		Critical Remaining Work Milestone		Summary		Page 13 of 16					TASK filter: EMA REPORT - Remaining TJP.					© Oracle Corporation				

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56									
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024	2025	2026	2027	2028			
			</																	

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24				BESS - Remaining TJP							06-Mar-24 15:56									
Activity ID	Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources		Predecessors	Successors	2024	2025	2026	2027	2028				
	EMAT1130	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	EMAT1120	EMAT1135									N/R 12-7-23 MR RFP - Long Lea
	COMMAT1040	N/R 12-7-23 MR RFP - GE JMUX	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	COMMAT1030	COMMAT1050									N/R 12-7-23 MR RFP - GE JMUX
	PCMAT1050	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	PCMAT1040	PCMAT1060									N/R 12-7-23 MR RFP - Long Lea
	DT0814	N/R 12-7-23 MR Construction Contracts Awarded -	0d		31-Dec-25 A	Comple		PURC			DT00									N/R 12-7-23 MR Construction Co
	DMAT1030	N/R 12-7-23 MR RFP - Long Lead Item #1	0d	17-Dec-25 A	31-Dec-25 A	Comple		PURC	Purchasing Department	DMAT1020	DMAT1040									N/R 12-7-23 MR RFP - Long Lea
	COMMAT1050	N/R 12-7-23 MR ES Selects Material Vendor - GE JMUX	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	COMMAT1040	COMMAT1060									N/R 12-7-23 MR ES Selects/Mat
	PCMAT1060	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	PCMAT1050	PCMAT1070									N/R 12-7-23 MR ES Provides En
	DMAT1040	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	14-Jan-26 A	Comple		PURC	Purchasing Department	DMAT1030	DMAT1050									N/R 12-7-23 MR ES Provides En
	A-DFR1130	Procure DFR Mat N/R	20d	22-Dec-25	19-Jan-26	Not Started		PURC	Purchasing Department	A-DFR1120										Procure DFR Mat N/R
	DT0203	N/R 12-7-23 MR Constr Auth Notice (CAN) completed [go/ no go]	0d		20-Jan-26 A	Comple		PM			DT00									N/R 12-7-23 MR Constr Auth No
	DT00	N/R 12-7-23 MR T-00 Construction Start Distribution (Outage Start if needed)	0d	21-Jan-26 A		Comple		PM		DT1811, DT2605, DT0805UG, DT0814, DT0602, DT0404, DT0303	DCN1030, DCN1020, DCN1010									N/R 12-7-23 MR T-00 Constructi
	EMAT1135	N/R 12-7-23 MR ES Provides Eng Specs.To Bidders - Long Lead Item #1	0d	01-Jan-26 A	21-Jan-26 A	Comple		PURC	Purchasing Department	EMAT1130	EMAT1140									N/R 12-7-23 MR ES Provides Er
	COMMAT1060	N/R 12-7-23 MR ES Issues P.O. - GE JMUX	0d	15-Jan-26 A	28-Jan-26 A	Comple		PURC	Purchasing Department	COMMAT1050										N/R 12-7-23 MR ES Issues P.O.
	PCMAT1070	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	15-Jan-26 A	28-Jan-26 A	Comple		PURC	Purchasing Department	PCMAT1060	PCMAT1080									N/R 12-7-23 MR Bid Period - Lo
	DMAT1050	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	15-Jan-26 A	28-Jan-26 A	Comple			Purchasing Department	DMAT1040	DMAT1060									N/R 12-7-23 MR Bid Period - Lo
	PCMAT1080	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	29-Jan-26 A	11-Feb-26 A	Comple		PURC	Purchasing Department	PCMAT1070	PCMAT1090									N/R 12-7-23 MR ES Reviews S
	DMAT1060	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	29-Jan-26 A	11-Feb-26 A	Comple		PURC	Purchasing Department	DMAT1050	BT0408, DMAT1070									N/R 12-7-23 MR ES Reviews S
	EMAT1140	N/R 12-7-23 MR Bid Period - Long Lead Item #1	0d	22-Jan-26 A	18-Feb-26 A	Comple		PURC	Purchasing Department	EMAT1135	EMAT1150									N/R 12-7-23 MR Bid Period - L
	PCMAT1090	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	12-Feb-26 A	25-Feb-26 A	Comple		PURC	Purchasing Department	PCMAT1080	PCMAT1100									N/R 12-7-23 MR ES Selects M
	DMAT1070	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	12-Feb-26 A	25-Feb-26 A	Comple		PURC	Purchasing Department	DMAT1060	DMAT1080									N/R 12-7-23 MR ES Selects M
	EMAT1150	N/R 12-7-23 MR ES Reviews Submitted Bids - Long Lead Item #1	0d	19-Feb-26 A	04-Mar-26 A	Comple		PURC	Purchasing Department	EMAT1140	BT0408, EMAT1160									N/R 12-7-23 MR ES Reviews S
	DT0602	N/R 12-7-23 MR Units of Property issued to Plant Accounting	0d		06-Mar-26 A	Comple	PM	PM		FINCO1010	DT00									N/R 12-7-23 MR Units of Prop
	PCMAT1100	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	26-Feb-26 A	11-Mar-26 A	Comple		PURC	Purchasing Department	PCMAT1090										N/R 12-7-23 MR ES Issues P.
	DMAT1080	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	26-Feb-26 A	11-Mar-26 A	Comple		PURC	Purchasing Department	DMAT1070	DMAT1090									N/R 12-7-23 MR ES Issues P.
	EMAT1160	N/R 12-7-23 MR ES Selects Material Vendor - Long Lead Item #1	0d	05-Mar-26 A	18-Mar-26 A	Comple		PURC	Purchasing Department	EMAT1150	EMAT1170									N/R 12-7-23 MR ES Selects M
	DMAT1090	N/R 12-7-23 MR Material Vendor Submits Shop Drawings - Long Lead Item #1	0d	12-Mar-26 A	25-Mar-26 A	Comple		PURC	Purchasing Department	DMAT1080	DMAT1100									N/R 12-7-23 MR Material Ven
	EMAT1170	N/R 12-7-23 MR ES Issues P.O. / ES Approves CR - Long Lead Item #1	0d	19-Mar-26 A	01-Apr-26 A	Comple		PURC	Purchasing Department	EMAT1160										N/R 12-7-23 MR ES Issues P
	DMAT1100	N/R 12-7-23 MR ES Reviews Shop Drawings - Long Lead Item #1	0d	26-Mar-26 A	08-Apr-26 A	Comple		ENG.ST.LII	Michael Hickey	DMAT1090	DMAT1110									N/R 12-7-23 MR ES Reviews
Actual Work Remaining Work		Critical Remaining Work Milestone		Summary		Page 15 of 16					TASK filter: EMA REPORT - Remaining TJP.					© Oracle Corporation				

Hyde Park BESS - Battery Energy Storage System Solution 03-06-24			BESS - Remaining TJP								06-Mar-24 15:56											
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTM	Resources	Predecessors	Successors	2024		2025		2026		2027		2028	
	DMAT1110	N/R 12-7-23 MR ES Approves Shop Drawings - Long Lead Item #1		0d	09-Apr-26 A	22-Apr-26 A	Comple		ENG.ST.LII	Michael Hickey	DMAT1100	DMAT1120	N/R 12-7-23 MR ES Approves Shop Drawings - Long Lead Item #1									
	DMAT1120	N/R 12-7-23 MR Material Vendor Manufactures Long Lead Item #1		0d	23-Apr-26 A	06-May-26 A	Comple		PURC	Purchasing Department	DMAT1110	DMAT1130	N/R 12-7-23 MR Material Vendor Manufactures Long Lead Item #1									
	DMAT1130	N/R 12-7-23 MR Material Vendor Delivers Long Lead Item #1		0d	07-May-26 A	20-May-26 A	Comple	M-DELIV	PURC	Purchasing Department	DMAT1120	DT0401	N/R 12-7-23 MR Material Vendor Delivers Long Lead Item #1									

[illegible]

 Actual Work
 Critical Remaining Work
 Summary
 Remaining Work
  Milestone

[illegible]

[illegible]

Hyde Park BESS - Battery Energy Storage System Solution - Traditional				00: EMA - Remaining (MLJ)						08-Jun-26 11:34																	
Activity ID		Activity Name			Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources																
												2026				2027				2028				2029			
													Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3		
		80089926.03.03.23.3 Soil and Groundwater Management Plan (SGMP)			77	06-Oct-26	29-Jan-27					29-Jan-27, 80089926.03.03.23.3 Soil and Groundwater Management P															
		ENV1400 Recieve Geotech Boring Plan Prior to Developing Permit			0		06-Oct-26	Not Started		ENG.ENV.ENV		◆ Recieve Geotech Boring Plan Prior to Developing Permit															
		ENV1401 Determine Need for Soil & Groundwater Sampling			5	07-Oct-26	14-Oct-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Determine Need for Soil & Groundwater Sampling															
		ENV1402 Develop Soil & Groundwater Samplin Plan			7	15-Oct-26	23-Oct-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Develop Soil & Groundwater Samplin Plan															
		ENV1403 Conduct Field Work for Soil & Groundwater Characterization			60	26-Oct-26	22-Jan-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	█ Conduct Field Work for Soil & Groundwater Characterization															
		ENV1404 Develop SGMP			14	04-Jan-27	22-Jan-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Develop SGMP															
		ENV1405 Eversource Review SGMP			5	25-Jan-27	29-Jan-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Eversource Review SGMP															
		ENV1406 Finalize SGMP			5	25-Jan-27	29-Jan-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Finalize SGMP															
		80089926.03.03.24 Plume Analysis			120	07-Oct-26	01-Apr-27					01-Apr-27, 80089926.03.03.24 Plume Analysis															
		80089926.03.03.24.1 Conceptual Site Model (Sampling Locations & Permit)			120	07-Oct-26	01-Apr-27					01-Apr-27, 80089926.03.03.24.1 Conceptual Site Model (Sampling															
		PA1010 Plume Analysis - Initial Planning & Mobilization			10	07-Oct-26	21-Oct-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Plume Analysis - Initial Planning & Mobilization															
		PA1020 Plume Analysis - Field Sampling			10	22-Oct-26	04-Nov-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Plume Analysis - Field Sampling															
		PA1030 Plume Analysis - Laboratory Analysis			20	05-Nov-26	04-Dec-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Plume Analysis - Laboratory Analysis															
		PA1040 Plume Analysis - Data Interpritation & Modeling			60	07-Dec-26	04-Mar-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	█ Plume Analysis - Data Interpritation & Modeling															
		PA1050 Plume Analysis - Report Draft & Review			20	05-Mar-27	01-Apr-27	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Plume Analysis - Report Draft & Review															
		80089926.03.04 Local Approvals & Permits			42	15-Sep-26	16-Nov-26					16-Nov-26, 80089926.03.04 Local Approvals & Permits															
		80089926.03.04.4 Stormwater Permit - BWSC			42	15-Sep-26	16-Nov-26					16-Nov-26, 80089926.03.04.4 Stormwater Permit - BWSC															
		ENV1010 Permit Drawings and Stormwater Management Report Provided by Design Engineers w/MA-Registered			0		15-Sep-26	Not Started		ENG.ENV.ENV		◆ Permit Drawings and Stormwater Management Report Provided by Design Enginee															
		ENV1020 Permit Application Preperation			5	16-Sep-26	22-Sep-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Permit Application Preperation															
		ENV1030 DRAFT Permit Application Issued for Internal Review			0		22-Sep-26	Not Started		ENG.ENV.ENV		◆ DRAFT Permit Application Issued for Internal Review															
		ENV1040 Eversource Review			5	23-Sep-26	29-Sep-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Eversource Review															
		ENV1050 Finalize Permit Application			2	30-Sep-26	01-Oct-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Finalize Permit Application															
		ENV1070 File Permit Application			0		01-Oct-26	Not Started		ENG.ENV.ENV		◆ File Permit Application															
		ENV1080 Agency Review - Stormwater Permit - BWSC			30	02-Oct-26	16-Nov-26	Not Started		ENG.ENV.ENV	Christopher Newhall,	▮ Agency Review - Stormwater Permit - BWSC															
		ENV1090 Issuance of Permit			0		16-Nov-26	Not Started		ENG.ENV.ENV		◆ Issuance of Permit															
		80089926.04 SITING APPROVALS / PERMITS / PROJECT SERVICES			494	02-Aug-23	18-May-28					18-May-28, 80089926.04 SITING															
		SIT1010 Eng'g Services for Siting Support (Surveys, prep dwgs, field feasibility studies)			60	24-Mar-25,	28-Aug-26	In Progress		SITING	Leah Zambemardi	█ Eng'g Services for Siting Support (Surveys, prep dwgs, field feasibility studies)															
		3670 Document Development and Internal Review for Application Filing to the ISD/BPDA/ZBA			81	20-Feb-25,	29-Sep-26	In Progress	SITING	SITING	Leah Zambemardi	█ Document Development and Internal Review for Application Filing to the ISD/BPD															
		SIT1000 Siting Approvals / Permits / Project Services - WBS Summary			494	02-Aug-23	18-May-28	In Progress		SITING		█ Siting Approvals / Permits / Project															
		80089926.04.02 Regional Approvals			493	09-Jun-26	18-May-28					18-May-28, 80089926.04.02 Reg															
		80089926.04.02.01 ISO Technical Approval (I.3.9)			493	09-Jun-26	18-May-28					18-May-28, 80089926.04.02.01 I															
		80089926.04.02.01.02 ISO I.3.9			493	09-Jun-26	18-May-28					18-May-28, 80089926.04.02.01.0															
		80089926.04.02.01.02.1 System Impact Study			152	09-Jun-26	14-Jan-27					14-Jan-27, 80089926.04.02.01.02.1 System Impact Study															
		ISO1250 PSCADD Study - Distribution Syst Planning			30	09-Jun-26	20-Jul-26	Not Started		ENG.SYS.SYSP	Amsa Mangga	▮ PSCADD Study - Distribution Syst Planning															
		ISO1240 Perform System Impact Study (SIS) Short Circuit Study			100	20-Aug-26	14-Jan-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	█ Perform System Impact Study (SIS) Short Circuit Study															
		80089926.04.02.01.02.2 Soln Pkg & Estimate			63	15-Jan-27	15-Apr-27					15-Apr-27, 80089926.04.02.01.02.2 Soln Pkg & Estimate															
		ISO1130 Solution Pkg revised, -25/ +25 Estimate Complete - Distribution Syst Planning			0		15-Apr-27	Not Started		ENG.SYS.SYSP		◆ Solution Pkg revised, -25/ +25 Estimate Complete - Distribution Sys															
		ISO1120 Revise Solution Pkg, -25/ +25 Estimate - Distribution Syst Planning			63	15-Jan-27	15-Apr-27	Not Started		ENG.SYS.SYSP	Amsa Mangga	█ Revise Solution Pkg, -25/ +25 Estimate - Distribution Syst Planning															
		80089926.04.02.01.02.4 Reliability Committee			36	16-Apr-27	07-Jun-27					07-Jun-27, 80089926.04.02.01.02.4 Reliability Committee															
		ISO1180 Reliabilty Committee (RC) Review/ Recommendation			21	16-Apr-27	14-May-27	Not Started		ENG.SYS.SYSP	Melanie Bennett	▮ Reliabilty Committee (RC) Review/ Recommendation															
		ISO1190 Final Report/ Presentation			15	17-May-27	07-Jun-27	Not Started		ENG.SYS.SYSP	Melanie Bennett	▮ Final Report/ Presentation															
		80089926.04.02.01.02.5 P&C NPCC Directory 4			231	23-Jun-27	18-May-28					18-May-28, 80089926.04.02.01.0															
		NPCC1090 Approve Application Diagram, One Lines, BOM, Floor Plan (Preliminary P&C IFC)			1	23-Jun-27	23-Jun-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Approve Application Diagram, One Lines, BOM, Floor Plan (P															
		NPCC1000 Compile Support Drawings Package - NPCC Directory 4 Submittal			5	24-Jun-27	30-Jun-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Compile Support Drawings Package - NPCC Directory 4 Subt															
		NPCC1010 Complete D4 Form (PSRF) - NPCC Directory 4 Submittal			2	01-Jul-27	02-Jul-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Complete D4 Form (PSRF)- NPCC Directory 4 Submittal															
		NPCC1020 Complete Cover Letter- NPCC Directory 4 Submittal			3	05-Jul-27	07-Jul-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Complete Cover Letter- NPCC Directory 4 Submittal															
		NPCC1030 Develop + Provide Power Point Presentation - NPCC Directory 4 Submittal			5	08-Jul-27	14-Jul-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Develop + Provide Power Point Presentation - NPCC Direct															
		NPCC1040 Reivew & Comment - NPCC Directory 4 Submittal			3	15-Jul-27	19-Jul-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Reivew & Comment - NPCC Directory 4 Submittal															
		NPCC1050 Submit NPCC Directory 4 Submittal			1	20-Jul-27	20-Jul-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Submit NPCC Directory 4 Submittal															
		NPCC1060 Receive Approval (30 Day Lead Time) - NPCC Directory 4 Submittal			30	21-Jul-27	31-Aug-27	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Receive Approval (30 Day Lead Time) - NPCC Director															
		NPCC1080 Begin P/C Construction			1	18-May-28	18-May-28	Not Started		ENG.ST.P&C.P&	Benjamin A. Franco	▮ Begin P/C Construction															
Remaining Level of Effort Actual Work Critical Remaining Work Actual Level of Effort Remaining Work Milestone				Page 5 of 11						TASK filter: NOT CLOSED.																	
										© Oracle Corporation																	

Hyde Park BESS - Battery Energy Storage System Solution - Traditional		00: EMA - Remaining (MLJ)							08-Jun-26 11:34															
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources	2026			2027				2028				2029		
												Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
		80089926.05.06.01 P&C Engineering Detailed Design		141	01-Dec-26	22-Jun-27						22-Jun-27, 80089926.05.06.01 P&C Engineering Detailed Design												
OP1010		P&C Validation of PAF's Initial Assessment Go/No-Go		2	01-Dec-26	02-Dec-26	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco	P&C Validation of PAF's Initial Assessment Go/No-Go													
OP1020		P&C Updates OP-24 Appendix B		4	03-Dec-26	08-Dec-26	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco	P&C Updates OP-24 Appendix B													
OP1030		P&C Submits OP-24 Appendix B to ISO-NE		1	09-Dec-26	09-Dec-26	Not Started	OP-24	ENG.ST.P&C.P&I	Benjamin A. Franco	P&C Submits OP-24 Appendix B to ISO-NE													
PC1100		Engineer Develop P&C Detail Design		50	16-Mar-27	24-May-27	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco	Engineer Develop P&C Detail Design													
PC1110		ES Engineering Review P&C Detail Design		20	25-May-27	22-Jun-27	Not Started	ES-DTL	ENG.ST.P&C.P&I	Benjamin A. Franco	ES Engineering Review P&C Detail Design													
		80089926.05.06.011 P&C Engineering Pre-IFC Design		70	23-Jun-27	29-Sep-27						29-Sep-27, 80089926.05.06.011 P&C Engineering Pre-IFC Design												
CSD1030		PE Initiate CASCADE Forms (Appendix A) for Construction Package		5	11-Aug-27	17-Aug-27	Not Started	AMK.EG	ENG.ST.P&C.P&I	Benjamin A. Franco	PE Initiate CASCADE Forms (Appendix A) for Construction Package													
		80089926.05.06.011.1 P&C Relay Settings Development		35	11-Aug-27	29-Sep-27						29-Sep-27, 80089926.05.06.011.1 P&C Relay Settings Development												
PC1400		Complete P&C Relay Settings		35	11-Aug-27	29-Sep-27	Not Started	ES-RSET	ENG.ST.P&C.P&I	Benjamin A. Franco	Complete P&C Relay Settings													
		80089926.05.06.011.03 P&C Engineering IFC Design		35	23-Jun-27	10-Aug-27						10-Aug-27, 80089926.05.06.011.03 P&C Engineering IFC Design												
PC1320		Engineer Develop P&C IFC		20	23-Jun-27	20-Jul-27	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco	Engineer Develop P&C IFC													
PC1330		ES Engineering Review & Issue P&C IFC		15	21-Jul-27	10-Aug-27	Not Started	ES-IFC	ENG.ST.P&C.P&I	Benjamin A. Franco	ES Engineering Review & Issue P&C IFC													
		80089926.05.07 Telecom Engineering		172	07-Dec-26	10-Aug-27						10-Aug-27, 80089926.05.07 Telecom Engineering												
		80089926.05.07.01 Telecomm Engineering Detailed Design		25	16-Mar-27	19-Apr-27						19-Apr-27, 80089926.05.07.01 Telecomm Engineering Detailed Design												
COM1040		Engineer Develop Communications Detail Design		15	16-Mar-27	05-Apr-27	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal	Engineer Develop Communications Detail Design													
COM1000		ES Engineering Review Communications Detail Design		10	06-Apr-27	19-Apr-27	Not Started	ES-DTL	ENG.ST.SUBTEC	Syed Zohair Iqbal	ES Engineering Review Communications Detail Design													
		80089926.05.07.011 Telecomm Engineering Pre-IFC Design		172	07-Dec-26	10-Aug-27						10-Aug-27, 80089926.05.07.011 Telecomm Engineering Pre-IFC Design												
COM1020		Comm Constructability Review		21	07-Dec-26	06-Jan-27	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal	Comm Constructability Review													
COM1050		Engineer Develop Communications IFC		20	23-Jun-27	20-Jul-27	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal	Engineer Develop Communications IFC													
COM1030		ES Engineering Review & Issue Communications IFC		10	21-Jul-27	03-Aug-27	Not Started	ES-IFC	ENG.ST.SUBTEC	Syed Zohair Iqbal	ES Engineering Review & Issue Communications IFC													
CSD1040		PE Initiate CASCADE Forms (Appendix A) for Construction Package		5	04-Aug-27	10-Aug-27	Not Started	AMK.EG	ENG.ST.SUBTEC	Syed Zohair Iqbal	PE Initiate CASCADE Forms (Appendix A) for Construction Package													
		80089926.05.08 Lead Commissioning Engineer		91	18-Aug-27	28-Dec-27						28-Dec-27, 80089926.05.08 Lead Commissioning Engineer												
		80089926.05.08.01 LCE Commissioning Plan		91	18-Aug-27	28-Dec-27						28-Dec-27, 80089926.05.08.01 LCE Commissioning Plan												
A3360		PM to Request Lead Commissioning Engineer GO/NO-GO		2	18-Aug-27	19-Aug-27	Not Started	PM	PM	Adam S. Kelley	PM to Request Lead Commissioning Engineer GO/NO-GO													
A2930		Assign Lead Commissioning Engineer		2	20-Aug-27	23-Aug-27	Not Started		ENG.ST.SUBTEC	Carlos E. Ortiz	Assign Lead Commissioning Engineer													
A2950		Develop Test and Commissioning Plan - Balance of Plant, no Battery System Commissioning		55	17-Sep-27	07-Dec-27	Not Started		TEST	LEAD COMMISSIONING	Develop Test and Commissioning Plan - Balance of Plant, no Battery System Commissioning													
A3300		STE assist compiling & review Vendor's Acceptance Test Procedures		15	08-Dec-27	28-Dec-27	Not Started	ES-COM	ENG.ST.SUBTEC	Carlos E. Ortiz	STE assist compiling & review Vendor's Acceptance Test Procedures													
		80089926.05.11 Standards		363	15-Sep-26	23-Feb-28						23-Feb-28, 80089926.05.11 Standards												
		80089926.05.11.3 Commissioning		110	18-Aug-27	25-Jan-28						25-Jan-28, 80089926.05.11.3 Commissioning												
STD1110		Develop Installation / Commissioning Standard		80	18-Aug-27	13-Dec-27	Not Started		SG	John J. Porazinski	Develop Installation / Commissioning Standard													
STD1150		Commissioning Standards Issued for Comments		10	14-Dec-27	27-Dec-27	Not Started		SG	John J. Porazinski	Commissioning Standards Issued for Comments													
STD1120		Commissioning Standards Review & Comments		10	28-Dec-27	10-Jan-28	Not Started		SG	John J. Porazinski	Commissioning Standards Review & Comments													
STD1130		Commissioning Standards Issue for Approval		5	11-Jan-28	18-Jan-28	Not Started		SG	John J. Porazinski	Commissioning Standards Issue for Approval													
STD1140		Commissioning Standards Sign-off		5	19-Jan-28	25-Jan-28	Not Started		SG	John J. Porazinski	Commissioning Standards Sign-off													
		80089926.05.11.4 Work Methods		140	04-Aug-27	23-Feb-28						23-Feb-28, 80089926.05.11.4 Work Methods												
STD1160		Develop Installation / Work Methods Standard		80	04-Aug-27	29-Nov-27	Not Started		SG	John J. Porazinski	Develop Installation / Work Methods Standard													
STD1200		Work Methods Standards Issued for Comments		20	30-Nov-27	27-Dec-27	Not Started		SG	John J. Porazinski	Work Methods Standards Issued for Comments													
STD1170		Work Methods Standards Review & Comments		20	28-Dec-27	25-Jan-28	Not Started		SG	John J. Porazinski	Work Methods Standards Review & Comments													
STD1180		Work Methods Standards Issue for Approval		10	26-Jan-28	08-Feb-28	Not Started		SG	John J. Porazinski	Work Methods Standards Issue for Approval													
STD1190		Work Methods Standards Sign-off		10	09-Feb-28	23-Feb-28	Not Started		SG	John J. Porazinski	Work Methods Standards Sign-off													
		80089926.05.11.5 Fire Protection		165	15-Sep-26	12-May-27						12-May-27, 80089926.05.11.5 Fire Protection												
STD1210		Develop Installation / Fire Protection Standard		120	15-Sep-26	10-Mar-27	Not Started		SG	John J. Porazinski	Develop Installation / Fire Protection Standard													
STD1250		Fire Protection Standards Issued for Comments		10	11-Mar-27	24-Mar-27	Not Started		SG	John J. Porazinski	Fire Protection Standards Issued for Comments													
STD1220		Fire Protection Standards Review & Comments		20	25-Mar-27	21-Apr-27	Not Started		SG	John J. Porazinski	Fire Protection Standards Review & Comments													
STD1230		Fire Protection Standards Issue for Approval		10	22-Apr-27	05-May-27	Not Started		SG	John J. Porazinski	Fire Protection Standards Issue for Approval													
STD1240		Fire Protection Standards Sign-off		5	06-May-27	12-May-27	Not Started		SG	John J. Porazinski	Fire Protection Standards Sign-off													
		80089926.06 PROCUREMENT LONG LEAD MATERIAL		516	25-Nov-24	20-Jun-28						20-Jun-28, 80089926.06 PROCUREMENT LONG LEAD MATERIAL												
PLLM1000		Procurement Long Lead Material - WBS Summary		516	25-Nov-24	20-Jun-28	In Progress		MAE.PURCH		Procurement Long Lead Material - WBS Summary													
		80089926.06.02 Electrical Long Lead Material		516	18-May-26	20-Jun-28						20-Jun-28, 80089926.06.02 Electrical Long Lead Material												
		80089926.06.02.2 Material Procurement - Battery System - Proc Item #1		386	18-May-26	16-Dec-27						16-Dec-27, 80089926.06.02.2 Material Procurement - Battery System - Proc Item #1												

Hyde Park BESS - Battery Energy Storage System Solution - Traditional			00: EMA - Remaining (MLJ)							08-Jun-26 11:34													
Activity ID		Activity Name	Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources														
										2026			2027				2028				2029		
										Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	
	EMAT1190	BATTERY SYSTEM - ES Reviews Design Drawings - Material Procurement	6	18-May-26	15-Jun-26	In Progress	ES-INTM	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	BATTERY SYSTEM - ES Reviews Design Drawings - Material Procurement													
	EMAT1200	BATTERY SYSTEM - ES Approves Shop Drawings - Material Procurement	20	16-Jun-26	13-Jul-26	Not Started	ES-RECI	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	BATTERY SYSTEM - ES Approves Shop Drawings - Material Procurement													
	EMAT1210	BATTERY SYSTEM - Fab / Manufacture - 15 months	300	14-Jul-26	20-Sep-27	Not Started		VEND	Vendor - Nomad	BATTERY SYSTEM - Fab / Manufacture - 15 months													
	EMAT1230	BATTERY SYSTEM - Delivery	60	21-Sep-27	16-Dec-27	Not Started	M-DELIV	VEND	Vendor - Nomad	BATTERY SYSTEM - Delivery													
	80089926.06.02.3 Material Procurement - Shelter Aisle Gear - (SWGR) Proc Item #3			513	26-May-26	15-Jun-28				15-Jun-28, 80089926.06.02.3													
	EMAT1360	SWITCHGEAR - ES Reviews Bids - Shelter Aisle Gear - Proc Item #3	1	26-May-26	08-Jun-26	In Progress		PROC	Purchasing Department	SWITCHGEAR - ES Reviews Bids - Shelter Aisle Gear - Proc Item #3													
	BATS1030	SWITCHGEAR - ES Bid Evaluation - Proc Item #3	10	11-Jun-26	24-Jun-26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald	SWITCHGEAR - ES Bid Evaluation - Proc Item #3													
	EMAT1390	SWITCHGEAR - ES Selects Contractor & Issue DR - Proc Item #3	15	25-Jun-26	15-Jul-26	Not Started	DR-M	ENG.ST.SS.ELEC	Shelley Hunter	SWITCHGEAR - ES Selects Contractor & Issue DR - Proc Item #3													
	EMAT1644	SWITCHGEAR - Technical Specifications Conformed and Sent to Procurement	20	25-Jun-26	22-Jul-26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald, I	SWITCHGEAR - Technical Specifications Conformed and Sent to Procurement													
	EMAT1654	SWITCHGEAR - DR Approved/Contracts Sent to Supplier for Review/Signature	15	16-Jul-26	05-Aug-26	Not Started		ENG.EPC.CC	Shelley Hunter	SWITCHGEAR - DR Approved/Contracts Sent to Supplier for Review/Signature													
	EMAT1310	SWITCHGEAR - ES Awards P.O Vendor Dwgs - Proc Item #3	10	06-Aug-26	19-Aug-26	Not Started		ENG.EPC.CC	Shelley Hunter	SWITCHGEAR - ES Awards P.O Vendor Dwgs - Proc Item #3													
	EMAT1300	SWITCHGEAR - Material Vendor Submits Design Drawings - Material Procurement - Proc Item #3	120	20-Aug-26	12-Feb-27	Not Started	MV-INTD	VEND	SWITCHGEAR Material	SWITCHGEAR - Material Vendor Submits Design Drawings - Material Procurement													
	EMAT1320	SWITCHGEAR - ES Reviews Design Drawings - Material Procurement - Proc Item #3	10	16-Feb-27	01-Mar-27	Not Started	ES-INTM	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	SWITCHGEAR - ES Reviews Design Drawings - Material Procurement													
	EMAT1330	SWITCHGEAR - ES Approves Shop Drawings - Material Procurement - Proc Item #3	10	02-Mar-27	15-Mar-27	Not Started	ES-RECI	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	SWITCHGEAR - ES Approves Shop Drawings - Material Procurement													
	EMAT1290	SWITCHGEAR - Fab / Manufacture - Proc Item #3 - 83 weeks	300	16-Mar-27	17-May-28	Not Started		VEND	SWITCHGEAR Material	SWITCHGEAR - Fab / Manufacture													
	EMAT1340	SWITCHGEAR - Delivery - Proc Item #3	20	18-May-28	15-Jun-28	Not Started	M-DELIV	VEND	SWITCHGEAR Material	SWITCHGEAR - Delivery - Proc Item #3													
	80089926.06.02.1 Material Procurement - GSU Transformer - Proc Item #6			485	21-Jul-26	20-Jun-28				20-Jun-28, 80089926.06.02.1													
	EMAT1604	GSU BATTERY TRANSFORMER - Engineering Issues Tech Spec, SOW & Dwgs Inputs to Package - P	20	21-Jul-26	17-Aug-26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald	GSU BATTERY TRANSFORMER - Engineering Issues Tech Spec, SOW & Dwgs Inputs to Package - P													
	EMAT1614	GSU BATTERY TRANSFORMER - ES Assembles Bid Package - Proc Item #6	10	18-Aug-26	31-Aug-26	Not Started		PROC	Glenn M. Cote	GSU BATTERY TRANSFORMER - ES Assembles Bid Package - Proc Item #6													
	EMAT1624	GSU BATTERY TRANSFORMER - Bid Period - Proc Item #6	0		28-Oct-26	Not Started	CS-BID	PURC		GSU BATTERY TRANSFORMER - Bid Period - Proc Item #6													
	A3032534	GSU BATTERY TRANSFORMER - Issue RFP - Proc Item #6	40	01-Sep-26	28-Oct-26	Not Started		PROC	Glenn M. Cote	GSU BATTERY TRANSFORMER - Issue RFP - Proc Item #6													
	EMAT1634	GSU BATTERY TRANSFORMER - ES Reviews Bids - Proc Item #6	10	29-Oct-26	12-Nov-26	Not Started		PROC	Purchasing Department	GSU BATTERY TRANSFORMER - ES Reviews Bids - Proc Item #6													
	BATS1190	GSU BATTERY TRANSFORMER - ES Bid Evaluation - Proc Item #6	15	29-Oct-26	19-Nov-26	Not Started		ENG.ST.SS.ELEC	Patrick J. Fitzgerald, I	GSU BATTERY TRANSFORMER - ES Bid Evaluation - Proc Item #6													
	EMAT1592	GSU BATTERY TRANSFORMER - ES Issue DR /PO - Proc Item #6	20	20-Nov-26	18-Dec-26	Not Started	DR-M	ENG.ST.SS.ELEC	Shelley Hunter	GSU BATTERY TRANSFORMER - ES Issue DR /PO - Proc Item #6													
	EMAT1664	GSU BATTERY TRANSFORMER - Material Vendor Submits Design Drawings - Material Procurement -	60	21-Dec-26	18-Mar-27	Not Started	MV-INTD	VEND	SWITCHGEAR Material	GSU BATTERY TRANSFORMER - Material Vendor Submits Design Drawings - Material Procurement													
	EMAT1674	GSU BATTERY TRANSFORMER - ES Reviews Design Drawings - Material Procurement - Proc Item #3	10	19-Mar-27	01-Apr-27	Not Started	ES-INTM	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	GSU BATTERY TRANSFORMER - ES Reviews Design Drawings - Material Procurement													
	EMAT1684	GSU BATTERY TRANSFORMER - ES Approves Shop Drawings - Material Procurement - Proc Item #3	10	02-Apr-27	15-Apr-27	Not Started	ES-RECI	ENG.ST.SS.ELEC	Patrick J. Fitzgerald	GSU BATTERY TRANSFORMER - ES Approves Shop Drawings - Material Procurement													
	EMAT1594	GSU BATTERY TRANSFORMER - Fab & Delivery - Proc Item #6 - 15 Months	300	16-Apr-27	20-Jun-28	Not Started	M-DELIV	VEND	TRANSFORMER Material	GSU BATTERY TRANSFORMER - Fab & Delivery - Proc Item #6 - 15 Months													
	80089926.061 PROCUREMENT MATERIAL - BALANCE OF MATERIAL			260	06-Aug-26	17-Aug-27				17-Aug-27, 80089926.061 PROCUREMENT MATERIAL - BALANCE OF MATERIAL													
	PBOM1000	Procurement Material - Balance of Material - WBS Summary	260	06-Aug-26	17-Aug-27	Not Started		MAE.PURCH		Procurement Material - Balance of Material - WBS Summary													
	80089926.061.01 Civil Balance of Material			121	16-Sep-26	12-Mar-27				12-Mar-27, 80089926.061.01 Civil Balance of Material													
	CMAT1020	Civil (BG) Material DR Complete	0		16-Sep-26	Not Started		ENG.ST.SS.CIVL		Civil (BG) Material DR Complete													
	CMAT1000	Prepare Civil Material DR	1	16-Sep-26	16-Sep-26	Not Started		ENG.ST.SS.CIVL	Joseph R. Sperry	Prepare Civil Material DR													
	CMAT1010	Civil Material Fab & Delivery	120	17-Sep-26	12-Mar-27	Not Started		VEND	Vendor - Material - T	Civil Material Fab & Delivery													
	80089926.061.1 Distribution Balance of Material			110	15-Sep-26	24-Feb-27				24-Feb-27, 80089926.061.1 Distribution Balance of Material													
	80089926.061.1.09 Distribution - Manholes			55	15-Sep-26	03-Dec-26				03-Dec-26, 80089926.061.1.09 Distribution - Manholes													
	PCMAT1160	MANHOLES - Issue Spec.& BOM to Vendor	15	15-Sep-26	05-Oct-26	Not Started		ENG.D.ENG.ENG	Jenna M. Cloutier	MANHOLES - Issue Spec.& BOM to Vendor													
	PCMAT1180	MANHOLES - Vendor Submit Updated Quote	10	06-Oct-26	20-Oct-26	Not Started		VEND	Vendor - Material - T	MANHOLES - Vendor Submit Updated Quote													
	PCMAT1170	MANHOLES - Material Vendor Manufactures	20	21-Oct-26	18-Nov-26	Not Started		VEND	Vendor - Material - T	MANHOLES - Material Vendor Manufactures													
	PCMAT1210	MANHOLES - Material Vendor Delivers	10	19-Nov-26	03-Dec-26	Not Started		VEND	Vendor - Material - T	MANHOLES - Material Vendor Delivers													
	80089926.061.1.1 Distribution - Cable - Proc Item #5			110	15-Sep-26	24-Feb-27				24-Feb-27, 80089926.061.1.1 Distribution - Cable - Proc Item #5													
	PCMAT1240	CABLE - Order through Maximo	10	15-Sep-26	28-Sep-26	Not Started		PROC	Purchasing Department	CABLE - Order through Maximo													
	PCMAT1230	CABLE - Material Vendor Manufactures - Proc Item #5	90	29-Sep-26	09-Feb-27	Not Started		VEND	Vendor - Material - T	CABLE - Material Vendor Manufactures - Proc Item #5													
	PCMAT1250	CABLE - Material Vendor Delivers - Proc Item #5	10	10-Feb-27	24-Feb-27	Not Started		VEND	Vendor - Material - T	CABLE - Material Vendor Delivers - Proc Item #5													
	80089926.061.1.2 Distribution - Conduit			55	15-Sep-26	03-Dec-26				03-Dec-26, 80089926.061.1.2 Distribution - Conduit													
	PCMAT1260	CONDUIT - Issue Spec.& BOM to Vendor	15	15-Sep-26	05-Oct-26	Not Started		ENG.D.ENG.ENG	Jenna M. Cloutier	CONDUIT - Issue Spec.& BOM to Vendor													
	PCMAT1280	CONDUIT - Vendor Submit Updated Quote	10	06-Oct-26	20-Oct-26	Not Started		VEND	Vendor - Material - T	CONDUIT - Vendor Submit Updated Quote													
	PCMAT1270	CONDUIT - Material Vendor Manufactures	20	21-Oct-26	18-Nov-26	Not Started		VEND	Vendor - Material - T	CONDUIT - Material Vendor Manufactures													
	PCMAT1290	CONDUIT - Material Vendor Delivers	10	19-Nov-26	03-Dec-26	Not Started		VEND	Vendor - Material - T	CONDUIT - Material Vendor Delivers													
	80089926.061.02 Electrical Balance of Material			260	06-Aug-26	17-Aug-27				17-Aug-27, 80089926.061.02 Electrical Balance of Material													
80089926.061.02.4 Material Procurement - AC Station Service Transformer - Proc Item #2			260	06-Aug-26	17-Aug-27				17-Aug-27, 80089926.061.02.4 Material Procurement - AC Station Service Transformer - Proc Item #2														

Remaining Level of Effort

Actual Work

Critical Remaining Work

Actual Level of Effort

Remaining Work

Milestone

Milestone

Page 8 of 11

TASK filter: NOT CLOSED.

© Oracle Corporation

Hyde Park BESS - Battery Energy Storage System Solution - Traditional			00: EMA - Remaining (MLJ)							08-Jun-26 11:34														
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources	2026			2027				2028				2029		
											Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	
		EMAT1500 AC STATION SERVICE TRANSFORMER - ES Issue DR /PO - Proc Item #2		10	06-Aug-26	19-Aug-26	Not Started		ENG.EPC.CC	Shelley Hunter	AC STATION SERVICE TRANSFORMER - ES Issue DR /PO - Proc Item #2													
		EMAT1450 AC STATION SERVICE TRANSFORMER - Fab & Delivery - Proc Item #2 - 50 Weeks		250	20-Aug-26	17-Aug-27	Not Started		VEND	TRANSFORMER MANUFACTURING	AC STATION SERVICE TRANSFORMER - Fab & Delivery													
		80089926.061.10 Telecomm Material		115	07-Dec-26	20-May-27					20-May-27, 80089926.061.10 Telecomm Material													
		COMMAT1000 TELECOMM - Issue Spec. Communications		5	07-Dec-26	11-Dec-26	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal	TELECOMM - Issue Spec. Communications													
		COMMAT1070 TELECOMM -Material Vendor Manufactures Communications		100	14-Dec-26	06-May-27	Not Started		VEND	Vendor - Material - T	TELECOMM -Material Vendor Manufactures Communications													
		COMMAT1080 TELECOMM -Material Vendor Delivers Communications		10	07-May-27	20-May-27	Not Started	M-DELIV	VEND	Vendor - Material - T	TELECOMM -Material Vendor Delivers Communications													
		80089926.062 PROCUREMENT OF SERVICE CONTRACTS		403	16-Mar-26	10-Jan-28					10-Jan-28, 80089926.062 PROCUREMENT OF SERVICE CONTRACTS													
		PSC1000 Procurement of Service Contracts - WBS Summary		403	16-Mar-26	10-Jan-28	In Progress		MAE.PURCH		Procurement of Service Contracts - WBS Summary													
		80089926.062.01 Engineer of Choice - Direct Award or Bid - Service Contracts		16	27-May-26	29-Jun-26					29-Jun-26, 80089926.062.01 Engineer of Choice - Direct Award or Bid - Service Contracts													
		80089926.062.01.2 EOC - Engineer of Choice Process (Prelim, Detailed & IFC) - Civil, Substation Design, P&C		16	01-Jun-26 A	29-Jun-26					29-Jun-26, 80089926.062.01.2 EOC - Engineer of Choice Process (Prelim, Detailed & IFC													
		TMP1250 Civil, Substation Design, P&C Engineering of Choice (EOC) - Proposal Evaluations		5	01-Jun-26 A	12-Jun-26	In Progress		ENG.ST.SS.CIVL	Rosa Sojan	Civil, Substation Design, P&C Engineering of Choice (EOC) - Proposal Evaluations													
		TMP1230 Civil, Substation Design, P&C Engineering of Choice (EOC) - EOC Committee Reviews at Monthly Meeting		0		23-Jun-26	Not Started		ENG.ST.SS.CIVL		Civil, Substation Design, P&C Engineering of Choice (EOC) - EOC Committee Reviews at Monthly Meeting													
		TMP1260 Civil, Substation Design, P&C Engineering of Choice (EOC) - DR & PO Issued to Vendor		5	23-Jun-26	29-Jun-26	Not Started		PROC	Purchasing Department	Civil, Substation Design, P&C Engineering of Choice (EOC) - DR & PO Issued to Vendor													
		80089926.062.01.11 Visual Rendering Services for Siting Support		10	27-May-26	19-Jun-26					19-Jun-26, 80089926.062.01.11 Visual Rendering Services for Siting Support													
		TMP1340 Siting Engineer of Choice (EOC) - Proposal Evaluations		5	27-May-26	12-Jun-26	In Progress		ENG.ST.SS.ELEC	Patrick J. Fitzgerald	Siting Engineer of Choice (EOC) - Proposal Evaluations													
		FCN2150 Siting Engineer of Choice (EOC) - ES Awards P.O. / Aprv CR		5	15-Jun-26	19-Jun-26	Not Started		ENG.EPC.CC	Shelley Hunter	Siting Engineer of Choice (EOC) - ES Awards P.O. / Aprv CR													
		80089926.062.2 Strategic Contractor		100	18-Aug-27	10-Jan-28					10-Jan-28, 80089926.062.2 Strategic Contractor													
		A5240 Strategic Sourcing Partner (SSP) - Contractor to Submit Final Pricing Information		40	18-Aug-27	14-Oct-27	Not Started		PROC	Purchasing Department	Strategic Sourcing Partner (SSP) - Contractor to Submit Final Pricing Information													
		A5180 Strategic Sourcing Partner (SSP) - Issue FNTP (60 days Post Siting Approval)		60	15-Oct-27	10-Jan-28	Not Started		ENG.EPC.CC	Shelley Hunter	Strategic Sourcing Partner (SSP) - Issue FNTP (60 days Post Siting Approval)													
		80089926.062.02 Constructor of Choice - Direct Award or Bid		60	14-Jul-26	06-Oct-26					06-Oct-26, 80089926.062.02 Constructor of Choice - Direct Award or Bid													
		80089926.062.02.6 Civil Construction Service / Contracts		60	14-Jul-26	06-Oct-26					06-Oct-26, 80089926.062.02.6 Civil Construction Service / Contracts													
		ACN2120 Civil Geotech Constructor of Choice (COC) - ES Assembles Bid Package		10	14-Jul-26	27-Jul-26	Not Started		PROC	Purchasing Department	Civil Geotech Constructor of Choice (COC) - ES Assembles Bid Package													
		ACN2090 Civil Geotech Constructor of Choice (COC) - ES Issues RFP / Bid Period - Civil Contractor		20	28-Jul-26	24-Aug-26	Not Started		PROC	Purchasing Department	Civil Geotech Constructor of Choice (COC) - ES Issues RFP / Bid Period - Civil Contractor													
		ACN2100 Civil Geotech Constructor of Choice (COC) - ES Reviews & Evaluates Bids - Civil Contractor		10	25-Aug-26	08-Sep-26	Not Started		ENG.ST.SS.CIVL	Joseph R. Sperry	Civil Geotech Constructor of Choice (COC) - ES Reviews & Evaluates Bids - Civil Contractor													
		ACN2160 Civil Geotech Constructor of Choice (COC) - ES Selects Contractor - Civil Contractor		10	09-Sep-26	22-Sep-26	Not Started		PROC	Purchasing Department	Civil Geotech Constructor of Choice (COC) - ES Selects Contractor - Civil Contractor													
		ACN2150 Civil Geotech Constructor of Choice (COC) - ES Awards P.O. / Aprv CR - Civil Contractor		10	23-Sep-26	06-Oct-26	Not Started		PROC	Purchasing Department	Civil Geotech Constructor of Choice (COC) - ES Awards P.O. / Aprv CR - Civil Contractor													
		COH1110 Civil Geotech Constructor of Choice (COC) - Construction Contracts Awarded		0		06-Oct-26	Not Started		PURC		Civil Geotech Constructor of Choice (COC) - Construction Contracts Awarded													
		80089926.062.04 Testing Service - Direct Award or Bid		61	23-Jun-27	16-Sep-27					16-Sep-27, 80089926.062.04 Testing Service - Direct Award or Bid													
		80089926.062.04.3 OVERALL SYSTEM COMMISSIONING SERVICES (LCE) - CONTRACTOR OF CHOICE		61	23-Jun-27	16-Sep-27					16-Sep-27, 80089926.062.04.3 OVERALL SYSTEM COMMISSIONING SERVICES (LCE) - CONTRACTOR OF CHOICE													
		A3470 TEST SERVICES - COC - Eversource Provides Spec, Tech Docs		10	23-Jun-27	06-Jul-27	Not Started		ENG.ST.SUBTEC	Carlos E. Ortiz	TEST SERVICES - COC - Eversource Provides Spec, Tech Docs													
		A3032314 TEST SERVICES - COC - Procurement Prepare Package		10	07-Jul-27	20-Jul-27	Not Started		ENG.ST.SUBTEC	Carlos E. Ortiz	TEST SERVICES - COC - Procurement Prepare Package													
		A3032524 TEST SERVICES - COC - Issue RFP		0		20-Jul-27	Not Started		PURC		TEST SERVICES - COC - Issue RFP													
		A3032324 TEST SERVICES - COC - Issue RFP - Bid Period		20	21-Jul-27	17-Aug-27	Not Started		PROC	Purchasing Department	TEST SERVICES - COC - Issue RFP - Bid Period													
		A3480 TEST SERVICES - COC - COC Evaluates Bid and Provides Quote		1	18-Aug-27	18-Aug-27	Not Started		PROC	Purchasing Department	TEST SERVICES - COC - COC Evaluates Bid and Provides Quote													
		A3490 TEST SERVICES - COC - Eversource Reviews Quote from COC		10	19-Aug-27	01-Sep-27	Not Started		PROC	Purchasing Department	TEST SERVICES - COC - Eversource Reviews Quote from COC													
		A3500 TEST SERVICES - COC - Eversource Awards Contract - Submit DR / PO Issued		10	02-Sep-27	16-Sep-27	Not Started		PROC	Purchasing Department	TEST SERVICES - COC - Eversource Awards Contract - Submit DR / PO Issued													
		80089926.07 CONSTRUCTION		213	28-Feb-28	28-Dec-28					28-Dec-28, 80089926.07 CONSTRUCTION													
		CON1010 Construction Start Milestone		0	28-Feb-28		Not Started	SOC	PM		Construction Start Milestone													
		DCN1020 Distribution - Constr Compl		0		27-Sep-28	Not Started		O&M		Distribution - Constr Compl													
		CFM Construction Finish Milestone		0		27-Sep-28	Not Started		PM		Construction Finish Milestone													
		ISD EQUIPMENT IN SERVICE DATE (requires FOOB constraint)		0		28-Dec-28*	Not Started	ISD	PM		EQUIPMENT IN SERVICE DATE (requires FOOB constraint)													
		CON1000 Construction - WBS Summary		213	28-Feb-28	28-Dec-28	Not Started		CONTR		Construction - WBS Summary													
		80089926.07.03 Site Development		110	28-Feb-28	01-Aug-28					01-Aug-28, 80089926.07.03 Site Development													
		80089926.07.03.01 Temp Site Access Road and Site Clearing		45	28-Feb-28	28-Apr-28					28-Apr-28, 80089926.07.03.01 Temp Site Access Road and Site Clearing													
		BCN2210 Site Prep, Access Roads		45	28-Feb-28	28-Apr-28	Not Started		CONTR	Construction Resources	Site Prep, Access Roads													
		80089926.07.03.02 Distribution Street & Access Road		100	13-Mar-28	01-Aug-28					01-Aug-28, 80089926.07.03.02 Distribution Street & Access Road													
		BCN2230 Distribution Ductbank @ Access Road		30	13-Mar-28	21-Apr-28	Not Started		CONTR	Construction Resources	Distribution Ductbank @ Access Road													
		BCN2240 Distribution Ductbank @ Street		70	24-Apr-28	01-Aug-28	Not Started		CONTR	Construction Resources	Distribution Ductbank @ Street													
		80089926.07.03.03 Equipment Foundations		45	28-Feb-28	28-Apr-28					28-Apr-28, 80089926.07.03.03 Equipment Foundations													
		BCN2090 Station - Foundations & Ductbanks		45	28-Feb-28	28-Apr-28	Not Started		CONTR	Construction Resources	Station - Foundations & Ductbanks													
		80089926.07.04 Substation Construction		110	24-Apr-28	27-Sep-28					27-Sep-28, 80089926.07.04 Substation Construction													
Remaining Level of Effort				Actual Work		Critical Remaining Work																		
Actual Level of Effort				Remaining Work		Milestone																		
Page 9 of 11										TASK filter: NOT CLOSED.														
										© Oracle Corporation														

Hyde Park BESS - Battery Energy Storage System Solution - Traditional			00: EMA - Remaining (MLJ)						08-Jun-26 11:34																
Activity ID		Activity Name		Remaining Duration	Start	Finish	Activity Status	Activity Flag	DEPT - DEPARTMENT	Resources															
											2026			2027				2028				2029			
												Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	
	PC1200	Vendor Engineering Verifies P&C Mark Ups and Updates As-Built Drawing Package. Submits Final Pack		30	16-Nov-28	29-Dec-28	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco															
	PC1300	Eversource Engineering Verifies Quality of Submitted P&C Drawings		20	01-Jan-29	26-Jan-29	Not Started		ENG.ST.P&C.P&I	Benjamin A. Franco															
	PC1310	P&C Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Field		5	29-Jan-29	02-Feb-29	Not Started	ES-ASB	ENG.ST.P&C.P&I	Benjamin A. Franco															
	PC1500	FE Files P&C Design Drawings at Station		30	21-Mar-29	01-May-29	Not Started	ES-FLSA	ENG.ST.P&C.P&I	Benjamin A. Franco															
	80089926.10.07 Telecomm Close Out			151	28-Sep-28	01-May-29																			
	COMM1000	Prepare Communication Field Mark Ups		60	28-Sep-28	22-Dec-28	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal															
	COMM1100	Eversource Engineering Prepares & Sends Electrical Field Mark Up Packge to Vendor		5	26-Dec-28	01-Jan-29	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal															
	COMM1200	Vendor Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Package. Submit:		30	02-Jan-29	12-Feb-29	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal															
	COMM1300	Eversource Engineering Verifies Communication Mark Ups and Updates As-Built Drawing Package		20	13-Feb-29	12-Mar-29	Not Started		ENG.ST.SUBTEC	Syed Zohair Iqbal															
	COMM1400	Communications Transmittal, Approve Final As-Built Package, and EDC Files and Issue Drawings to Fie		5	13-Mar-29	19-Mar-29	Not Started	ES-ASB	ENG.ST.SUBTEC	Syed Zohair Iqbal															
	COMM1500	Operations Files Communication Design Drawings at Station		30	21-Mar-29	01-May-29	Not Started	ES-FLSA	ENG.ST.SUBTEC	Syed Zohair Iqbal															
	80089926.10.10 Work Order / Financial Close Out			10	10-Feb-28	24-Feb-28																			
	FINCO1000	PM/ Engr/ Plant Acct Develop Prelim UOP & UOR List(s)		5	10-Feb-28	16-Feb-28	Not Started	PM	PM	Adam S. Kelley															
	FINCO1010	Review UOP, compl forms, fwd to Plant Acctng, enter in wrkbk		5	17-Feb-28	24-Feb-28	Not Started		PM	Adam S. Kelley															
	80089926.10.11 Project Management Close Out			40	28-Sep-28	24-Nov-28																			
	CSD1050	Construction Rep Finalize CASCADE Forms (within 14 days following ISD)		10	28-Sep-28	12-Oct-28	Not Started	AMK.CR	PM	Adam S. Kelley															
	CSD1060	Asset Management Complete Asset Update(s) in CASCADE		30	13-Oct-28	24-Nov-28	Not Started	AMK.AM	ENG.AM	Brandon E. Cruz															

ATTACHMENT C



**Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review**

Project Title: Hyde Park Battery Energy Storage System	☐ Prior Constructability Review Performed?
	☐ Program Related Site-Specific Review?
Project Numbers(s): 21334	Project Constructability Phase: ☒ Conceptual Engineering Design ☐ Preliminary Engineering Design ☐ Detailed Engineering Design
Date(s) of Constructability Review: 10/31/2023	
07/20/2023	
Project Scope Statement: Installation of a 10MW/20MWh utility-scale battery storage system (BESS) along Cummings Highway near Hyde Park Substation #496 in Hyde Park, MA. This constructability review is being based off an initial project walkdown and tabletop review and is being completed prior to Conceptual Engineering for the BESS development. More encompassing reviews will be completed as the project progresses through engineering stages.	

Department Review/Walkdown Attendees*:

Required [X]	Attended [X]	Department	Name	Required [X]	Attended [X]	Department	Name
☐	☒	Project Management	A.Kelley	☐	☒	S/S Engineering	P. Fitzgerald
☐	☐	Asset Management		☐	☒	P&C Engineering	E. Revi
☐	☒	Constr. Management	R. Taves, J. Didomenico	☐	☒	Capital Projects Eng.	J. Zicko, L. Williams
☐	☐	S/S Technical / LCE		☐	☒	Telecom	B. Khoury
☐	☐	S/S Operations**		☐	☒	Environmental	C. Newhall
☐	☐	Field Engineering		☐	☐	Real Estate	
☐	☐	Line Engineering		☐	☐	ROW T&D Rights & Survey	
☐	☐	Vegetation Management		☐	☐	Survey Engineering	
☐	☒	Siting	Leah Zambernardi	☐	☐	Cost Estimating	
☐	☐	Safety		☐	☒	Project Services	E. Snedeker
☐	☐	SCADA		☐	☐	Material Procurement	

* Sign-off in the table above for the Detailed Engineering Design Constructability Review indicates that all Action Items/Concerns from prior Constructability Reviews have been mitigated/resolved as documented in this form. Not every department is required to attend walkdowns for every project but should be consulted for project impact considerations.

** Internal Eversource Substation Operations must be included in the review even if external construction resources are planned

[Project Manager] certifies that this review was performed to completeness, that all applicable departments were informed of the project and have had opportunity to provide input to this review:



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

6/5/24

Name / Signature

Date

Documents available for use prior to Constructability Review					
[X]	Document	[X]	Document	[X]	Document
☒	Initial Scope -Birdseye site plan -General Arrangement	☐	Preliminary Drawings -Civil Site Drawing -Electrical Drawing -Section Drawings	☐	Prints in hand for marking up -Panel Front/Read View -Control Enclosure Layout -Floor / Wall Plans
☐	Detailed Scope Document	☒	Proposed Schedule	☒	Construction Sequence/Timeline
☐	Previous Notes / Comments / Observations	☐	Major Material / Equipment List	☐	Replace / Reuse of equipment

Action Items and Resolutions:

Item No.	Const. Review Action Item/Concern	Design Resolution/Mitigation:
1		
2		
3		
4		
5		

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
A	Engineering & Asset Management		



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
1	Scope of Work - Is scope adequately developed and sufficiently detailed for intended deliverable? - Any complimentary projects or other adjacent obsolete equipment for inclusion? - Any overlapping projects that could impact work/outages? - Consider Physical Installation, P&C Scope, Civil Scope, Engineering studies, Design standard adherence - Any remote station impacts? - Engineering Estimates for Design and Technical Support? (EE and EOC) Total Project	☐	Scope of work has been completed for both internal Eversource engineering and external Battery Energy Storage System vendor. Next steps following partial funding will be to progress engineering deliverables up to Preliminary Engineering where many of the items in the section will be addressed.
2	Materials Identified/Ordered -Any concerns with BOM/Major Materials? Review BOM lead times and project need dates -Specialty Major Materials: Identify Specialty Equipment & Materials potentials - temporary/replacement - long lead items? -Any Materials needed to facilitate construction access?	☐	Major material lead times for batteries, grounding transformers, and inverters are estimated to be at 12 months +/- . This duration has been built into the project schedule including standard procurement practices based on funding schedule.
3	Control House - Review Unmapped equipment interferences, Control/Battery Enclosure Layout, HVAC System (if equipment added), space for new or modified equipment, AC / DC station breaker availability and service loading, Panel Space, Expansion, one-line and/or riser type panelboard drawings, other.	☐	Indicative pricing effort has provided us with multiple equipment layouts for the site. Further engineering will be required to finalize layout, but the site is able to accommodate the installation of the equipment in multiple locations.
4	Ratings & Dimensions for Major Equipment. Any concerns upon field review? Equipment Rating One Line (and/or equipment thermal ratings tables) - Review for any errors in field	☒	No site concerns. As engineering progresses, we must ensure that the new access road can handle the equipment.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
5	Wall, Fence, Gate, and Foundation Considerations -Any permanent / temporary alterations or relocations required? -Pre-Project Sound Measurements Needed? Sound Walls? -Take measurements for Foundation quantities, dimensions and concrete volumes -CIP Review/Ballistic needs. Any Physical Security Requirement Consideration? -Assess condition of existing yard equipment foundations and viability for reuse (if proposed) -Review station expansion areas (if needed) for structures, old foundations, etc. -Any Steel Structural Plan Considerations? -Any impacts to construction plan?	☐	Site is currently fenced, and the future Battery Energy Storage System and associated equipment will also be fenced. Sound studies will be completed further along in the engineering process to ensure system adheres to dB requirements.
6	Below Grade / Enclosure Raceway - If possible, evaluate existing raceway and review spacing requirements. Review for any subsurface investigation needs or potential below grade obstructions. GPR/test pits needed? Geotechnical Investigation - is soil boring or resistivity investigation needed?	☐	See item 7 below.
7	Underground Obstructions -Drawing review concerns? -Is preconstruction or test excavation needed? -Is reuse of underground control cables / Conduits /replace Non-Shielded Cable feasible (if applicable) -Are spare conduits available? Are they useable? -Review if any areas will need to be plated for increased load bearing for equipment access (i.e., large Xfmr, etc.)	☐	Underground obstructions and conditions to be determined through required soil borings. It is understood that the site is a former landfill for construction debris and efforts will be made during the design process to limit excavation when possible.
8	Grounding Plan (Ground Grid) - Will footprint impact layout or other equipment? Identify locations to improve grounding. Impacts to foundations and ground grid plans. Observations on wetlands or ability to perform tests	☐	This is a new location, so a new ground grid will be installed in the location of the BESS. Coordination with the foundation plan and other equipment will be done to ensure the proper ground grid size. No site conditions pose any challenges for the ground grid.



Constructability Review Form: M6-PM-2012 - Eversource

Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
9	Nomenclature / Switching One-Line Diagram -Any obvious deviations from diagrams provided?	☒	No
10		☒	No
11	IEC 61850 Substation Automation and Communication Requirements - Any Impacts/Retrofit needed?	☐	No
12	Any Upgrades to ES Standards - Is retrofit needed to bring up to ES standards? Can that be done?	☐	Capital Projects Engineering has written technical specifications used in the Battery Energy Storage System Request for Proposal. These technical specifications are being reviewed and cross referenced with a draft document that the Eversource Standards group has drafted.
B	Construction		
1	Any safety hazards specific to the facility? i.e., Low Bus, Deterioration, etc.	☐	No
2	Prior known Site-Specific Conditions -Common or Obvious station Issues -Missing scope items that could impact costs?	☐	Unknown subsurface obstructions are being carried as a risk until soil boring is completed.
3	Identify/Consider Required Depts for Construction? -Who is constructor: Internal vs External Resources -S/S Ops, GO, Test, Field Engineering, etc. -Specialty Contractors Required? -Required Oversight Needed - Estimated level of CR and LCE support needed	☐	Outside vendor will complete Construction of Battery Energy Storage System. Civil work for underground distribution circuit extension to be completed by outside vendor while cable pulling and termination to be completed by internal resources.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
4	Site Plan, Access, and Temporary Construction Facilities: -Confirm maintenance access for proposed new or replaced equipment -Verify clear footprint for yard equipment / Property Plan -General access to site, can areas outside substation be utilized? -Substation Entry Plan (driveway and drive path modifications) - Identification of Unique Requirements, i.e. (access bridges, station service, etc.) -Site Development Plan (grading and drainage) with Cut/Fill -Existing Rock Conditions? Potential Impacts (rock quantity)? -Any police details needed for deliveries, restricted hours, and/or need for extended hour approvals -Any Sanitary Civil Requirements? (Septic tanks, etc.) -Review turning radius for delivery trucks (if critical) -Any Substation Access and site/yard access Limitations? -Office Trailer Needs: Facilities, Utilities, Restroom, Container, Dumpster Space -Weather impacts to Site (Snow Placement) -Identify Material Laydown and Security -Where will new vs removed equipment go? -Construction permit for trailers outside substation fence?	☐	Layout requirements have accommodated for both maintenance clearances between equipment as well as around equipment and fence perimeter. Existing access road will be improved and utilized to access location. Further site and grading plans to be developed as engineering progresses as well as a pre-construction plan that will address the other items.
5	Unique Equipment Considerations: Gas insulated substation, Gas insulated line, Gas insulated bus, Pass Breaker, Statcom, Sync Condenser	☒	This is Eversource's second BESS installation, and this should still be considered a piece of unique/specialty equipment to be installed.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
6	Outage & Energization Plan and Construction Sequence: <i>Review electrical working clearances for equipment during construction, Detailed look at protection zone changes, Locked sequence or random outages, Outage durations minimized. Considerations:</i> -Approach Distances – Outage Requirements to maintain electrical clearances throughout proposed installation -Distribution Clearances -Clearance Holder Assignments -Drawings vs Field Conditions -Outage boundaries and interconnections -Phased approach needs for additional mobilizations -Verify outage sequence supports the planned construction sequence and timeline. (Lead Commissioning Engineer "LCE", Substation Operations, and Outage Planner input) -Any design alterations or considerations to Eliminate SCLL Conditions, Reduce Recall Time, Reduce Outage durations, Special Conditions -Additional outage requirements: Railroad Catenaries, Fiber outage, SCADA outage, Relay permissions -Seasonal or weather conditions impacts	☐	Will be completed further along in the engineering process.
7	Equipment Testing & Acceptance -Grounding plans / Resources? Will we need grounds hung? -Hi-pot testing and where will the equipment be set up?	☐	Will be completed further along in the engineering process.
8	Need for mobile generator and where connected (if needed)	☒	Mobile generators not required for construction. If construction is delayed then spot generation could be considered.
9	Need for mobile battery and where connected (if needed)	☒	Mobile batteries not required for construction. Mobile batteries could be considered as a partial solution to assist with schedule compression mitigation.
10	OT work required or Notable Permit Conditions (special hours)?	☐	TBD
11	Coordination of boundaries/seams (Identification of point of demarcation for	☐	Will be completed further along in the engineering process.



Constructability Review Form: **M6-PM-2012 - Eversource**
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
	connection to existing equipment and/or for coordination with other municipalities)		
C	Environmental		
1	Soil Pre-Characterization or Grab Sampling Completed/Planned	☐	Soil borings and characterization planned.
2	Desktop investigation and EAD Review -Required Permits -Stormwater Considerations -Dewatering requirements -Wetland and buffer review	☐	EAD completed. Below are known permitting requirements at this time: -DEP Waste Management Program Post-Closure Use Permit for authorization to construct on closed landfill facility. -If soil is to be disposed offsite from Eversource property, soil samples will need to be taken to characterize the soil and a soil management plan will need to be developed to get approval from a disposal facility or landfill. - Requires annual Tier II reporting to MassDEP for batteries. - If project is expected to disturb ≥ 1 acre of land, then will require submittal of eNOI for coverage under the EPA 2022 Construction General Permit, incl. development of a SWPPP and associated inspections are required.
3	Risk or presence of PCB's/Asbestos/Lead/SF6	☐	Possible risk. Additional communication with DEP required along with review of landfill closure documents.
4	Transformer Spill Containment Recommended?	☐	Required spill containment will be determined with further engineering.
5	Expanding Footprint / New Buildings or future Expansion Plans? (Enviro Impact)	☐	Equipment will be ground mounted. A building will not be constructed.
6	Environmental oversight considerations? (Inspection type/Interval)	☐	Oversight will be required but length and frequency will be determined at permit application and approval stages.
7	Environmental mitigations - Any matting or seasonal needs? Review D&M Plan with CR.	☒	Not anticipated at this time.
D	Siting, Project Services, and ROW T&D Rights & Survey		
1	Any regulatory concerns or approvals needed? - Any concerns noted that could be mitigated by scope/design alterations? - Likely to require visual screening? What considerations (feasibility or location)? -Zoning Relief and Permits required?	☐	Views of BESS from abutting neighborhood and concerns for losing open land could be mitigated by scope/design alterations. Visual screening will likely be required, siting BESS closer to shopping plaza & away from neighborhoods. Screening/buffering likely required for abutting residential neighborhood. Zoning Relief and Permits are required. City of Boston Article 80 & Zoning Variance.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
2	Other siting considerations?	☐	Neighborhood is concerned about losing open land. Also interested in green energy and energy savings solutions.
3	Neighbor/Abutter impacts: - Any noise, visual impacts, or sensitive abutters - Any changes to lighting or additional lighting needed?	☐	Visual impact of equipment to be reviewed with City of Boston and community and accommodations to be made during engineering phase for any required screening of faux façade.
4	Review for roadway, gate access, trailers, equip height, noise, weekend/night work, or any potential public disturbance related to work	☐	Increased traffic will be seen during construction. Post construction, minimal traffic can be expected to the location for routine maintenance purposes.
5	Installation within Eversource Rights - Is Rights and Restrictions Review required?	☐	Eversource is owner of property
6	Expanding Footprint / New Buildings / Fence line or future Expansion Plans?	☐	Location of Battery Energy Storage System to take into consideration future development of the parcel for other electrical needs.
7	Proposed Equipment on same property? - Any Encroachments	☐	No encroachments anticipated
F	Protection & Controls		
1	Panel Graphic additions and/or Modifications	☐	Not applicable
2	Space on existing panels for new equipment? (front & rear of panel fronts)	☐	Not applicable
3	Panel fronts conform to drawings	☐	Not applicable
4	Access to bring new panels into control enclosure	☐	Not applicable
5	Nameplate list and location of nameplates	☐	Will be provided as engineering progresses
6	Verifies that relays and meters that will be an interconnected project work are as shown on the drawings.	☐	As engineering progresses drawings will be reviewed to ensure proper relaying and autotransfer scheme is achieved to allow the BESS to be supplying energy
7	Temporary protection, relays, control cable needed? How long will they be required?	☐	None anticipated at this time.
8	Will relay setting changes be needed? Any additional outage requirements for changes?	☐	At this stage of the engineering, it is not a requirement to do any additional relay setting changes as this will be interconnecting onto an existing distribution circuit. As the engineering progresses, the impact of the BESS interconnection on existing protection schemes will need to be assessed to determine the need for any remote setting changes.



Constructability Review Form: M6-PM-2012 - Eversource
Substation Constructability Review

	Key Review Item Description	N/A or No Concern	Review Notes, Concerns, or Additional Info Needed
G	Telecom Engineering		
1	Review telecom equipment in existing control enclosure	☐	Not Applicable
2	Review path to get telecom circuits into the station and paths within the control house. Determine whether new underground duct or overhead tray work is needed.	☒	Fiber exists at the station and ADSS is available at 427 Cummins Highway if required. Wired or wireless options will be reviewed during the engineering phases.
3	Review current or proposed Telecom equipment power connections and validate with P&C any need for system 1 and system 2 separation.	☐	No need for system separation
4	Review current Telecom fiber diversity and validate with P&C any fiber diversity requirements.	☐	
5	Review and list any remote station impacts/updates	☐	Not Applicable
6	Confirm whether correct interface equipment is available (i.e.) for fiber termination or multiplexor termination	☐	This will be a new installation, and equipment will be specified to be installed
7	Confirm whether existing communications cabinets are enough to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
8	Review space for any required additional communications cabinets which may be required to support additional telecommunications equipment installed	☐	New installation, will require new cabinets
9	Review third party Telecommunications at substation and confirm whether additional capacity or Ground Potential Rise protection is required	☐	

Photos

(Ex. Equipment control cabinet, Breakers, MODs, CCVTs, PTs, Transformers)

ATTACHMENT D1 - D2

BESS High Level Scope

- Mid-term alternative before long-term solution is implemented
- Located off Cummings Hwy in Hyde Park

	Quantity	Total
Battery Size	1 installation	10 MW / 20 MWh
Inverter Size	6 x 2.5 MVA inverters 2 x 7.5 MVA sets	15 MVA



☐ **Manhole Installation**

- (3) concrete manholes with inside dimension of 6'x10'x8'

☐ **Conduit Installation**

- (1,270') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

☐ **Primary Underground Cable Removal**

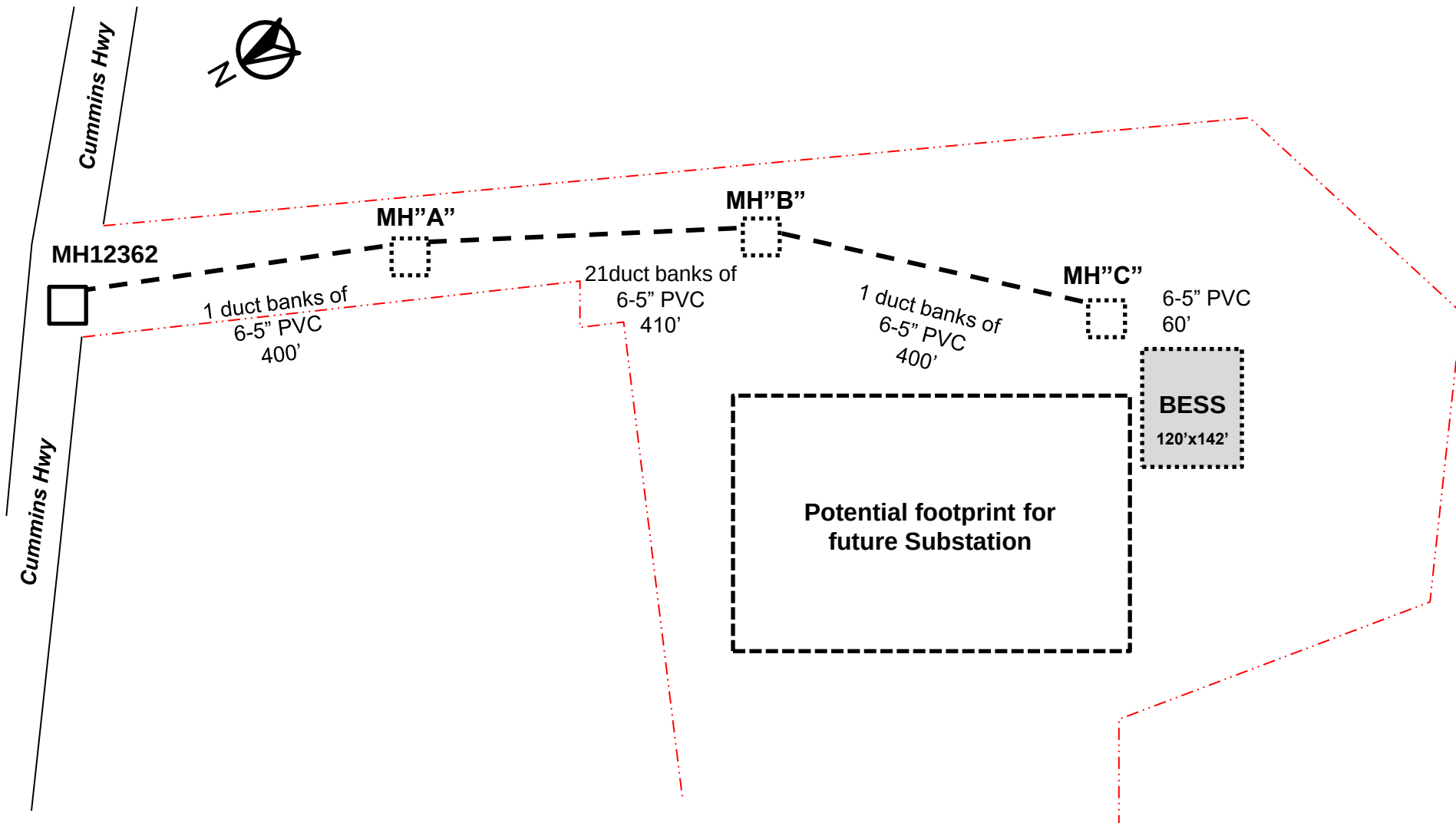
- (13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

☐ **Primary Underground Cable Installation**

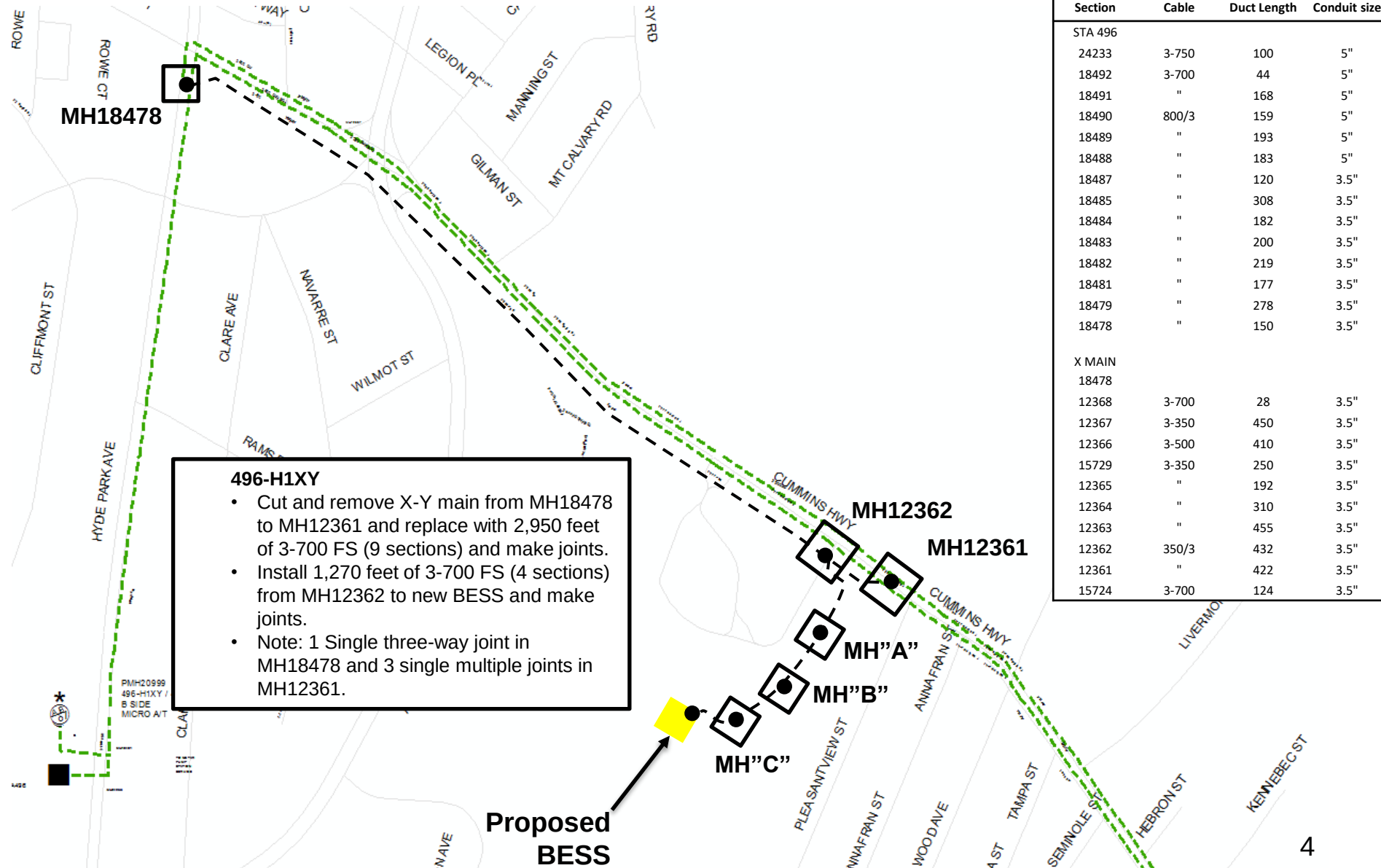
- (16,395') of 3-700 FS 15kV primary cable (47 sections)

Total Cost Estimate = \$4,459,417.00

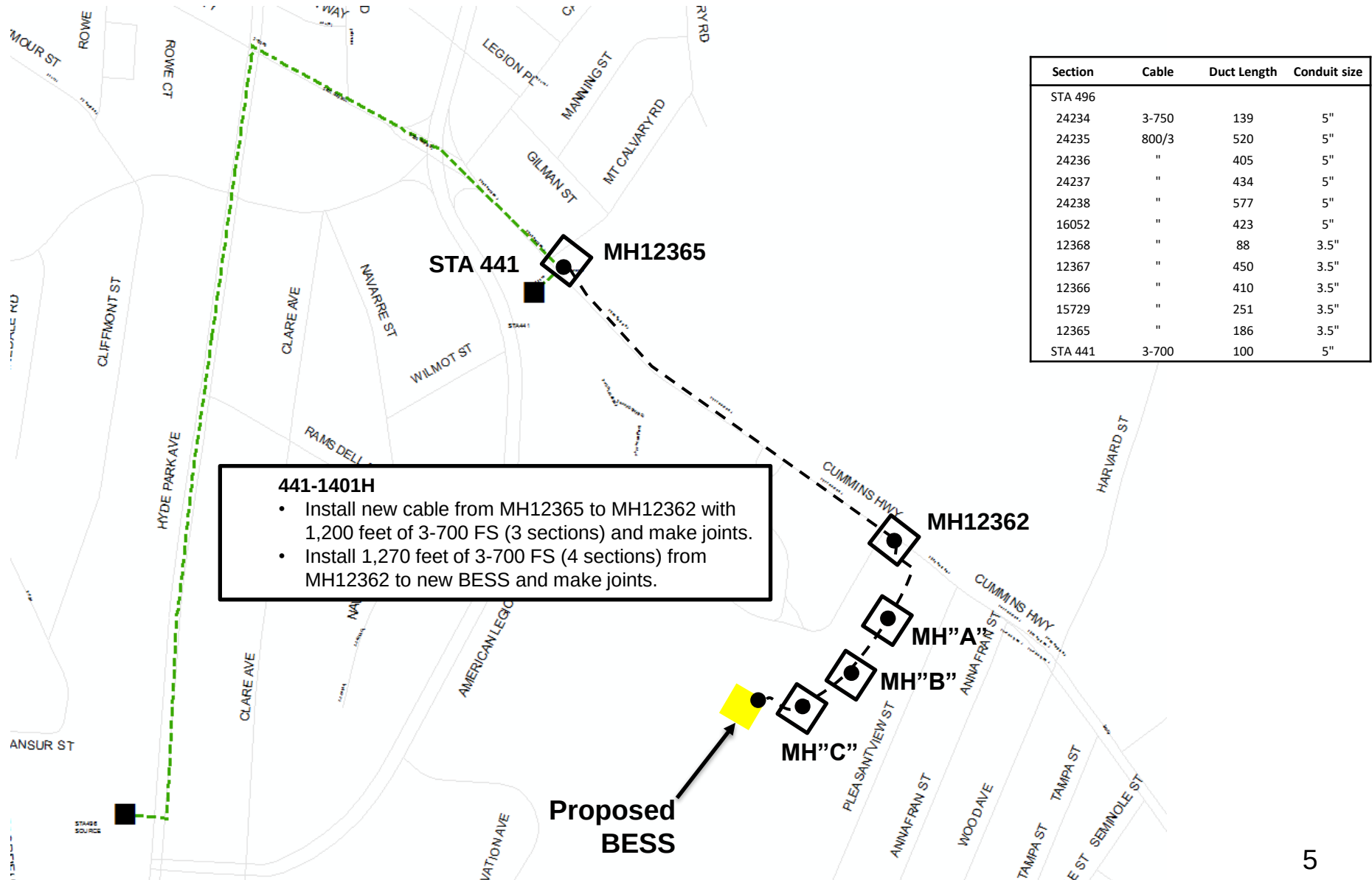
BESS Preliminary Design



BESS – 496-H1XY (Bus 1)



BESS – 441-1401H (Bus 2)



BESS – 362-1207H (Bus 3)

362-1207H

- Cut and remove cable from MH15729 to MH12361 (5 sections).
- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

MH15729

MH12362

MH12361

MH"A"

MH"B"

MH"C"

Proposed
BESS

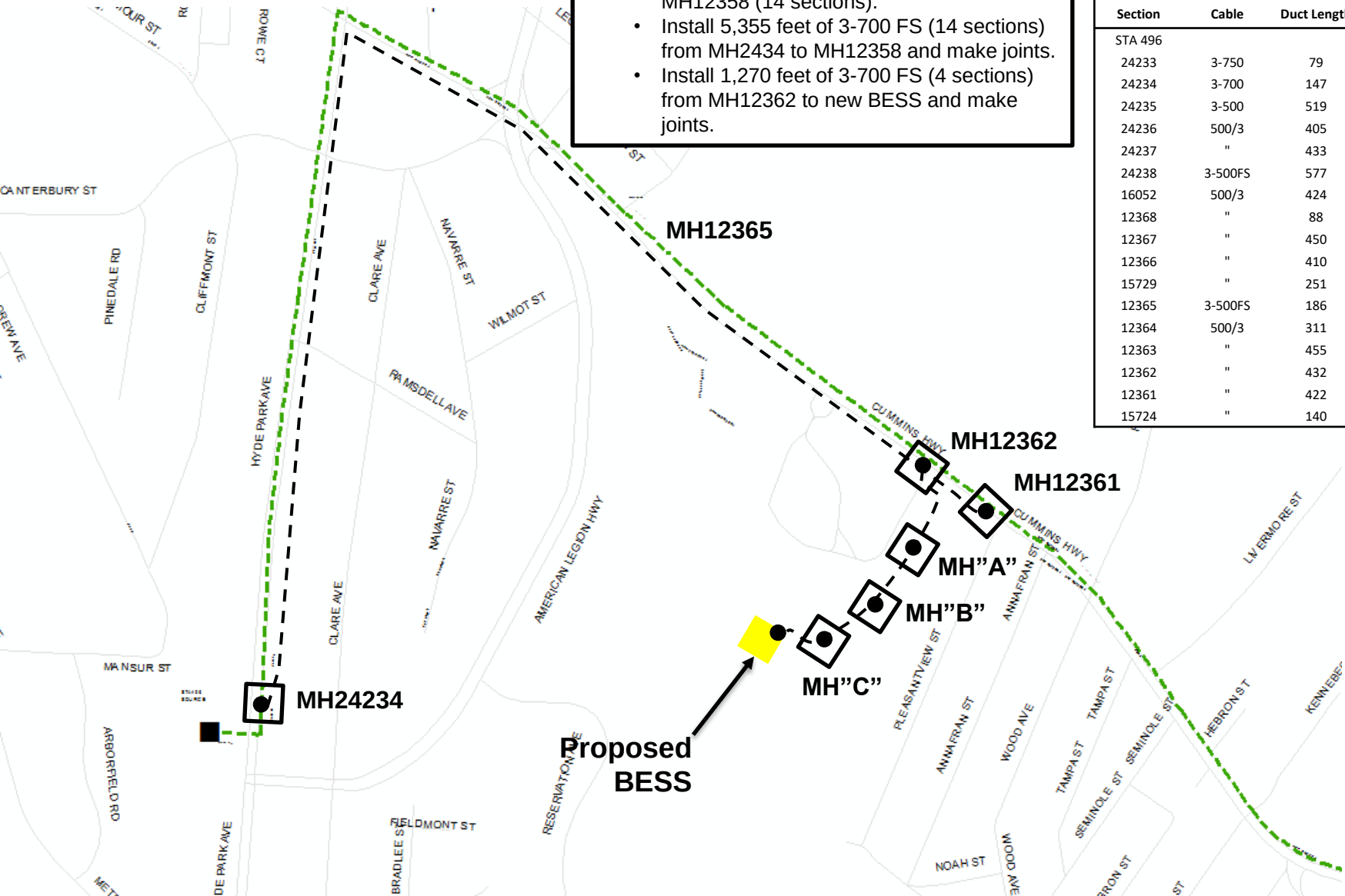
Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	125	5"
28091	3-700	390	5"
28092	"	500	5"
28093	"	500	5"
28094	"	500	5"
28095	"	500	5"
28096	"	450	5"
28097	"	500	3.5"
12366	"	150	3.5"
15729	3-700	251	3.5"
12365	500/3	185	3.5"
12364	"	310	3.5"
12363	3-500	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"

BESS – 362-1209H (Bus 4)

362-1209H

- Cut and remove cable from MH2434 to MH12358 (14 sections).
- Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	79	5"
24234	3-700	147	5"
24235	3-500	519	5"
24236	500/3	405	5"
24237	"	433	5"
24238	3-500FS	577	5"
16052	500/3	424	5"
12368	"	88	3.5"
12367	"	450	3.5"
12366	"	410	3.5"
15729	"	251	3.5"
12365	3-500FS	186	3.5"
12364	500/3	311	3.5"
12363	"	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"
15724	"	140	3.5"



BESS Preliminary Scope of Distribution Work

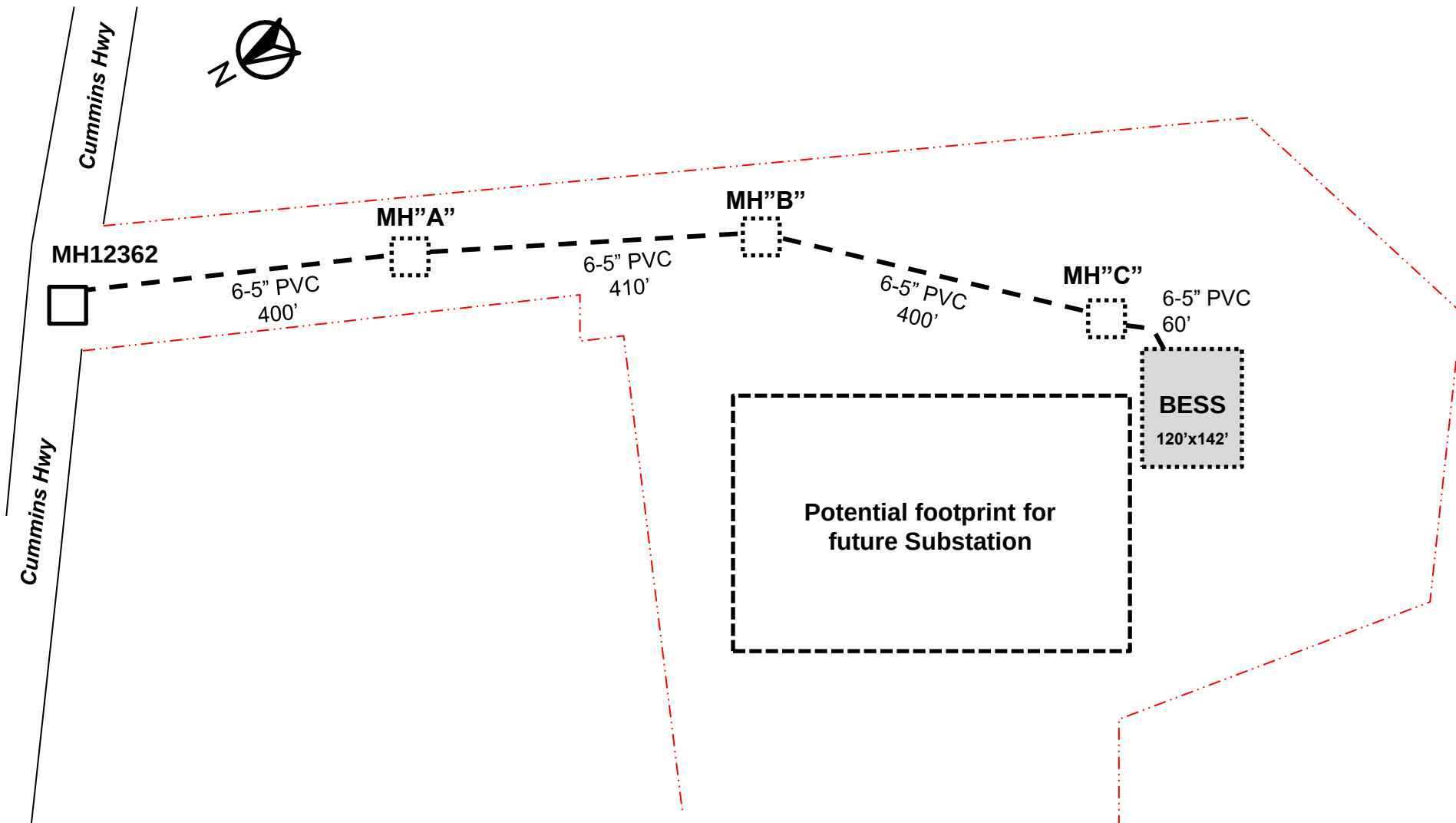
	Conduit	Manholes	Cable Install	Cable Removal	Straight Joint	Single 3-way	Single Multiple
-	1,270'	3	-	-	-	-	-
496-H7	-	-	1,270'	-	9	-	3
441-1401H	-	-	2,670'	186'	21	1	3
362-1207H	-	-	3,080'	1,810'	24	-	3
362-1209H	-	-	2,920'	1,650'	18	1	3
Total	1,270'	3	9,940'	3,646'	72	2	12

Notes:

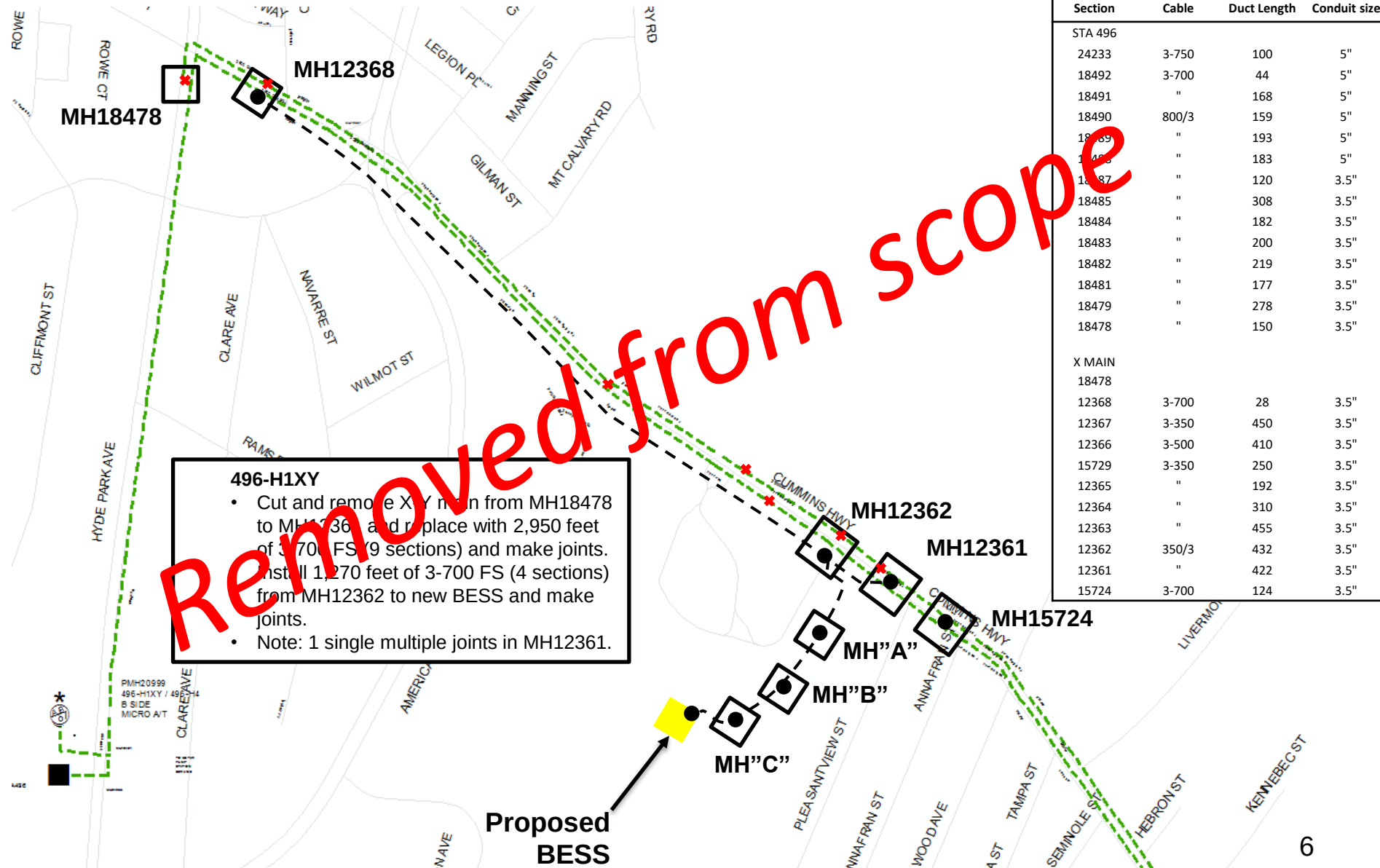
Conduit to be installed will be at least 6-5" PVC type "EB" encased in concrete for primary cable.

Manhole size will have inside dimensions of at least: 6' x 10' 8'.

BESS Preliminary Design



BESS – 496-H1XY (Bus 1)

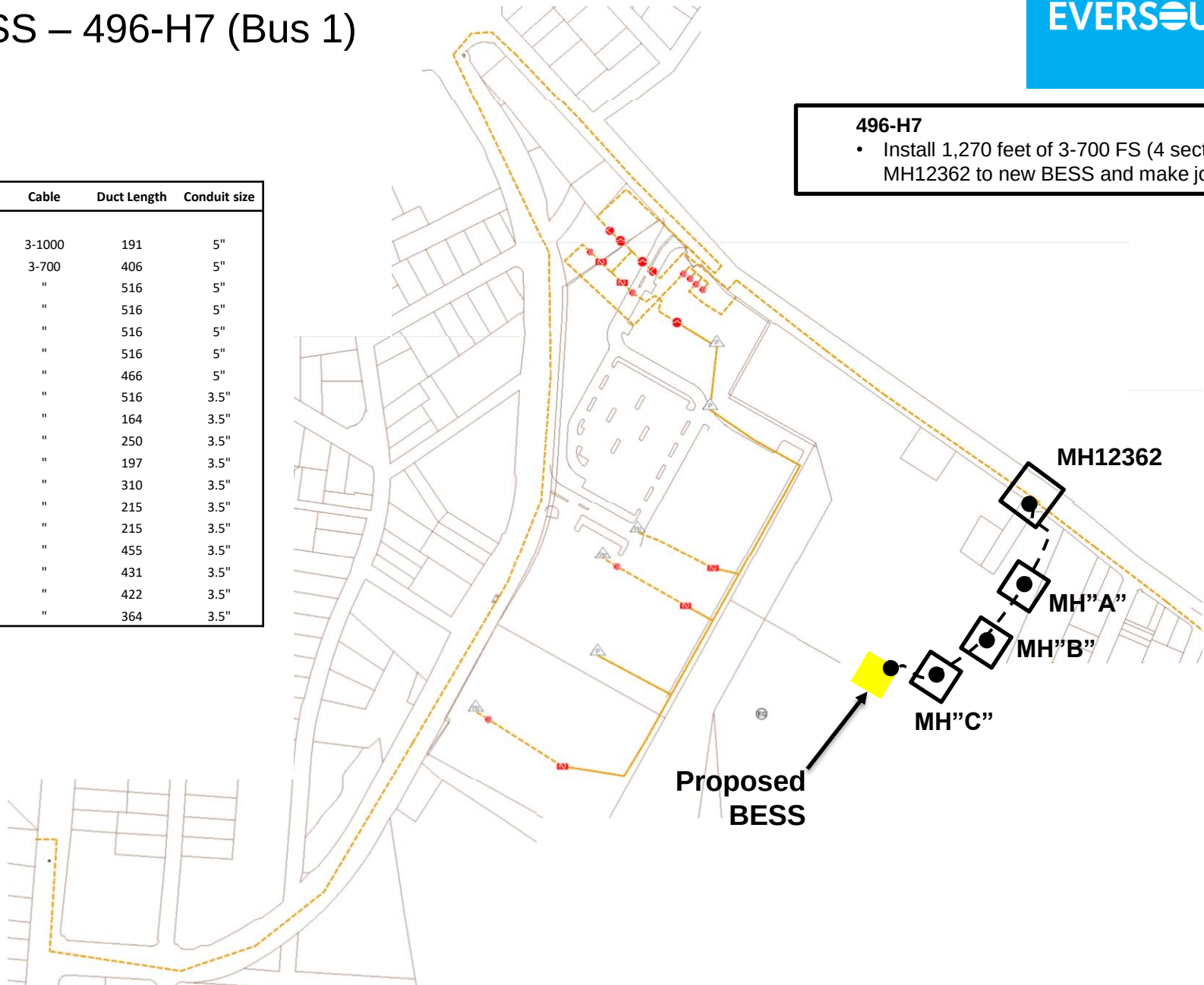


BESS – 496-H7 (Bus 1)

Section	Cable	Duct Length	Conduit size
STA 496			
24234	3-1000	191	5"
28091	3-700	406	5"
28092	"	516	5"
28093	"	516	5"
28094	"	516	5"
28095	"	516	5"
28096	"	466	5"
28097	"	516	3.5"
12366	"	164	3.5"
15729	"	250	3.5"
12365	"	197	3.5"
12364	"	310	3.5"
22934	"	215	3.5"
12364	"	215	3.5"
12363	"	455	3.5"
12362	"	431	3.5"
12361	"	422	3.5"
12360	"	364	3.5"

496-H7

- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.



Proposed
BESS

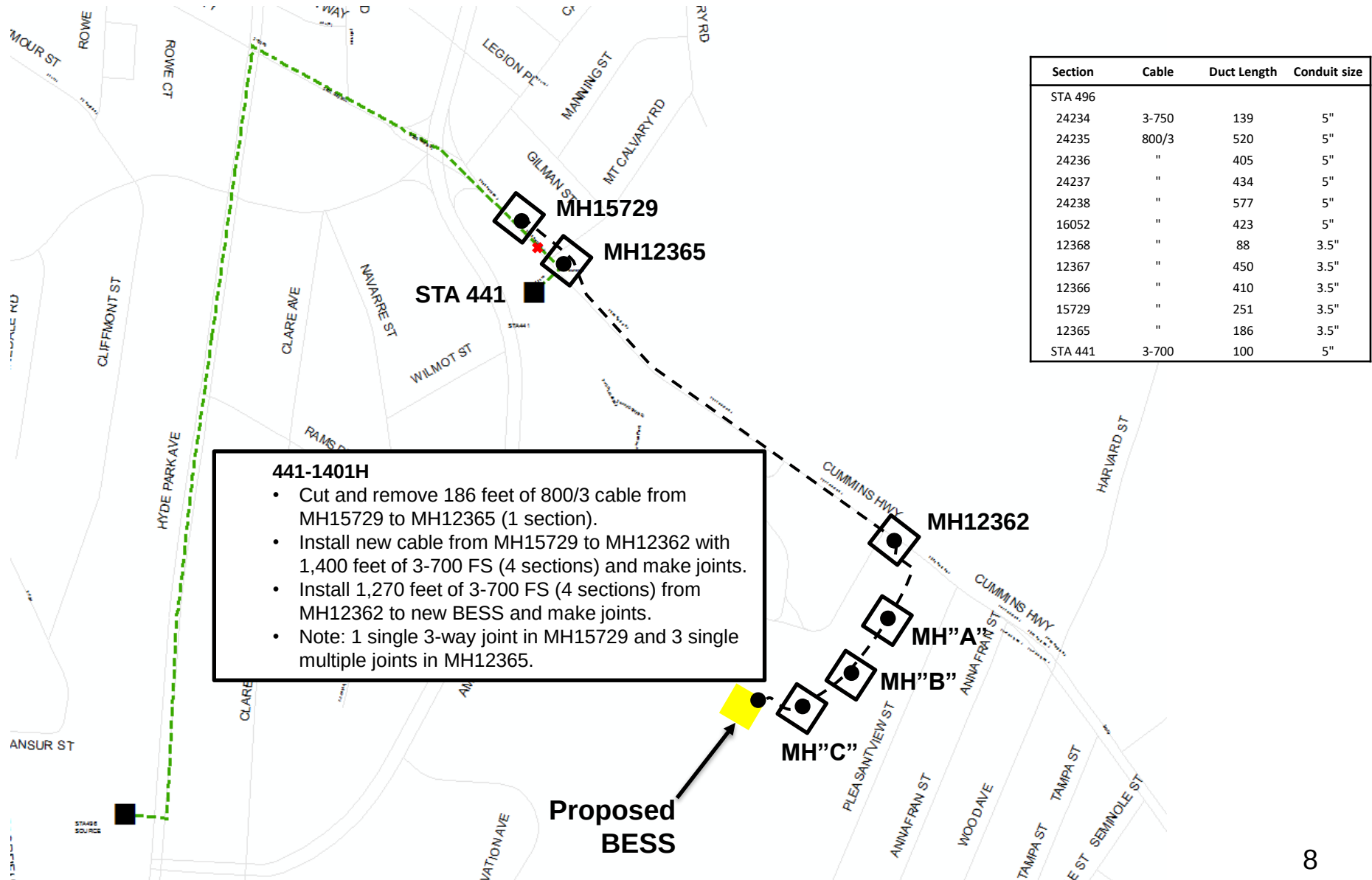
MH12362

MH"A"

MH"B"

MH"C"

BESS – 441-1401H (Bus 2)

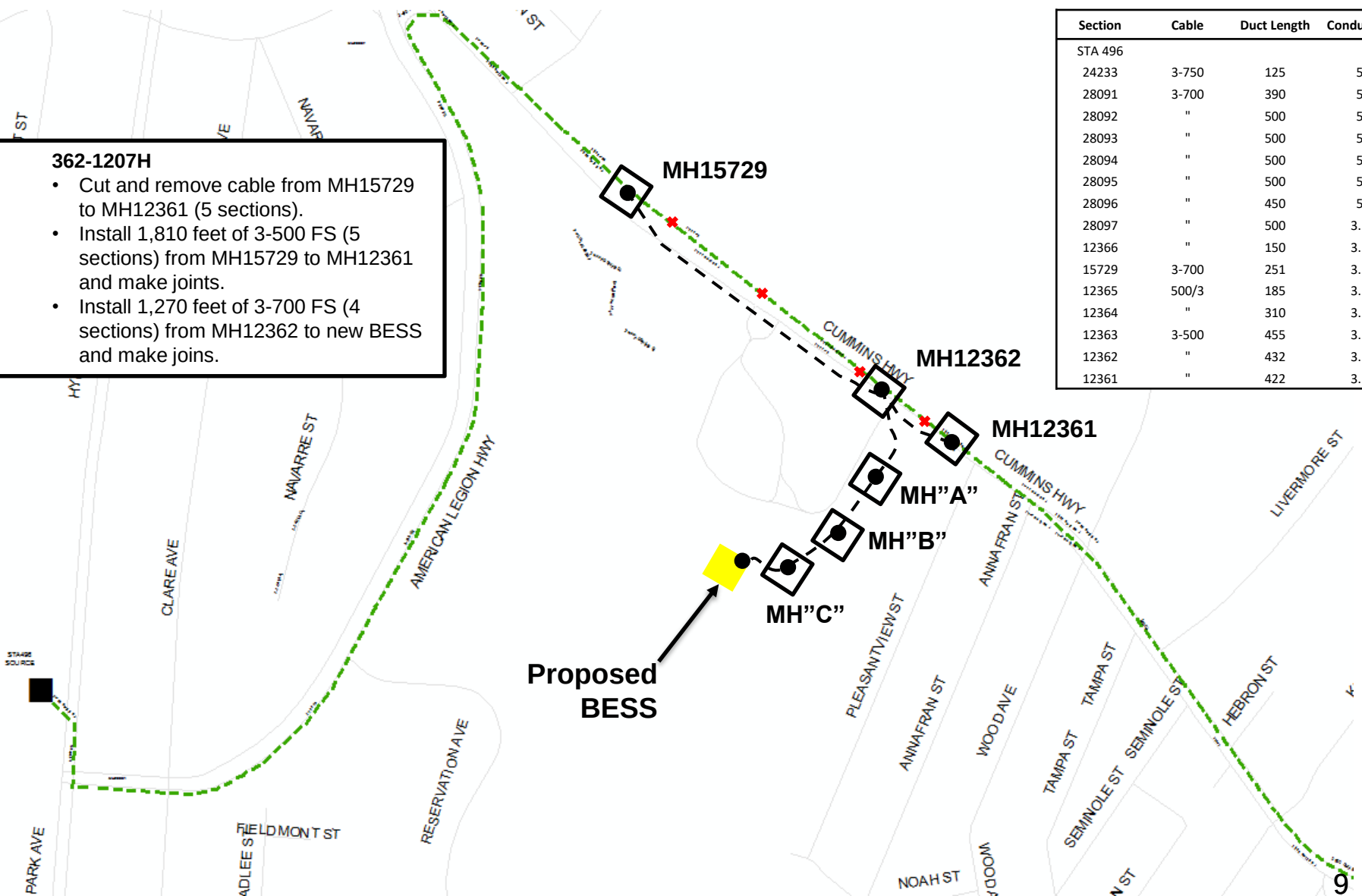


BESS – 362-1207H (Bus 3)

362-1207H

- Cut and remove cable from MH15729 to MH12361 (5 sections).
- Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

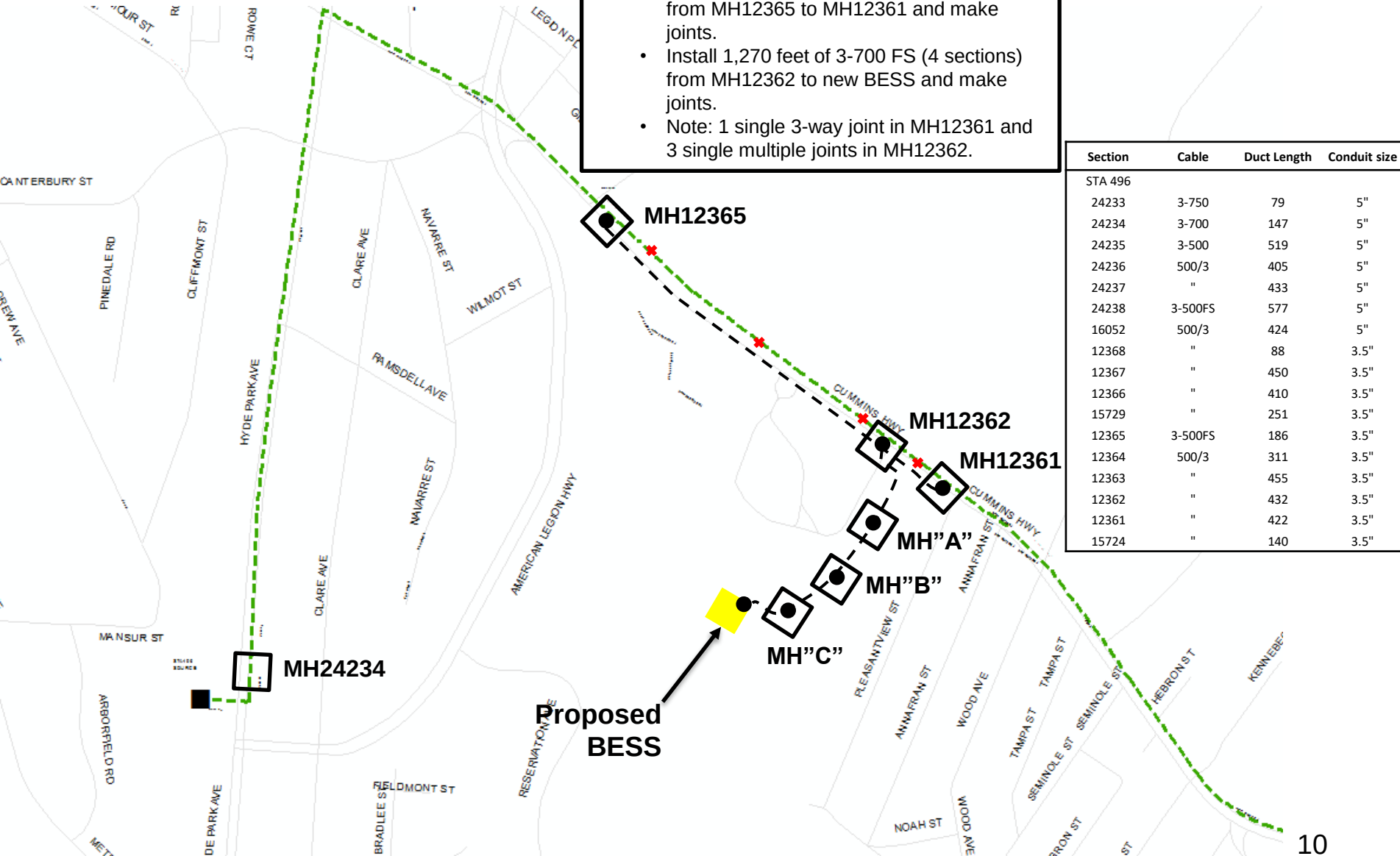
Section	Cable	Duct Length	Conduit size
STA 496			
24233	3-750	125	5"
28091	3-700	390	5"
28092	"	500	5"
28093	"	500	5"
28094	"	500	5"
28095	"	500	5"
28096	"	450	5"
28097	"	500	3.5"
12366	"	150	3.5"
15729	3-700	251	3.5"
12365	500/3	185	3.5"
12364	"	310	3.5"
12363	3-500	455	3.5"
12362	"	432	3.5"
12361	"	422	3.5"



BESS – 362-1209H (Bus 4)

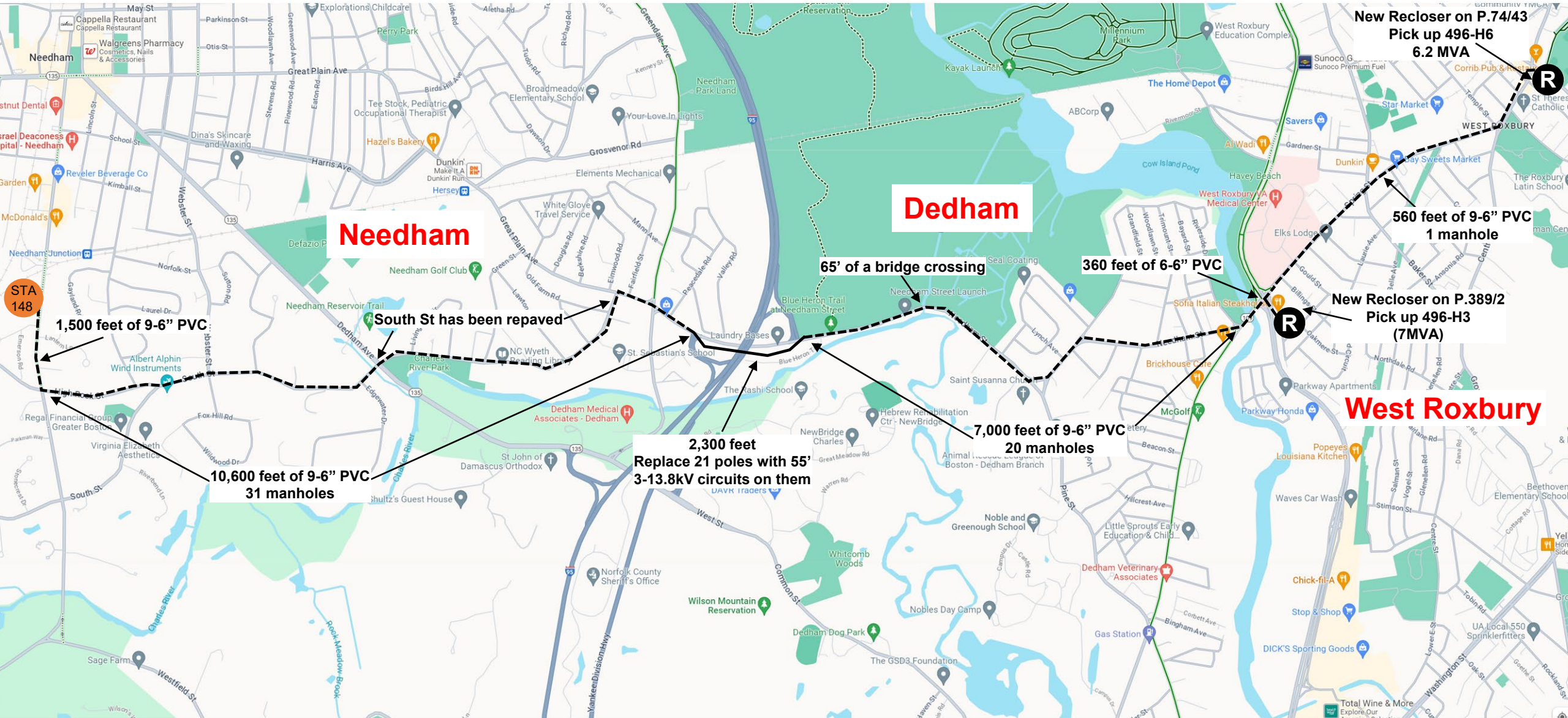
362-1209H

- Cut and remove cable from MH12365 to MH12361 (4 sections).
- Install 1,650 feet of 3-700 FS (4 sections) from MH12365 to MH12361 and make joints.
- Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.
- Note: 1 single 3-way joint in MH12361 and 3 single multiple joints in MH12362.



ATTACHMENT E

Conduit and Route from STA 148 to 496-H3 & 496-H6



Scope of Work to relieve STA 496 via STA 148.

19,600 feet of 9-6" PVC conduit

52 manholes

360 feet of 6-6" PVC conduit

350 feet of 2-5" PVC conduit

21 (55 foot poles)

To Relieve 496-H3 (7 MVA)

19,900 feet of 3-700FS

2,450 feet of 477 ACSR

2 New Reclosers

2 New Underslung

2 new 45 foot poles

To Relieve 496-H6 (6.2 MVA)

25,900 feet of 3-700FS

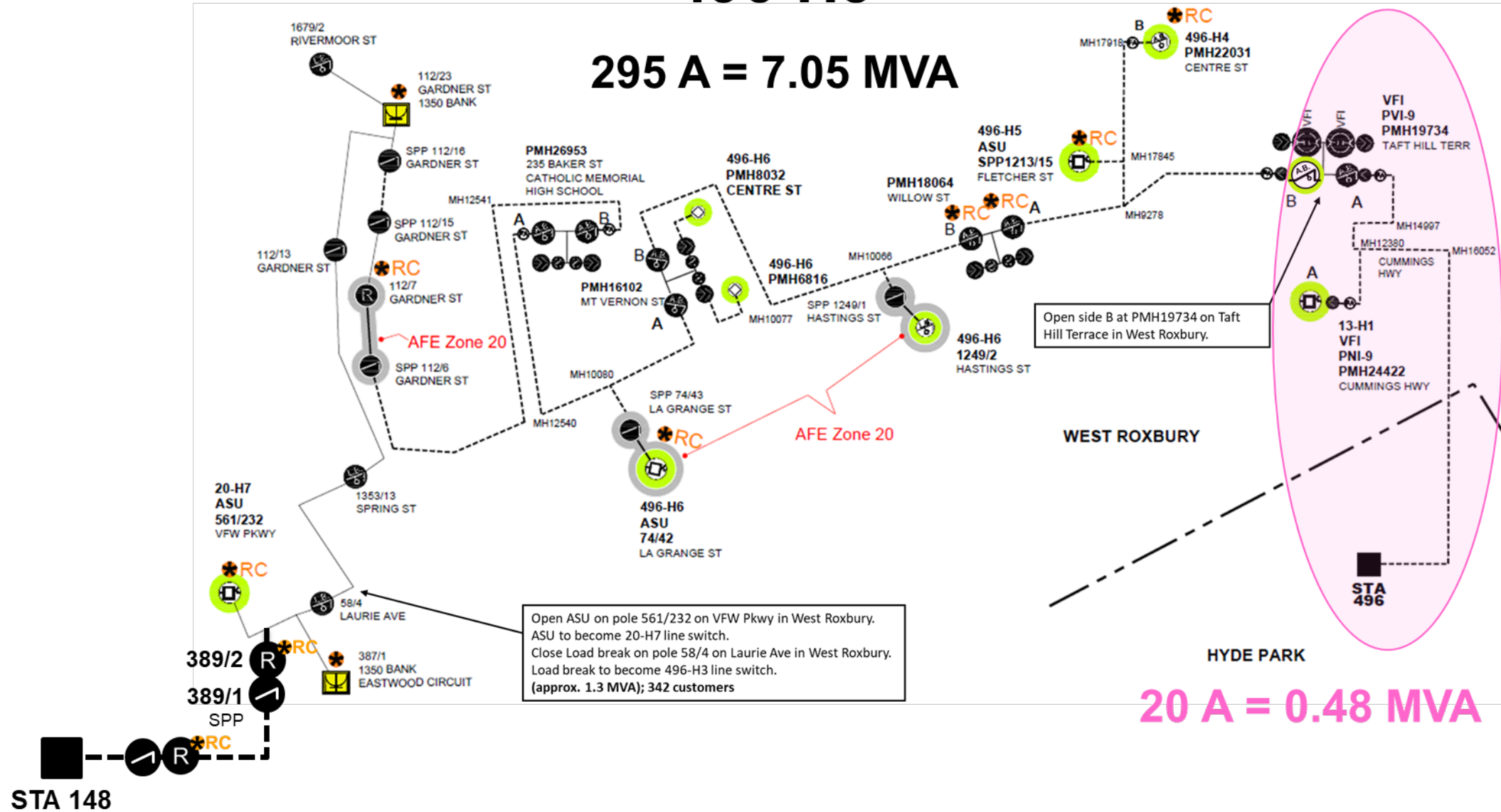
2,300 feet of 477 ACSR

2 New Reclosers

2 new Underslung

496-H3

295 A = 7.05 MVA

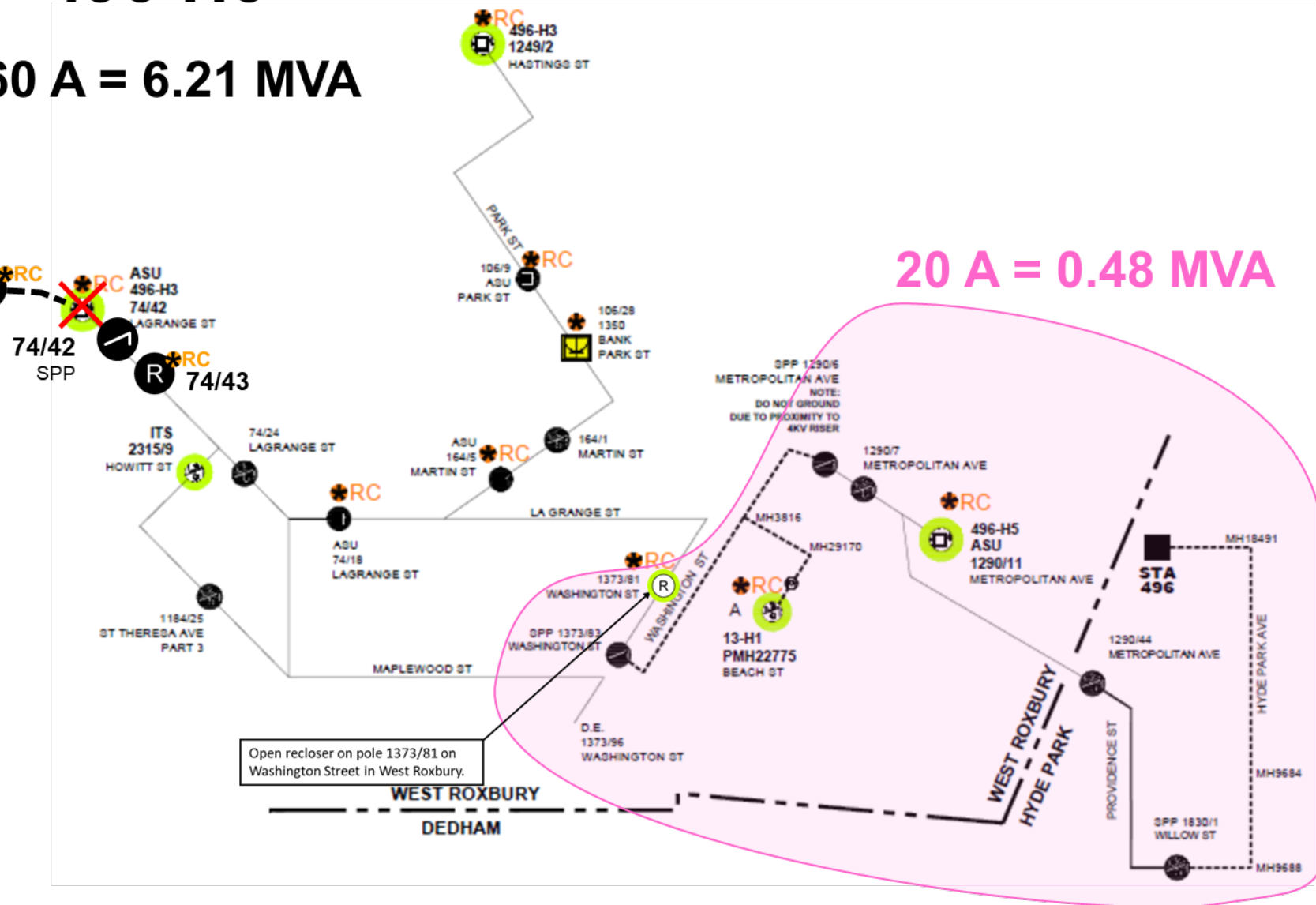


496-H6

260 A = 6.21 MVA

STA 148

20 A = 0.48 MVA



ATTACHMENT F



Hyde Park Substation

Battery Energy Storage System (BESS)

Distribution System Impact Study: Steady State and Dynamic Analysis

Report

System Planning MA

Prepared By: Mariam Chatha, Souresh Mukherjee and Syed Ali

Revision History

Date	Rev.	Description
	1	Report
6/10/2026	2	Report

TABLE OF CONTENTS

EXECUTIVE SUMMARY.....	3
RISK OF ISLANDING (ROI) & TOV SCREENINGS	
1.1 Risk of Islanding (ROI) screening	6
1.2 TOV screening.....	9
POWER FLOW STUDY EVALUATION	
1.3 Steady State Power Flow Analysis	17
1.4 Thermal Ratings	19
1.5 Reverse power flow Evaluation	21
1.6 Voltage Flicker Study	22
1.7 Short Circuit Analysis	22

Executive Summary

This report includes the dynamic analysis performed for the Battery Energy Storage System (BESS) being interconnected at four 13.8 kV Hyde Park distribution feeders at Cummins Highway in Boston on Eversource owned property. The four feeders are 496-H7, 441-1401, 362-1209H and 362-1207H supplied from Hyde Park Substation. There will be a 2.5MW/5MWH battery on each of the four feeders making the project consist of a total of 10 MW/20MWH Battery Storage. The Hyde Park BESS will be operated in parallel with Eversource's electrical grid. The BESS shall be comprised of a system of battery racks, inverters, control system, pad mounted transformers, and free standing breakers. The BESS will be located and interconnected to Eversource's existing 13.8 kV distribution infrastructure adjacent to the BESS facility.

The batteries would be dispatched with a peak load shave functionality. They will charge during the light load and discharge during the peak load in order to lower the substation peak. The substation peak exceeds the firm capacity of the station and is forecasted to keep rising due to step loads and EV charging load. The Battery can ride through multi-day peak load events. BESS can be connected to Baker St#110 side in the future through 25kV or 4kV feeders to support Baker Substation if needed, which will require underground cable construction.

Below is the preliminary distribution scope of work for the BESS interconnection to the four feeders.

Manhole Installation

-(3) concrete manholes with inside dimension of 6'x10'x8'

Conduit Installation

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

-(1,210') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes Note: this second duct bank is required only if the substation is constructed

-(60') 6-5" PVC type EB encased in concrete from MH12362 to BESS via 3 new manholes

Primary Underground Cable Removal

-(13,065') of 15kV primary cable: 350/3, 3-350 FS, 3-500 FS, etc. (37 sections)

Primary Underground Cable Installation

-(16,395') of 3-700 FS 15kV primary cable (47 sections)

Below is the scope of work for connecting the BESS to feeder 496-H7.

-Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 441-1401H.

-Install new cable from MH12365 to MH12362 with 1,200 feet of 3-700 FS (3 sections) and make joints.

-Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1207H.

-Cut and remove cable from MH15729 to MH12361 (5 sections).

-Install 1,810 feet of 3-500 FS (5 sections) from MH15729 to MH12361 and make joints.

-Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

Below is the scope of work for connecting the BESS to feeder 362-1209H.

-Cut and remove cable from MH2434 to MH12358 (14 sections).

-Install 5,355 feet of 3-700 FS (14 sections) from MH2434 to MH12358 and make joints.

-Install 1,270 feet of 3-700 FS (4 sections) from MH12362 to new BESS and make joints.

To study the impact of the BESS on the Hyde Park distribution system, a transient dynamic analysis of the preferred alternative was performed in the PSCAD (Power Systems Computer Aided Design) tool. The four Hyde Park 13.8 kV distribution feeders along with the proposed BESS systems were modeled¹ in PSCAD by using circuit load and impedance data collected from steady-state Synergi models. The dynamic Risk of Islanding (ROI) analysis, and the Transient Overvoltage (TOV) analysis were done to verify compliance with IEEE 1547-2018 Standard.

The ROI analysis determined the BESS inverter's response to unintended island conditions. According to the IEEE standard, for an unintended islanding condition, the distributed energy

¹ The BESS systems on the distribution circuits were modelled with a generic inverter model, not specific to any one manufacturer. After a vendor for the BESS is selected, the dynamic study will have to be re-run with the manufacturer specific inverter model.

resource (which in this case is the BESS) shall detect the island, cease to energize the island, and trip within 2 seconds of the formation of the island.

When interconnecting distributed energy resource there is a concern that during step changes in load (such as tripping of an upstream device), the proposed inverters may cause transient over voltages in excess of 1.2 per unit, which can potentially cause damage to the customer's equipment and/or utility equipment. The TOV analysis was conducted to verify if the expected transient overvoltage conditions caused by the BESS inverters during step changes in load were within standard limits.

The ROI and the TOV analyses were performed on the normal (N-0) configuration and several contingency (N-1) configuration scenarios. The N-1 operational scenarios included one of the four feeder outages, a transformer outage, and a bus outage. The BESS system passed the ROI test for all scenarios tested. The BESS system also does not contribute to any TOV conditions for any of the tested scenarios.

Risk of Islanding (ROI) Screening

Based on the suggested guidelines for anti-islanding screening from Sandia National Laboratories and Eversource's distributed energy resources (DER) screening for ROI assessment (shown in Figure 1), additional requirements of ROI are necessary. Further study was performed to determine if the BESS inverters are reacting appropriately and not cause violations.

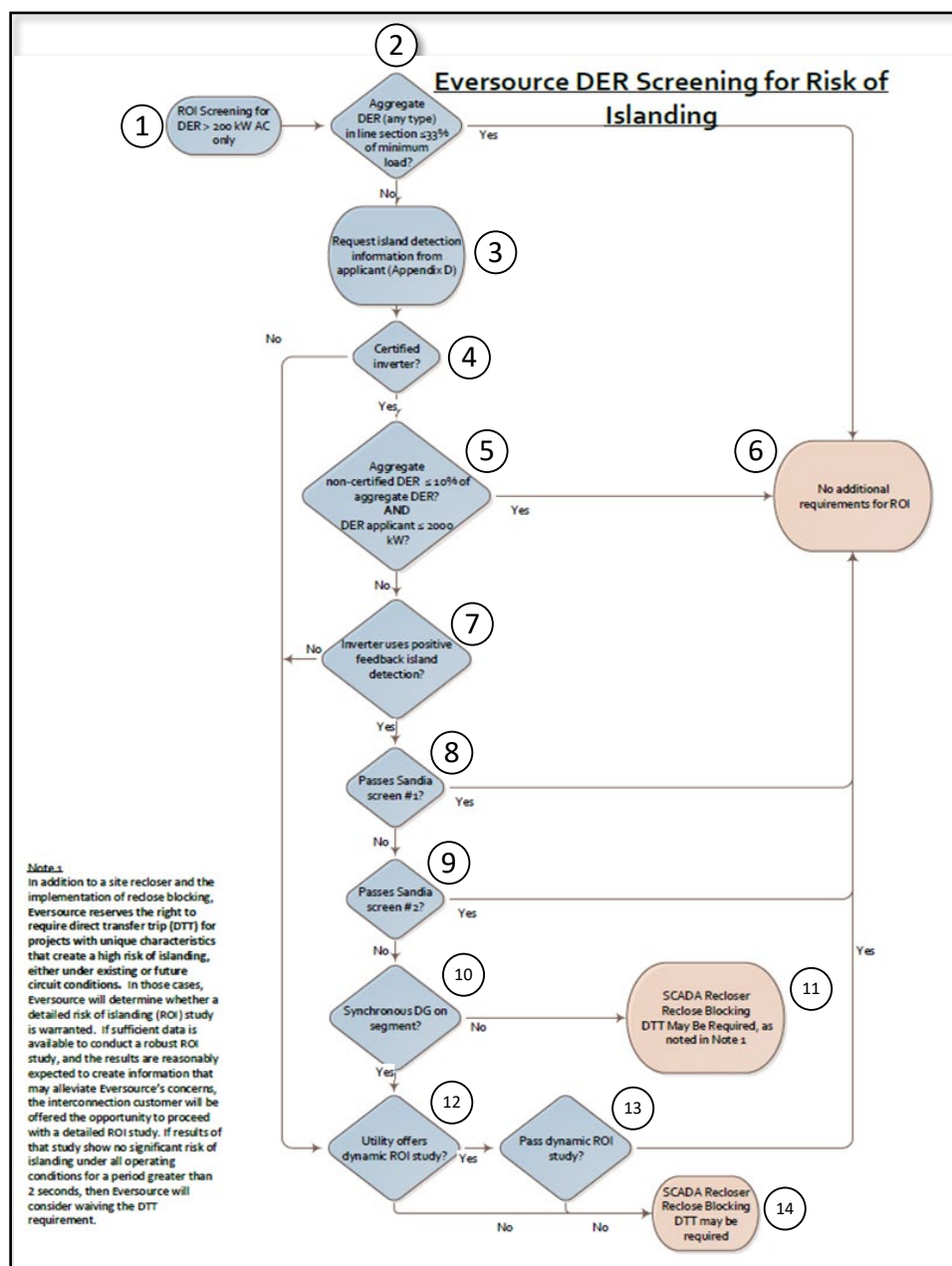


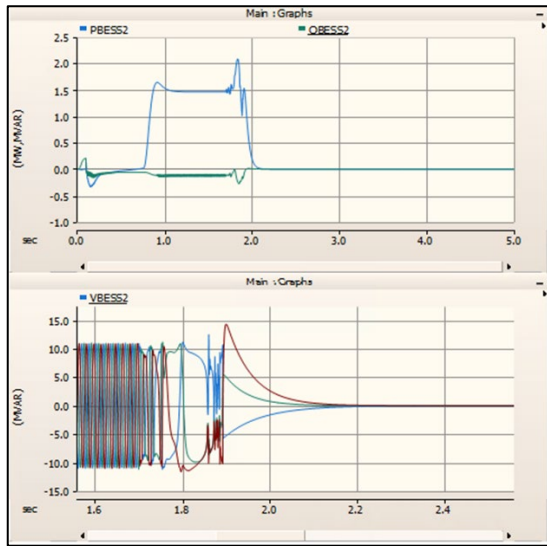
FIGURE 1. RISK OF ISLANDING (ROI) SCREENING PROCESS

During the dynamic study, in PSCAD, a detailed risk of islanding analysis was performed. The BESS system passed the ROI test for all four feeder interconnections.

Feeder	DER Interconnection	ROI Test
441-1401H	2.5 MW BESS	PASS
496-H7	2.5 MW BESS	PASS
362-1207H	2.5 MW BESS	PASS
362-1209H	2.5 MW BESS	PASS

TABLE 1: HYDE PARK BESS DYNAMIC STUDY – ROI RESULTS

PSCAD analysis shows that the BESS inverter trips within 2 seconds of unintentional island formation.



**FIGURE 2: ROI TEST FOR FEEDER 441-1401H
496-H7**

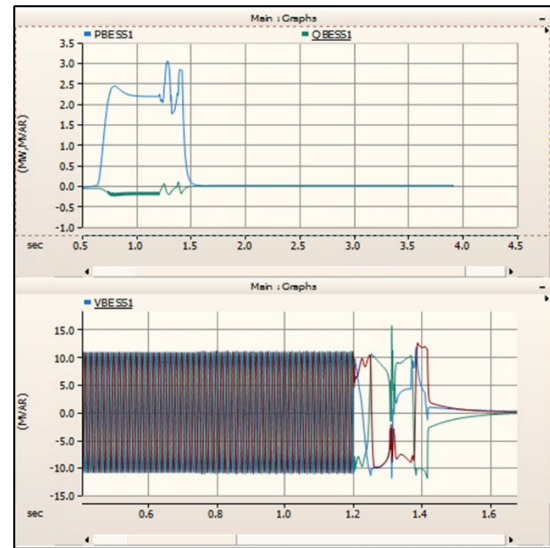


FIGURE 3: ROI TEST FOR FEEDER

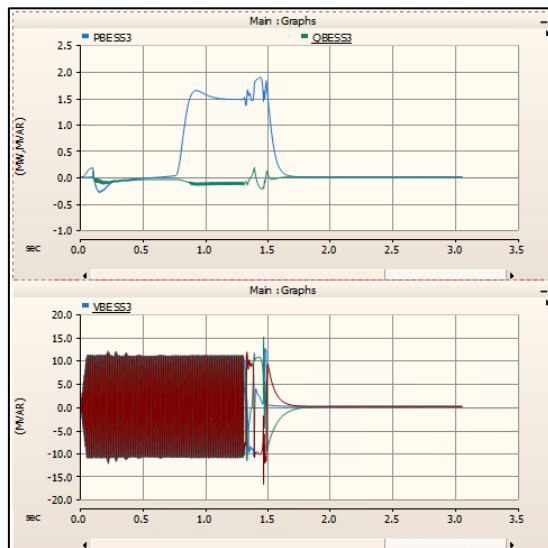


FIGURE 4: ROI TEST FOR FEEDER 362-1207H

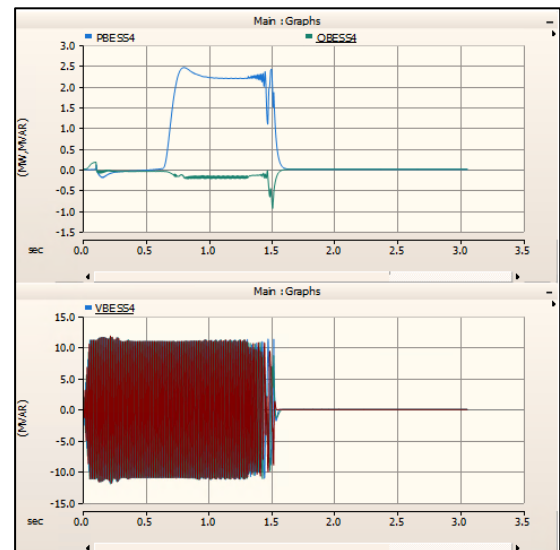


FIGURE 5: ROI TEST FOR FEEDER 362-1209H

Transient Overvoltage (TOV) Screening

If aggregate DER is greater than or equal to 115% of minimum gross load of the circuit, then there is a risk of Transient Overvoltage (TOV) at the feeder level and/or substation level. The same PSCAD dynamic study required for Risk of Islanding is further necessary to evaluate the TOV concerns. For dynamic transient overvoltage study, circuit load and impedance data are collected from steady-state Synergi models and used to build dynamic feeder models. The BESS inverters were modelled using generic PSCAD models. All other generators on the distribution circuits are modelled with a generic inverter model, not specific to any one manufacturer. The TOV analysis includes both Load Rejection Overvoltage (LROV) and Ground Fault Overvoltage (GFOV) studies.

Load Rejection Overvoltage (LROV) occurs when an upstream circuit breaker or recloser opens, which can cause an overvoltage on the isolated circuit due to stored energy in line and transformer inductances. LROV is modelled in PSCAD by operating all generation at 100% output in a minimum load case and opening the feeder circuit breaker. The voltage at the Point of Interconnection (POI) is measured.

Ground Fault Overvoltage (GFOV) is simulated by using the same minimum load, maximum generation scenario as the TOV scenario. A single-line-to-ground fault is initiated at the high side of the BESS interconnection transformer which persists until the upstream feeder breaker interrupts the fault after 10 cycles (0.16667 seconds). This is a conservative simulation and models a failure of protection selectivity.

Both LROV and GFOV durations must not exceed the following limits specified in IEEE 1547.

Transient Overvoltage Magnitude (per unit)	Maximum Allowable Duration (ms)
1.2	166
1.3	16
1.4	3
1.7	1.6
2.0	0

TABLE 2: IEEE 1547 STANDARD

Both the LROV and GFOV analyses were performed under the normal configuration and following contingency scenarios:

- i. Transformer #1 OOS

- ii. Bus #1 OOS
- iii. Bus #3 OOS

The normal configuration of the proposed BESS at Hyde Park station is shown on the one-line diagram in figure 6.

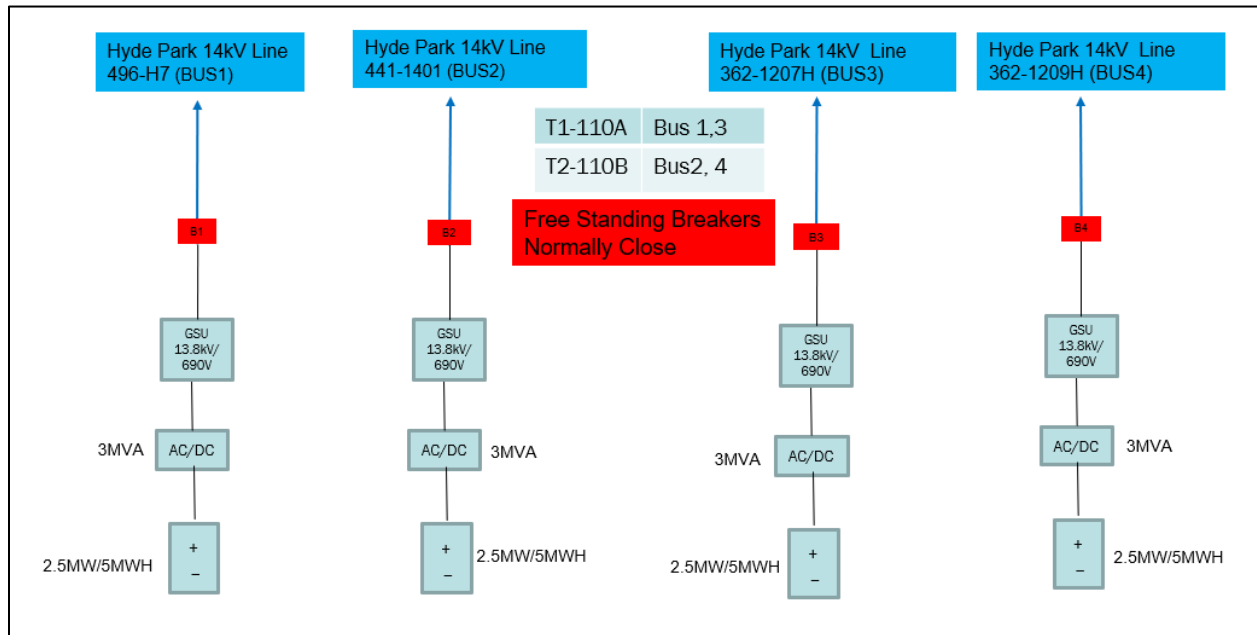


FIGURE 7: HYDE PARK REVISED BESS ONE-LINE DIAGRAM

Results at Project (BESS) POI under Normal Configuration (N-0):

Load Rejection Overvoltage (LROV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 441-1401H.

Pre-project					Post-project				
Feeder 441-1401H					Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.56	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 441-1401H BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 441-1401H.

Pre-project					Post-project				
Feeder 441-1401H					Feeder 441-1401H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 496-H7 BESS Interconnection

BESS does not contribute to load rejection overvoltage on feeder 496-H7.

Pre-project					Post-project				
Feeder 496-H7					Feeder 496-H7				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	2.10	1.80	0
1.3 pu	16	0	0	0	1.3 pu	16	1.66	1.36	0
1.4 pu	3	0	0	0	1.4 pu	3	1.26	0.98	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0.22	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 496-H7 BESS Interconnection

BESS does not contribute to ground fault overvoltage on feeder 496-H7.

Pre-project					Post-project				
Feeder 496-H7					Feeder 496-H7				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	24.54
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.52
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.20	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0.02
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results –Feeder 362-1207H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1207H.

Pre-project					Post-project				
Feeder 362-1207H					Feeder 362-1207H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	36.76	0	0
1.3 pu	16	0	0	0	1.3 pu	16	1.42	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to load rejection overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	25.84	0
1.3 pu	16	0	0	0	1.3 pu	16	0	7.88	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Results at Project (BESS) POI under Contingency Configurations (N-1):

(i) Transformer #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0	0.02
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.04	4.42	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(ii) Bus #1 Out of Service:

Load Rejection Overvoltage (LROV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0	0.08	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

(iii) Bus #3 Out of Service:

Ground Fault Overvoltage (GFOV) Results – Feeder 362-1209H Interconnection

BESS does not contribute to ground fault overvoltage on feeder 362-1209H.

Pre-project					Post-project				
Feeder 362-1209H					Feeder 362-1209H				
Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C	Voltage	IEEE 1547 - 2018 Guidelines	Phase A	Phase B	Phase C
PU	(ms)	(ms)	(ms)	(ms)	PU	(ms)	(ms)	(ms)	(ms)
1.2 pu	166	0	0	0	1.2 pu	166	0.02	4.20	0
1.3 pu	16	0	0	0	1.3 pu	16	0	0	0
1.4 pu	3	0	0	0	1.4 pu	3	0	0	0
1.7 pu	1.6	0	0	0	1.7 pu	1.6	0	0	0
2.0 pu	0	0	0	0	2.0 pu	0	0	0	0

Steady State Analysis Summary Results

Power Flow Study

The below graphs show that the four (4) 2.5MW/5 MWh battery units can reduce the peak load to approximately 95% of the Load Carrying Capacity of the Hyde Park Substation in year 2032 or approximately 148 MVA. The BESS would be interconnected to four (4) 14kV distribution feeders, 496-H7, 441-1401, 362-1209H and 362-1207H. One 2.5MW/5 MWh battery will be connected to each of the four feeders.

The project is designed to meet the 10 year forecast, where the highest load being is in 2032. Year 2032 is chosen as the year with the highest load forecast of 155 MVA for the Hyde Park Substation without any batteries installed. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load as shown below.

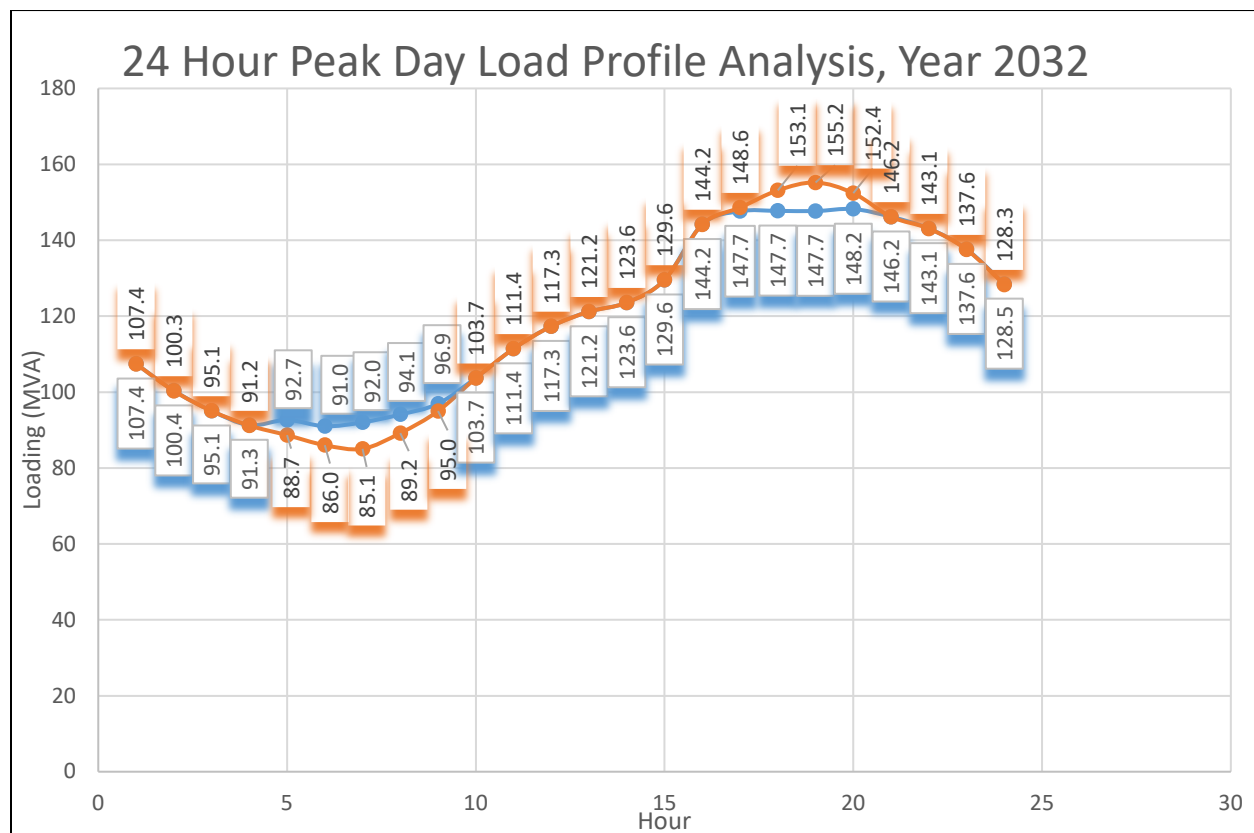


Figure 8

The orange line represents the Hyde Park Substation load without any batteries installed and the blue line represents the Substation load with the four batteries installed.

The below graph shows the resultant load profile for the substation during the peak load day under contingency assuming 13.17 MVA of distribution load transfers and the battery is operational. This decrease load to 141 MVA is due to a load transfer of 13.17 MVA which occurs during N-1 condition through distribution feeder ties and the decrease to 133.7 at 19:00 is due to the BESS.

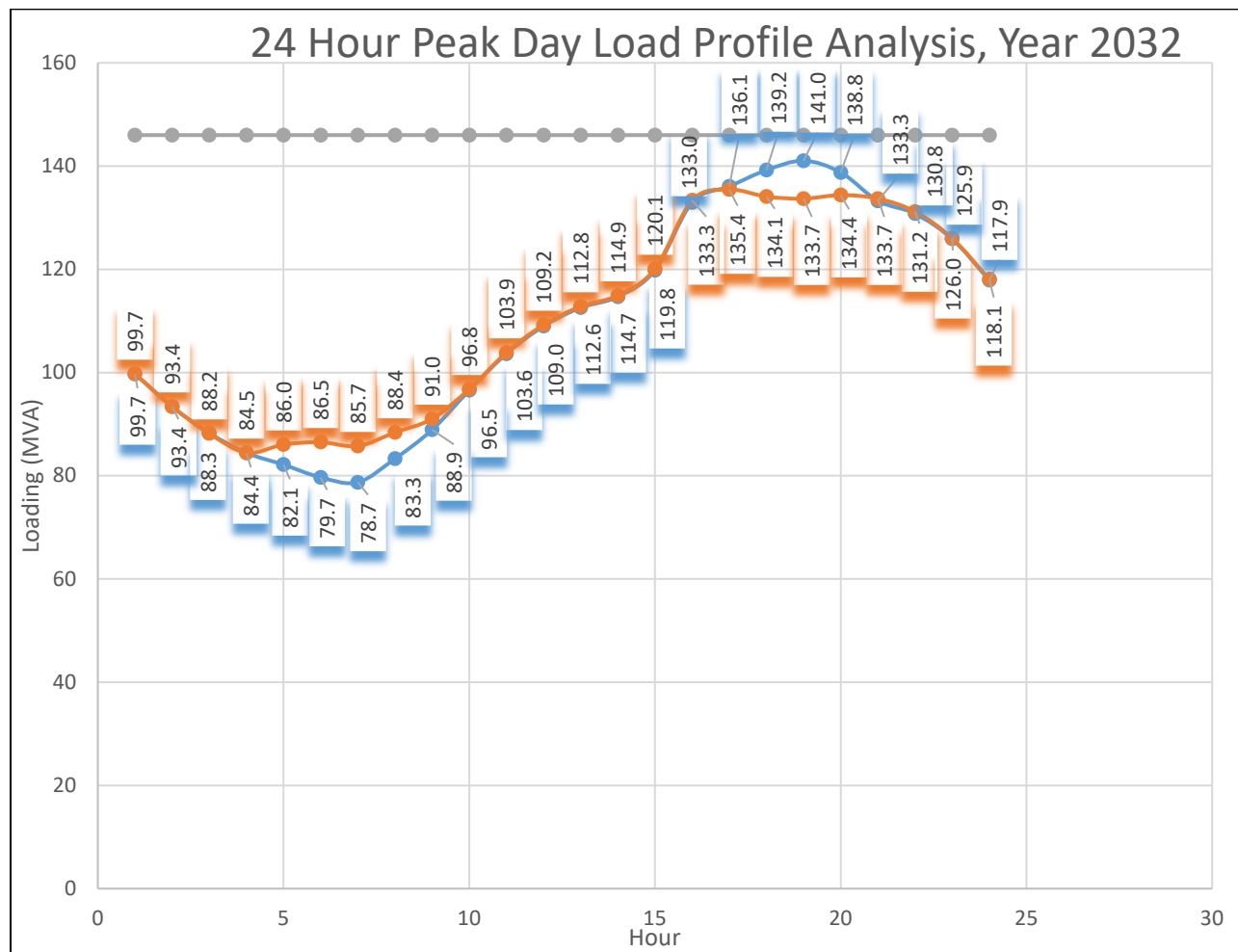


Figure 9

Thermal Rating Analysis

Below graph shows the 2032 24 hour peak day loading on each of the four feeders that the BESS is interconnected to and assuming the BESS is operational under both normal and contingency (loss of Hyde Park Substation transformer) operation. Each feeder loading is treated as independent of source outages. Therefore, the loading on each feeder remains unchanged (no additional load is transferred to or from those feeders) under contingency operating conditions. It shows the 24 hour peak load day profile for the four selected feeders in MVA based on the 2032 modeled loads. The batteries act as a load and charge from 04:00 to 09:00 and discharge from 17:00 to 21:00 to decrease the Substation peak load.

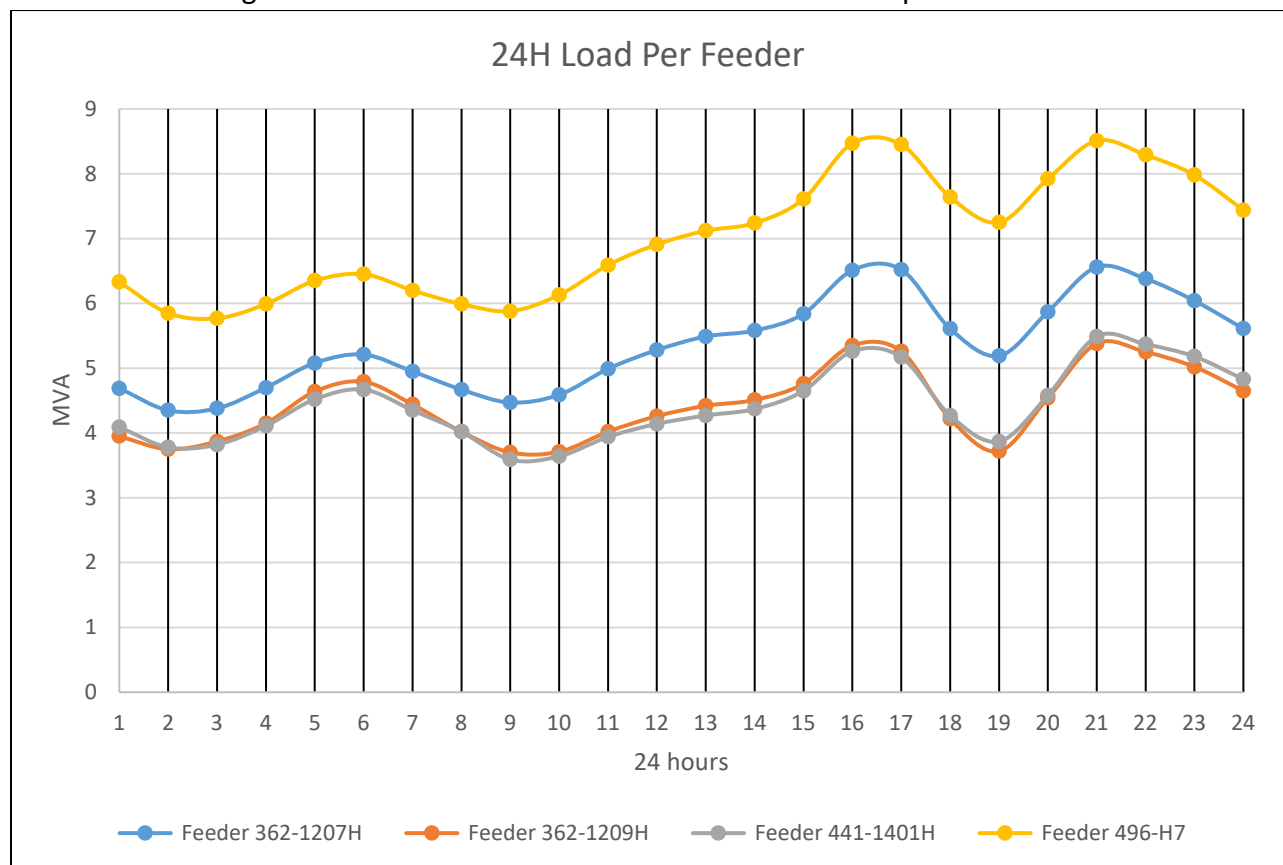


Figure 10

As shown by the below table, the circuit loading is below 80 percent of the circuit ratings when the battery is charging from 4:00 to 9:00 and acting as a load.

	Feeder_362_12 07H	Feeder_362_12 09H	Feeder_441_14 01H	Feeder_496 _H7
5:00 AM	4.9	4.4	4.4	6.4
6:00 AM	5.5	5.1	5.0	6.5
7:00 AM	5.5	5.0	4.9	6.2
8:00 AM	5.1	4.6	4.5	6.0
9:00 AM	4.6	4.0	3.8	5.9
80% of Circuit Rating (MVA)	6.72	6.72	8.72	9.28
Circuit Rating (MVA)	8.4	8.4	10.9	11.6

TABLE 4: CIRCUIT LOADING

The exact loading for the year 2032 cannot be predicted at this stage. However, based on the above loading data, there is a flexibility to charge the BESS on three of the above feeders instead of four if need be. Based on the above loading, feeder 441-1401 has the lowest amount of load and would be able to accommodate another 2.5 MW of BESS while feeder 496-H7 may not. System Planning will be monitoring and modifying the battery schedule based on the EV charging profile and its load on the Hyde Park distribution circuits.

The BESS also helps to bring down the peak load on these 14 kV feeders. The Hyde Park circuit 496-H7 historically runs 50-80% of its summer normal ampacity in the summer. The circuit 362-1207 supplied historically runs 60-80% of its summer normal in the summer months. The circuit 362-1209 historically runs 60-90% of its summer normal in the summer months.

Reverse Power Flow Evaluation

Due to the small size of the battery, there is no reverse power flow into the Hyde Park substation transformers. To evaluate the reverse power flow, the worst case scenario was analyzed where two batteries with a total size of 5MW/10MWH were connected to one feeder and the BESS was operating at a 100% output. This is assuming there is a contingency on one of the four 14 kV feeders and there would need to be two 2.5MW/5MWH BESS on one feeder. It is seen that the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery. Therefore, there is no reverse power flow because the battery would just decrease the load and is not large enough to cause reverse power flow into the electric grid.

For example, the below tables show the loading on the transformer supplying feeder 496-H7 before and after 5MW/10MWH BESS was connected to it and operating at 100%output.

Transformer: STA_496_110A_2

🔗 ⏪ ⏩ 🔍 📄 📑

Transformer

LTC / LDC

Settings

Tertiary

Net Protector

Rates

Info

Results

 View analysis results

Result	A/AB	B/BC	C/CA	Avg	Mx/T
General					
Ft Into	495578	465396	485737	482237	495578
Ft Away	---	---	---	---	0
Nom kV	ABC	ABCN	---	---	13.80
Taps and LTC					
Vrr	---	121.7	---	121.7	---
Amps Out	1599	1611	1615	1608	1615
Tap Pos	11	11	11	11	11
Volts and Amps					
Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
Volts Out	121.8	121.7	121.7	121.8	121.8
Amps Into	206	206	207	206	207
Pct pf	92.0	92.0	92.0	92.0	---
Imbalance	V,I,In	0.0%	0.3%	0 A	---
Power Flow					
MVA Into	13.70	13.70	13.70	13.70	41.10
MW Into	12.60	12.60	12.60	12.60	37.80
Mvar Into	5.38	5.38	5.38	5.38	16.14
MW Demand	12.46	12.55	12.58	12.53	37.59
Mvar Demand	3.46	3.47	3.47	3.47	10.40
Nom kvar Into	D,S,LC	0	0	0	---
Nom kW Into	D,S,LC	0	0	0	---

Power Flow Without BESS

Transformer: STA_496_110A_2



Transformer	View analysis results					
LTC / LDC	Result	A/AB	B/BC	C/CA	Avg	Mx/T
Settings	General					
Tertiary	Ft Into	495578	465396	485737	482237	495578
Net Protector	Ft Away	---	---	---	---	0
Rates	Nom kV	ABC	ABCN	---	---	13.80
Info	Taps and LTC					
Results	Vrr	---	121.8	---	121.8	---
	Amps Out	1412	1424	1427	1421	1427
	Tap Pos	11	11	11	11	11
	Volts and Amps					
	Cumul Volt Drop	0.0	0.0	0.0	0.0	0.0
	Volts Out	121.8	121.8	121.8	121.8	121.8
	Amps Into	182	182	183	182	183
	Pct pf	90.4	90.4	90.4	90.4	---
	Imbalance	V,I,In	0.0%	0.3%	0 A	---
	Power Flow					
	MVA Into	12.11	12.11	12.11	12.11	36.32
	MW Into	10.94	10.94	10.94	10.94	32.83
	Mvar Into	5.18	5.18	5.18	5.18	15.55
	MW Demand	10.81	10.91	10.94	10.89	32.66
	Mvar Demand	3.69	3.69	3.69	3.69	11.07
	Nom kvar Into	D,S,LC	0	0	0	---
	Nom kW Into	D,S,LC	0	0	0	---

Power Flow With BESS

As shown, the load on the transformer supplying that feeder had just decreased by 5 MW due to the battery and there is no reverse power flow.

Flicker

The flicker values comply with the Eversource's allowable limit of 2.0% at the distribution circuits. The voltage difference when the BESS is off and when the BESS is operating at 100% output was analyzed and was below 2% during normal and contingency scenarios. The worst case scenario where there are two BESS with a total size of 5 MW/10MWH operating at 100% output on one feeder was also analyzed. The voltage difference still remained below 2%. This voltage difference was analyzed at the peak load of 155 MVA.

Transformer LTC Impact

The tap position range from -16 to 16 and the voltage is maintained at 122 Volts at the 14 kV buses. When the BESS is connected, the voltage increases slightly due to the decrease in load. Therefore, the BESS may cause the LTC to move to a slightly lower tap position to maintain the voltage at 122 volts. Overall, due to the small size of the battery, it would not have much impact on the transformer LTC tap positions.

Short Circuit

Due to the small size of the battery, it would not have much impact on the short circuit current.